Supplementary material for "A Cheap Model and a Deterministic Checker: Fewer Silent Errors in Numeric Document Extraction from the Documents' Own Identities"

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This supplement holds the long per-run listings that the paper's appendix summarises: every run of every frozen version on every held-out set, every paired difference, the development grids, the per-run results of the private-equity case study and the version-by-version record of changes. Section, table and figure numbers prefixed with S are this supplement's; all others refer to the paper. Nothing here is new: each table is the one printed in the previous version of the paper's appendix, generated from the same result files.

S1 Private-equity case study: per-run results

Material moved from Appendix A of the paper.

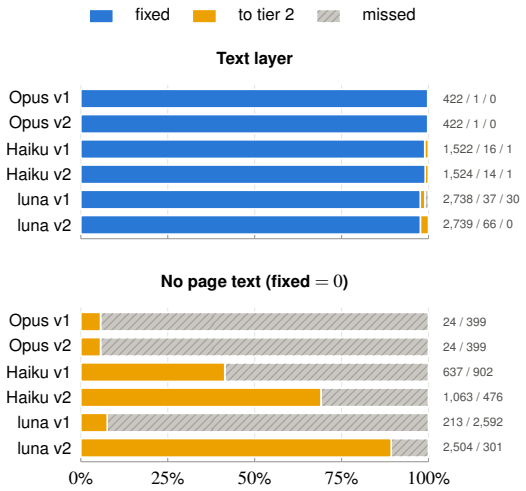


Figure S1.1: Outcomes on held-out set 2 (funds 31–39, 4,767 tier 0 errors) as a share of each model's errors, for frozen PE v1 and v2. Counts at right are fixed / tier 2 / missed with a text layer (top) and tier 2 / missed without page text (bottom). With a text layer both versions fix 97.6–99.8% of each model's errors.

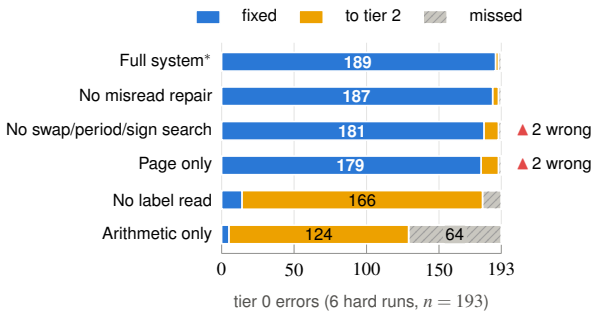


Figure S1.2: Ablations of PE v1's tier 1 on the six Fund R hard runs. Without the search, 2 label reads with the wrong sign are applied (▲). *Turning off the cited-spot check or the "not printed anywhere" flag gives the same counts. Wilson intervals are in Table S1.3.

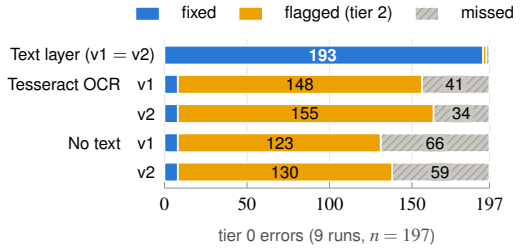


Figure S1.3: Tier 1 outcomes on Fund R (nine runs) by the page text available: the digital text layer, Tesseract OCR of the degraded scans, or none. Neither PE version made a wrong change.

Table S1.1: The five PE v1 defects found by the attack suite (text mode, 360 cases) and the v2 changes

v1 defect	v2 change
Label read took a percentage in a GP-letter sentence as a position's fair value (20 wrong changes)	No % or multiple reads for dollar fields; position reads need a Cost / Fair Value header; reads applied jointly under the guard (Section 3.2), withheld reads routed
Pages printed "in thousands" (15 wrong changes, 5 silent)	Page scale read and used in every page comparison; scaled pages confirm but never replace; copied thousands routed
A page typo "repaired" from another column of the same line (8 wrong changes)	Never misread-repair a value its own cell prints; "taken" spots by box overlap; swap and sign search also check the own line
Blank fields crashed an upstream check, so tier 1 never ran (9 crashes)	Nulls removed before checks; partial sums skipped; empty fields and starved checks routed
Signs never compared with the page (12 silent)	Page sign read; mismatches fixed from the page under the guard, or routed when unclear

S1.1 Per-run results, Fund R

Table S1.2: Claude runs, text layer, frozen v1
Errors are out of 252 numbers per run; v2 gives identical counts in text mode.

Level	Model	Errors	Fixed	To tier 2	Missed	Wrong changes
clean	Haiku	3	3	0	0	0
scan	Haiku / Sonnet / Opus	3 / 0 / 1	3 / 0 / 1	0 / 0 / 0	0 / 0 / 0	0
bad_scan	Haiku / Sonnet / Opus	58 / 21 / 18	57 / 21 / 17	1 / 0 / 0	0 / 0 / 1	0
headless	Haiku / Sonnet / Opus	67 / 21 / 8	66 / 21 / 7	1 / 0 / 0	0 / 0 / 1	0
6 hard runs		**193**	**189**	**2**	**2**	**0**
9 runs (with scan)		**197**	**193**	**2**	**2**	**0**
10 runs (with clean)		**200**	**196**	**2**	**2**	**0**

Table S1.3: Ablations, six hard runs (193 errors), frozen v1
Wilson 95% intervals are on the fixed share. An error moved closer or wrong can also be named by a tier 2 item, so rows can sum to more than 193.

Tier 1 configuration	Fixed (Wilson 95%)	Closer	Wrong changes	To tier 2	Missed
Full	189 (94.8–99.2%)	0	0	2	2
No label read	14 (4.4–11.8%)	2	0	166	13
No swap / period / sign search	181 (89.4–96.4%)	1	2	10	2
No misread-digit repair	187 (93.4–98.6%)	0	0	4	2
No page check at the cited spot	189	0	0	2	2
No "not printed anywhere" flag	189	0	0	2	2
Arithmetic only (search, no page)	5 (1.1–5.9%)	2	0	124	64
Page only (label read, no search or misread repair)	179 (88.2–95.6%)	1	2	12	2

Table S1.4: Outcomes by the page text tier 1 reads (9 runs, 197 errors): fixed / tier 2 / missed
Neither version made a wrong change.

Page text	v1	v2
Original digital text layer	193 / 2 / 2	193 / 2 / 2
Tesseract OCR of the degraded scans (default safeguards)	8 / 148 / 41	8 / 155 / 34
No text at all	8 / 123 / 66	8 / 130 / 59

Table S1.5: OCR and no-text, per run
Fixed / tier 2 / missed; v1 → v2 where they differ.

Run	No text	OCR
scan Haiku / Sonnet / Opus	3/0/0, 0/0/0, 0/0/1	3/0/0, 0/0/0, 0/1/0
bad_scan Haiku	3/36/19	3/49/6
bad_scan Sonnet	0/15/6 → 0/16/5	0/15/6 → 0/16/5
bad_scan Opus	1/10/7 → 1/12/5	1/10/7 → 1/12/5
headless Haiku	0/47/20 → 0/50/17	0/58/9 → 0/61/6
headless Sonnet	0/10/11 → 0/11/10	0/10/11 → 0/11/10
headless Opus	1/5/2	1/5/2

Table S1.6: gpt-6-luna on Fund R, text layer, frozen v1
Without page text, tier 1 fixes none of luna's Fund R errors (bad_scan: 41 to tier 2, 145 missed; headless: 35 to tier 2, 154 missed).

Level	Errors (blank)	Fixed	To tier 2	Missed	Wrong
scan	3 (0)	3	0	0	0
bad_scan	186 (85)	163	10	13	0
headless	189 (82)	170	10	9	0

S1.2 Per-run results, synthetic fund extractions

Table S1.7: Development funds (frozen v1)
Text layer; OCR and no-text rows are identical to each other.

Run	Numbers	Tier 0 errors	Text: fixed / tier 2 / missed	No text: tier 2 / missed	Tier 2 items (text / no text)
fund01 Haiku	720	333	329 / 3 / 1	209 / 124	12 / 88
fund01 Opus	720	110	109 / 1 / 0	44 / 66	8 / 6
fund03 Haiku	712	362	360 / 2 / 0	192 / 170	6 / 28
fund03 Opus	712	68	68 / 0 / 0	0 / 68	4 / 0
fund05 Haiku	744	340	338 / 2 / 0	144 / 196	11 / 53
fund05 Opus	744	105	103 / 2 / 0	1 / 104	11 / 1
fund07 Haiku	588	225	223 / 2 / 0	76 / 149	12 / 43
fund07 Opus	588	20	20 / 0 / 0	0 / 20	9 / 2
fund09 Haiku	620	248	240 / 6 / 2	97 / 151	13 / 56
fund09 Opus	620	42	42 / 0 / 0	0 / 42	6 / 0

Table S1.8: Held-out set 1, frozen v1

No run had a wrong or broken value in any mode.

Run	Numbers	Tier 0 errors	Text: fixed / tier 2 / missed	No text: fixed / tier 2 / missed	Tier 2 items (text / no text)
fund11 Haiku	720	288	285 / 3 / 0	0 / 70 / 218	14 / 27
fund11 Opus	720	103	102 / 1 / 0	0 / 8 / 95	8 / 2
fund11 luna	720	599	585 / 12 / 2	0 / 45 / 554	36 / 25
fund13 Haiku	576	210	208 / 2 / 0	4 / 14 / 192	10 / 13
fund13 Opus	576	31	31 / 0 / 0	0 / 0 / 31	6 / 2
fund13 luna	576	448	439 / 5 / 4	0 / 45 / 403	23 / 37
fund15 Haiku	748	349	343 / 6 / 0	4 / 109 / 236	11 / 133
fund15 Opus	748	93	92 / 1 / 0	0 / 31 / 62	8 / 12
fund15 luna	748	655	638 / 13 / 4	0 / 65 / 590	30 / 35
fund17 Haiku	632	180	179 / 1 / 0	0 / 23 / 157	6 / 12
fund17 Opus	632	30	30 / 0 / 0	0 / 0 / 30	3 / 0
fund17 luna	632	481	466 / 12 / 3	0 / 62 / 419	34 / 47
fund19 Haiku	752	317	316 / 1 / 0	0 / 240 / 77	11 / 69
fund19 Opus	752	67	67 / 0 / 0	0 / 54 / 13	7 / 9
fund19 luna	752	643	633 / 3 / 7	0 / 47 / 596	21 / 31

Tier 0 error kinds, development runs: misread or other 1,508, missing 229, misfiled from another field 94, misfiled from another period or document 20, wrong sign 2. Held-out set 1, Claude runs: misread or other 1,273, missing 318, misfiled from another field 53, misfiled from another period or document 16, wrong sign 8. luna: missing 2,185, misread or other 580, misfiled from another field 50, misfiled from another period or document 10, scale 1.

Table S1.9: Held-out set 2 by model, v1 and v2

Per-run rows summed by model; blanks in parentheses; f = fixed, t2 = to tier 2, m = missed. No run fixed anything without page text.

Model	Errors (blank)	Text v1: f / t2 / m	Text v2: f / t2 / m	No text v1: t2 / m	No text v2: t2 / m
Claude Opus	423 (0)	422 / 1 / 0	422 / 1 / 0	24 / 399	24 / 399
Claude Haiku	1,539 (395)	1,522 / 16 / 1	1,524 / 14 / 1	637 / 902	1,063 / 476
gpt-6-luna	2,805 (2,235)	2,738 / 37 / 30	2,739 / 66 / 0	213 / 2,592	2,504 / 301
All	**4,767 (2,630)**	**4,682 / 54 / 31**	**4,685 / 81 / 1**	**874 / 3,893**	**3,591 / 1,176**

Table S1.10: Held-out set 2 per run, v1 and v2

No run had a wrong or broken value in any mode.

Run	Numbers	Errors	Text v1: f / t2 / m	Text v2: f / t2 / m	No text v1: t2 / m	No text v2: t2 / m
fund31 Haiku	712	378	370 / 8 / 0	372 / 6 / 0	67 / 311	267 / 111
fund31 Opus	712	109	108 / 1 / 0	108 / 1 / 0	0 / 109	0 / 109
fund31 luna	712	611	599 / 7 / 5	599 / 12 / 0	51 / 560	488 / 123
fund33 Haiku	652	330	327 / 2 / 1	327 / 2 / 1	178 / 152	271 / 59
fund33 Opus	652	86	86 / 0 / 0	86 / 0 / 0	0 / 86	0 / 86
fund33 luna	652	516	502 / 11 / 3	502 / 14 / 0	41 / 475	471 / 45
fund35 Haiku	648	268	266 / 2 / 0	266 / 2 / 0	121 / 147	178 / 90
fund35 Opus	648	69	69 / 0 / 0	69 / 0 / 0	0 / 69	0 / 69
fund35 luna	648	524	510 / 6 / 8	510 / 14 / 0	61 / 463	470 / 54
fund37 Haiku	660	259	256 / 3 / 0	256 / 3 / 0	114 / 145	151 / 108
fund37 Opus	660	88	88 / 0 / 0	88 / 0 / 0	0 / 88	0 / 88
fund37 luna	660	554	540 / 8 / 6	540 / 14 / 0	40 / 514	493 / 61
fund39 Haiku	700	304	303 / 1 / 0	303 / 1 / 0	157 / 147	196 / 108
fund39 Opus	700	71	71 / 0 / 0	71 / 0 / 0	24 / 47	24 / 47
fund39 luna	700	600	587 / 5 / 8	588 / 12 / 0	20 / 580	582 / 18

Tier 0 error kinds, held-out set 2 (all runs): missing 2,630, misread or other 1,999, misfiled from another field 102, misfiled from another period or document 32, wrong sign 4.

S1.3 Label-read failure causes before and after the changes

Development funds, text mode. The rows sum finer-grained per-cause counts. A coarser tally that counts only lines with three numbers gives 531 for entity columns and 230 for repeated labels.

Cause	Before (1,055 unfixed)	After (21 unfixed)
Entity column (company or LP/GP/total) not known to the reader	536	0
Same label on several lines	227	0
Missing value skipped by the label pass	229	2
Label not found on page	15	0
Narrative field ("has invested $X of $Y"), excluded on purpose	14	14
No column, several numbers on the line (non-narrative)	29	1
Column not placed by the column picker	0	3
Right read, fix withheld by routing	5	1

S1.4 Planted errors on 30 synthetic funds (pre-freeze code)

	Plants	Fixed	Wrong change	Escalated	Missed
With page text					
mislabel-style, one at a time	797	797	0	0	0
mislabel-style, all at once	746	746	0	0	0
random swaps, one at a time	1,800	1,800	0	0	0
random swaps, 5 at once	1,500	1,500	0	0	0
adversarial: compensating / invisible / rival	51 / 37 / 122	all	0	0	0
Arithmetic only (no page text)					
mislabel-style, one at a time	797	0	0	771	26
mislabel-style, all at once	746	0	0	722	24
random swaps, one at a time	1,800	0	0	1,798	2
random swaps, 5 at once	1,500	0	0	1,497	3
adversarial: compensating	51	0	0	1	50
adversarial: invisible swap	37	0	0	1	36
adversarial: rival fix	122	0	0	122 (all "a person should pick")	0

Proposed fix in the packet (no page text)	Right fix named	Wrong fix named	Rivals (person picks)	Not flagged
mislabel-style, one at a time	670 / 797 = 84.1%	2	99	26
mislabel-style, all at once	587 / 746 = 78.7%	3	132	24
random swaps, one at a time	1,445 / 1,800 = 80.3%	1	351	2
random swaps, 5 at once	1,196 / 1,500 = 79.7%	0	301	3

Of the planted document issues on the correct extractions, all 16 GP-vs-FS *typo* issues were reported as document issues. Of 20 *mismatch* issues, 18 were reported as document issues and 2 as document issues plus a "person should pick" item. Of 18 *stale* issues, 14 were reported as document or document plus review, 3 as review only, and 1 also with a wrong-value suggestion on the narrative. All 15 revised capital accounts were correctly superseded.

S1.5 Adversarial results per category

Table S1.11: Adversarial outcomes in the PE case study, totals

The 360-case suite (funds 11–19) found PE v1's defects and is PE v2's tuning set; the fresh 430-case set (funds 31–39) was generated after the v2 freeze.

Suite	Mode	Code	Cases	Wrong change	Wrong values	Silent	Silent (blank)	Routed	Fixed	Clean	Crash
360 (tuning for v2)	text	v1	360	43	246	22	5	39	175	67	9
	text	v2	360	**0**	0	5	0	62	183	110	0
	no text	v1	360	0	0	98	16	127	0	110	9
	no text	v2	360	0	0	83	0	167	0	110	0
430 fresh (test)	text	v1	430	33	246	24	5	39	236	87	6
	text	v2	430	**0**	0	2	0	63	245	120	0
	no text	v1	430	0	0	108	24	172	0	120	6
	no text	v2	430	0	0	93	0	217	0	120	0

Table S1.12: v1 suite (360 cases, funds 11–19), text mode, v1 → v2 (tuning set)

Category	Wrong change	Silent	Silent (blank)	Routed	Fixed	Clean	Crash
(a) coordinated bending	0 → 0	0 → 0	0 → 0	3 → 3	37 → 37	0 → 0	0 → 0
(b) decoy numbers	0 → 0	3 → 3	0 → 0	10 → 10	27 → 27	0 → 0	0 → 0
(c) coincidental ties	0 → 0	0 → 0	0 → 0	10 → 10	20 → 20	0 → 0	0 → 0
(d) planted source errors	8 → 0	0 → 0	0 → 0	0 → 0	10 → 10	37 → 45	0 → 0
(e) label collisions	20 → 0	0 → 0	0 → 0	3 → 3	2 → 2	30 → 50	0 → 0
(f) sign / scale traps	15 → 0	19 → 2	0 → 0	13 → 23	13 → 20	0 → 15	0 → 0
(g) OCR digit confusions	0 → 0	0 → 0	0 → 0	0 → 0	40 → 40	0 → 0	0 → 0
(h) blanks / nulls	0 → 0	0 → 0	5 → 0	0 → 13	26 → 27	0 → 0	9 → 0

Table S1.13: Fresh set (430 cases, funds 31–39), v1 → v2

Cells show one value where v1 and v2 agree.

Mode	Category	Cases	Wrong change	Wrong values	Silent	Silent (blank)	Routed	Fixed	Clean	Crash
text	(a) coordinated bending	50	0	0	0	0	1	49	0	0
text	(b) decoy numbers	60	0	0	0	0	10	50	0	0
text	(c) coincidental ties	40	0	0	0	0	10	30	0	0
text	(d) planted source errors	60	6 → 0	6 → 0	0	0	0	10	44 → 50	0
text	(e) label collisions	60	12 → 0	12 → 0	0	0	3	2	43 → 55	0
text	(f) sign / scale traps	65	15 → 0	228 → 0	24 → 2	0	14 → 27	12 → 21	0 → 15	0
text	(g) OCR digit confusions	50	0	0	0	0	1	49	0	0
text	(h) blanks / nulls	45	0	0	0	5 → 0	0 → 11	34	0	6 → 0
no text	(a) coordinated bending	50	0	0	50	0	0	0	0	0
no text	(b) decoy numbers	60	0	0	14	0	46	0	0	0
no text	(c) coincidental ties	40	0	0	0	0	40	0	0	0
no text	(d) planted source errors	60	0	0	10	0	0	0	50	0
no text	(e) label collisions	60	0	0	0	0	5	0	55	0
no text	(f) sign / scale traps	65	0	0	19	0	31	0	15	0
no text	(g) OCR digit confusions	50	0	0	0	0	50	0	0	0
no text	(h) blanks / nulls	45	0	0	15 → 0	24 → 0	0 → 45	0	0	6 → 0

S1.6 Examples of erased document issues

Run	Issue	Field	Entered	Printed
fund09 Opus	mismatch	call commitment, 2025-04-15	10,000,000	6,000,000
fund07 Opus	stale	GP total invested, 2025-06-30	20,300,000	20,050,000
fund03 Opus	typo	GP fair value, Derta AI, 2025-06-30	5,850,000	5,849,100
fund11 Opus	mismatch	call commitment, 2025-10-03	10,000,000	6,000,000
fund19 Opus	mismatch	call commitment, 2025-07-15	4,000,000	6,500,000
fund37 Opus	mismatch	call commitment, 2025-02-21	2,000,000	6,000,000
fund35 Opus	typo	GP fair value, Kandex Ltd., 2026-06-30	19,433,021	19,433,221
fund33 luna	stale	GP total invested, 2026-03-31	120,000,000	96,150,000

Table S1.14: Development funds 01–09 (10 runs, 1,853 errors): fixed / tier 2 / missed
OCR is identical to no text. The label-read changes were made on these funds, so the v1 and v2 columns are development results.

Page text	Before changes	v1 (frozen)	v2
Text layer	798 / 784 / 271	1,832 / 18 / 3	1,832 / 20 / 1
No text	0 / 763 / 1,090	0 / 763 / 1,090	0 / 1,003 / 850

Table S1.15: Held-out set 1, frozen v1
Text: fixed / tier 2 / missed / wrong. No page text (= OCR): fixed / tier 2 / missed.

Model	Errors	Blank	Text	No text
Claude Opus	324	0	322 / 2 / 0 / 0	0 / 93 / 231
Claude Haiku	1,344	318	1,331 / 13 / 0 / 0	8 / 456 / 880
gpt-6-luna	2,826	2,185	2,761 / 45 / 20 / 0	0 / 264 / 2,562

Table S1.16: Numbers bent to tie (bad_scan runs)
"Own checks tie": every check the error feeds passes on the model's numbers. Planted document issues are "erased" if made consistent, "kept" if their checks still fail, and "lost" if they could not be evaluated.

Runs	Tier 0 errors	Own checks tie	Flagged	No check covers	Missing	Doc. issues	Erased	Kept	Lost
Development, Haiku + Opus (10)	1,853	828	781	15	229	30	19	11	0
Held-out 1, Haiku + Opus (10)	1,668	751	570	29	318	28	17	11	0
Held-out 1, gpt-6-luna (5)	2,826	280	311	50	2,185	14	3	8	3
Held-out 2, Haiku + Opus + luna (15)	4,767	1,124	917	96	2,630	39	21	13	5

Table S1.17: Intervals for PE headline rates
Wilson 95% intervals treat errors as independent; the cluster bootstrap resamples the 13 Fund R documents or the 5 funds.

Result	k/n	Rate (%)	Wilson	Cluster
Fund R hard runs, text, v1	189/193	97.9	94.8–99.2	94.2–100.0
Fund R OCR, 9 runs, flagged	156/197	79.2	73.0–84.3	64.6–89.2
Held-out 1, Claude, text, v1	1,653/1,668	99.1	98.5–99.5	98.7–99.6
Held-out 1, luna, text, v1	2,761/2,826	97.7	97.1–98.2	97.2–98.2

Table S1.18: One-sided 95% upper bounds on the wrong-change rate where none occurred
For $k = 0$ in n, the Clopper–Pearson bound is $1 - 0.05^{1/n} \approx 3/n$.

Result ($k = 0$)	n	Bound	Clusters (bound)
Fund R hard runs, text, v1	193 errors	1.54%	0/13 docs (20.6%)
Fund R OCR, 9 runs	197 errors	1.51%	0/13 docs (20.6%)
Held-out 1, Claude, v1	1,668 errors	0.18%	0/5 funds (45.1%)
Held-out 1, luna, v1	2,826 errors	0.11%	0/5 funds (45.1%)
Held-out 2, all models, v2	4,767 errors	0.06%	0/5 funds (45.1%)
correct values	5,349	0.06%	
30 funds, planted	4,843 plants	0.06%	0/30 funds (9.5%)
Fresh adversarial, v2, per mode	430 cases	0.69%	0/5 funds (45.1%)

S2 Frozen qcore6: every held-out set and every run

Material moved from Appendix C of the paper; the run protocol is described there.

Table S2.1: Frozen qcore6: SEC set 1 held-out poor scans (29 statements, 2,144 numbers)
One run per model, mode and identity setting. Fixed: wrong in the extraction, right at the end. Coverage: share of numbers in some identity used by the run. Frozen qcore5, re-run on the same inputs, gives exactly the hand-identity rows (not repeated).

Pipeline	Accuracy [95% CI]	Silent [95% CI]	Flagged (wrong)	Broken	Fixed	To human	Coverage
Haiku alone	94.92% [92.06, 97.47]	109 [53, 175]	0 (0)	0	0	0.0%	–
Haiku + qcore6, text	98.83% [97.94, 99.56]	0 [0, 0]	51 (25)	0	84	2.4%	85.6%
Haiku + qcore6, no text	94.92% [92.06, 97.47]	57 [27, 92]	155 (52)	0	0	7.2%	85.6%
Haiku + qcore6, OCR docTR	95.52% [92.97, 97.75]	47 [21, 78]	249 (49)	0	13	11.6%	85.6%
Haiku + qcore6, OCR Tesseract	94.92% [92.06, 97.47]	38 [15, 66]	391 (71)	0	0	18.2%	85.6%
Haiku + qcore6 discovered ids, text	98.83% [97.94, 99.56]	0 [0, 0]	38 (25)	0	84	1.8%	89.2%
Haiku + qcore6 discovered ids, no text	94.92% [92.06, 97.47]	80 [33, 138]	100 (29)	0	0	4.7%	87.5%
Haiku + qcore6 hand + discovered ids, text	98.83% [97.94, 99.56]	0 [0, 0]	49 (25)	0	84	2.3%	93.8%
Haiku + qcore6 hand + discovered ids, no text	94.92% [92.06, 97.47]	56 [27, 91]	175 (53)	0	0	8.2%	92.7%
Haiku + qcore6 zero-config, text	97.29% [95.96, 98.54]	0 [0, 0]	80 (58)	0	51	3.7%	84.7%
Haiku + qcore6 zero-config, no text	94.92% [92.06, 97.47]	80 [33, 138]	100 (29)	0	0	4.7%	87.5%
Opus alone	98.60% [97.39, 99.56]	30 [9, 56]	0 (0)	0	0	0.0%	–
Opus + qcore6, text	99.44% [98.64, 100.00]	0 [0, 0]	16 (12)	0	18	0.7%	85.6%
Opus + qcore6, no text	98.60% [97.39, 99.56]	13 [3, 26]	62 (17)	0	0	2.9%	85.6%
Opus + qcore6, OCR docTR	98.79% [97.64, 99.66]	10 [2, 21]	183 (16)	0	4	8.5%	85.6%
Opus + qcore6, OCR Tesseract	98.60% [97.39, 99.56]	9 [2, 19]	282 (21)	0	0	13.2%	85.6%
Opus + qcore6 discovered ids, text	99.44% [98.64, 100.00]	5 [0, 14]	8 (7)	0	18	0.4%	90.2%
Opus + qcore6 discovered ids, no text	98.60% [97.39, 99.56]	18 [5, 34]	51 (12)	0	0	2.4%	90.6%
Opus + qcore6 hand + discovered ids, text	99.44% [98.64, 100.00]	0 [0, 0]	31 (12)	0	18	1.4%	93.8%
Opus + qcore6 hand + discovered ids, no text	98.60% [97.39, 99.56]	12 [2, 25]	82 (18)	0	0	3.8%	92.5%
Opus + qcore6 zero-config, text	99.35% [98.55, 99.91]	5 [0, 14]	10 (9)	0	16	0.5%	88.5%
Opus + qcore6 zero-config, no text	98.60% [97.39, 99.56]	18 [5, 34]	51 (12)	0	0	2.4%	90.6%
GPT alone	75.23% [68.22, 81.76]	531 [375, 707]	0 (0)	0	0	0.0%	–
GPT + qcore6, text	94.50% [91.86, 96.79]	8 [1, 19]	186 (110)	0	413	8.7%	85.6%
GPT + qcore6, no text	75.23% [68.22, 81.76]	316 [208, 430]	554 (215)	0	0	25.8%	85.6%
GPT + qcore6, OCR docTR	76.54% [69.42, 83.20]	194 [116, 279]	689 (309)	2	30	32.1%	85.6%
GPT + qcore6, OCR Tesseract	75.28% [68.26, 81.83]	193 [116, 281]	873 (337)	0	1	40.7%	85.6%
GPT + qcore6 discovered ids, text	94.50% [91.90, 96.78]	10 [1, 24]	137 (108)	0	413	6.4%	75.4%
GPT + qcore6 discovered ids, no text	75.23% [68.22, 81.76]	484 [330, 658]	144 (47)	0	0	6.7%	60.7%
GPT + qcore6 hand + discovered ids, text	94.59% [92.09, 96.80]	8 [1, 19]	188 (108)	0	415	8.8%	89.3%
GPT + qcore6 hand + discovered ids, no text	75.23% [68.22, 81.76]	303 [201, 413]	586 (228)	0	0	27.3%	90.4%
GPT + qcore6 zero-config, text	76.73% [69.42, 83.57]	56 [15, 109]	533 (443)	0	32	24.9%	51.1%
GPT + qcore6 zero-config, no text	75.23% [68.22, 81.76]	484 [330, 658]	144 (47)	0	0	6.7%	60.7%

Table S2.2: Frozen qcore6: SEC set 2 held-out poor scans (60 statements, 4,424 numbers)
One run per model, mode and identity setting. Fixed: wrong in the extraction, right at the end. Coverage: share of numbers in some identity used by the run. Frozen qcore5, re-run on the same inputs, gives exactly the hand-identity rows (not repeated).

Pipeline	Accuracy [95% CI]	Silent [95% CI]	Flagged (wrong)	Broken	Fixed	To human	Coverage
Haiku alone	95.43% [92.82, 97.63]	202 [104, 321]	0 (0)	0	0	0.0%	–
Haiku + qcore6, text	98.62% [97.83, 99.30]	4 [0, 9]	104 (57)	0	141	2.4%	82.9%
Haiku + qcore6, no text	95.43% [92.82, 97.63]	105 [58, 162]	242 (97)	0	0	5.5%	82.9%
Haiku + qcore6, OCR docTR	95.61% [93.01, 97.81]	52 [28, 81]	584 (142)	0	8	13.2%	82.9%
Haiku + qcore6, OCR Tesseract	95.41% [92.80, 97.60]	54 [30, 83]	737 (149)	1	0	16.7%	82.9%
Haiku + qcore6 discovered ids, text	98.58% [97.77, 99.27]	8 [2, 15]	75 (55)	1	140	1.7%	83.7%
Haiku + qcore6 discovered ids, no text	95.43% [92.82, 97.63]	171 [80, 280]	129 (31)	0	0	2.9%	82.7%
Haiku + qcore6 hand + discovered ids, text	98.62% [97.83, 99.30]	4 [0, 9]	116 (57)	0	141	2.6%	90.3%
Haiku + qcore6 hand + discovered ids, no text	95.43% [92.82, 97.63]	102 [55, 159]	270 (100)	0	0	6.1%	89.5%
Haiku + qcore6 zero-config, text	96.34% [93.96, 98.22]	33 [2, 92]	256 (129)	0	40	5.8%	77.8%
Haiku + qcore6 zero-config, no text	95.43% [92.82, 97.63]	171 [80, 280]	129 (31)	0	0	2.9%	82.7%
Opus alone	98.44% [96.94, 99.67]	69 [14, 136]	0 (0)	0	0	0.0%	–
Opus + qcore6, text	99.41% [98.86, 99.84]	0 [0, 0]	57 (26)	0	43	1.3%	82.9%
Opus + qcore6, no text	98.44% [96.94, 99.67]	42 [10, 83]	108 (27)	0	0	2.4%	82.9%
Opus + qcore6, OCR docTR	98.42% [96.92, 99.65]	30 [6, 62]	385 (40)	1	0	8.7%	82.9%
Opus + qcore6, OCR Tesseract	98.44% [96.94, 99.67]	28 [4, 61]	541 (41)	0	0	12.2%	82.9%
Opus + qcore6 discovered ids, text	99.41% [98.86, 99.84]	0 [0, 0]	41 (26)	0	43	0.9%	85.2%
Opus + qcore6 discovered ids, no text	98.44% [96.94, 99.67]	51 [9, 103]	82 (18)	0	0	1.9%	86.8%
Opus + qcore6 hand + discovered ids, text	99.41% [98.86, 99.84]	0 [0, 0]	65 (26)	0	43	1.5%	90.0%
Opus + qcore6 hand + discovered ids, no text	98.44% [96.94, 99.67]	39 [8, 79]	129 (30)	0	0	2.9%	90.0%
Opus + qcore6 zero-config, text	98.89% [97.74, 99.81]	2 [0, 6]	55 (47)	0	20	1.2%	84.3%
Opus + qcore6 zero-config, no text	98.44% [96.94, 99.67]	51 [9, 103]	82 (18)	0	0	1.9%	86.8%
GPT alone	77.12% [70.51, 83.03]	1012 [733, 1343]	0 (0)	0	0	0.0%	–
GPT + qcore6, text	93.40% [90.61, 95.91]	27 [5, 58]	393 (265)	0	720	8.9%	82.9%
GPT + qcore6, no text	77.12% [70.51, 83.03]	541 [408, 692]	993 (471)	0	0	22.4%	82.9%
GPT + qcore6, OCR docTR	79.09% [72.30, 85.14]	355 [266, 456]	1187 (570)	0	87	26.8%	82.9%
GPT + qcore6, OCR Tesseract	77.22% [70.56, 83.12]	377 [292, 472]	1519 (631)	1	5	34.3%	82.9%
GPT + qcore6 discovered ids, text	93.54% [90.88, 95.96]	37 [12, 70]	319 (249)	0	726	7.2%	76.0%
GPT + qcore6 discovered ids, no text	77.12% [70.51, 83.03]	793 [589, 1024]	459 (219)	0	0	10.4%	63.1%
GPT + qcore6 hand + discovered ids, text	93.56% [90.88, 96.01]	25 [4, 54]	411 (260)	0	727	9.3%	89.0%
GPT + qcore6 hand + discovered ids, no text	77.12% [70.51, 83.03]	529 [398, 681]	1072 (483)	0	0	24.2%	88.3%
GPT + qcore6 zero-config, text	78.77% [72.03, 84.77]	133 [54, 246]	1072 (806)	0	73	24.2%	52.3%
GPT + qcore6 zero-config, no text	77.12% [70.51, 83.03]	793 [589, 1024]	459 (219)	0	0	10.4%	63.1%

Table S2.3: Frozen qcore6: SEC set 3 (new) held-out poor scans (66 statements, 5,206 numbers)
One run per model, mode and identity setting. Fixed: wrong in the extraction, right at the end. Coverage: share of numbers in some identity used by the run. Frozen qcore5, re-run on the same inputs, gives exactly the hand-identity rows (not repeated).

Pipeline	Accuracy [95% CI]	Silent [95% CI]	Flagged (wrong)	Broken	Fixed	To human	Coverage
Haiku alone	95.85% [94.51, 97.08]	216 [146, 298]	0 (0)	0	0	0.0%	–
Haiku + qcore6, text	98.21% [97.14, 99.08]	15 [5, 28]	171 (78)	0	123	3.3%	81.7%
Haiku + qcore6, no text	95.85% [94.51, 97.08]	123 [82, 170]	413 (93)	0	0	7.9%	81.7%
Haiku + qcore6, OCR docTR	96.04% [94.68, 97.28]	85 [55, 119]	840 (121)	2	12	16.1%	81.7%
Haiku + qcore6, OCR Tesseract	95.83% [94.49, 97.05]	80 [53, 112]	1056 (137)	1	0	20.3%	81.7%
Haiku + qcore6, OCR docTR + Opus re-read	96.04% [94.68, 97.28]	80 [52, 113]	862 (126)	2	12	16.6%	81.7%
Haiku + qcore6 discovered ids, text	98.18% [97.11, 99.05]	18 [6, 33]	194 (77)	0	121	3.7%	84.7%
Haiku + qcore6 discovered ids, no text	95.85% [94.51, 97.08]	170 [109, 244]	232 (46)	0	0	4.5%	77.7%
Haiku + qcore6 hand + discovered ids, text	98.18% [97.11, 99.05]	15 [5, 28]	210 (80)	0	121	4.0%	89.4%
Haiku + qcore6 hand + discovered ids, no text	95.85% [94.51, 97.08]	125 [83, 172]	411 (91)	0	0	7.9%	87.9%
Haiku + qcore6 zero-config, text	96.83% [95.58, 97.91]	43 [10, 90]	229 (122)	0	51	4.4%	72.7%
Haiku + qcore6 zero-config, no text	95.85% [94.51, 97.08]	170 [109, 244]	232 (46)	0	0	4.5%	77.7%
Opus alone	99.04% [98.36, 99.59]	50 [21, 89]	0 (0)	0	0	0.0%	–
Opus + qcore6, text	99.79% [99.60, 99.94]	5 [0, 12]	72 (6)	0	39	1.4%	81.7%
Opus + qcore6, no text	99.04% [98.36, 99.59]	39 [13, 75]	39 (11)	0	0	0.7%	81.7%
Opus + qcore6, OCR docTR	99.04% [98.36, 99.59]	29 [10, 54]	513 (21)	0	0	9.9%	81.7%
Opus + qcore6, OCR Tesseract	99.04% [98.36, 99.59]	25 [10, 43]	635 (25)	0	0	12.2%	81.7%
Opus + qcore6, OCR docTR + Opus re-read	99.04% [98.36, 99.59]	27 [9, 50]	525 (23)	0	0	10.1%	81.7%
Opus + qcore6 discovered ids, text	99.79% [99.60, 99.94]	5 [0, 12]	109 (6)	0	39	2.1%	87.3%
Opus + qcore6 discovered ids, no text	99.04% [98.36, 99.59]	40 [14, 76]	44 (10)	0	0	0.8%	85.7%
Opus + qcore6 hand + discovered ids, text	99.79% [99.60, 99.94]	5 [0, 12]	113 (6)	0	39	2.2%	90.4%
Opus + qcore6 hand + discovered ids, no text	99.04% [98.36, 99.59]	39 [13, 75]	49 (11)	0	0	0.9%	89.6%
Opus + qcore6 zero-config, text	99.35% [98.79, 99.79]	6 [1, 13]	133 (28)	0	16	2.6%	82.5%
Opus + qcore6 zero-config, no text	99.04% [98.36, 99.59]	40 [14, 76]	44 (10)	0	0	0.8%	85.7%
GPT alone	72.28% [66.90, 77.37]	1443 [1120, 1813]	0 (0)	0	0	0.0%	–
GPT + qcore6, text	89.28% [85.75, 92.55]	65 [35, 100]	717 (493)	2	887	13.8%	81.7%
GPT + qcore6, no text	72.28% [66.90, 77.37]	754 [582, 938]	1539 (689)	0	0	29.6%	81.7%
GPT + qcore6, OCR docTR	73.42% [67.92, 78.67]	532 [393, 684]	1904 (852)	0	59	36.6%	81.7%
GPT + qcore6, OCR Tesseract	72.32% [66.92, 77.42]	552 [423, 698]	2201 (889)	1	3	42.3%	81.7%
GPT + qcore6 discovered ids, text	89.07% [85.55, 92.30]	65 [33, 103]	684 (504)	0	874	13.1%	78.3%
GPT + qcore6 discovered ids, no text	72.28% [66.90, 77.37]	1205 [928, 1506]	467 (238)	0	0	9.0%	56.8%
GPT + qcore6 hand + discovered ids, text	89.09% [85.51, 92.38]	62 [32, 97]	757 (506)	2	877	14.5%	88.9%
GPT + qcore6 hand + discovered ids, no text	72.28% [66.90, 77.37]	755 [585, 938]	1541 (688)	0	0	29.6%	86.5%
GPT + qcore6 zero-config, text	73.24% [67.79, 78.37]	187 [100, 290]	1833 (1206)	0	50	35.2%	46.7%
GPT + qcore6 zero-config, no text	72.28% [66.90, 77.37]	1205 [928, 1506]	467 (238)	0	0	9.0%	56.8%

Table S2.4: Frozen qcore6: SEC sets 1–3 pooled (155 statements, 11,774 numbers)
One run per model, mode and identity setting. Fixed: wrong in the extraction, right at the end. Coverage: share of numbers in some identity used by the run. Frozen qcore5, re-run on the same inputs, gives exactly the hand-identity rows (not repeated).

Pipeline	Accuracy [95% CI]	Silent [95% CI]	Flagged (wrong)	Broken	Fixed	To human	Coverage
Haiku alone	95.52% [94.26, 96.65]	527 [388, 688]	0 (0)	0	0	0.0%	–
Haiku + qcore6, text	98.48% [97.89, 98.97]	19 [7, 33]	326 (160)	0	348	2.8%	82.8%
Haiku + qcore6, no text	95.52% [94.26, 96.65]	285 [213, 365]	810 (242)	0	0	6.9%	82.8%
Haiku + qcore6, OCR docTR	95.79% [94.53, 96.90]	184 [135, 237]	1673 (312)	2	33	14.2%	82.8%
Haiku + qcore6, OCR Tesseract	95.51% [94.24, 96.63]	172 [127, 222]	2184 (357)	2	0	18.5%	82.8%
Haiku + qcore6 discovered ids, text	98.45% [97.86, 98.94]	26 [13, 43]	307 (157)	1	345	2.6%	85.1%
Haiku + qcore6 discovered ids, no text	95.52% [94.26, 96.65]	421 [295, 565]	461 (106)	0	0	3.9%	81.3%
Haiku + qcore6 hand + discovered ids, text	98.46% [97.88, 98.95]	19 [7, 33]	375 (162)	0	346	3.2%	90.5%
Haiku + qcore6 hand + discovered ids, no text	95.52% [94.26, 96.65]	283 [210, 364]	856 (244)	0	0	7.3%	89.4%
Haiku + qcore6 zero-config, text	96.73% [95.64, 97.65]	76 [19, 155]	565 (309)	0	142	4.8%	76.8%
Haiku + qcore6 zero-config, no text	95.52% [94.26, 96.65]	421 [295, 565]	461 (106)	0	0	3.9%	81.3%
Opus alone	98.73% [98.07, 99.30]	149 [81, 230]	0 (0)	0	0	0.0%	–
Opus + qcore6, text	99.58% [99.33, 99.80]	5 [0, 12]	145 (44)	0	100	1.2%	82.8%
Opus + qcore6, no text	98.73% [98.07, 99.30]	94 [50, 146]	209 (55)	0	0	1.8%	82.8%
Opus + qcore6, OCR docTR	98.76% [98.10, 99.32]	69 [36, 108]	1081 (77)	1	4	9.2%	82.8%
Opus + qcore6, OCR Tesseract	98.73% [98.07, 99.30]	62 [31, 99]	1458 (87)	0	0	12.4%	82.8%
Opus + qcore6 discovered ids, text	99.58% [99.33, 99.80]	10 [2, 21]	158 (39)	0	100	1.3%	87.1%
Opus + qcore6 discovered ids, no text	98.73% [98.07, 99.30]	109 [56, 174]	177 (40)	0	0	1.5%	87.0%
Opus + qcore6 hand + discovered ids, text	99.58% [99.33, 99.80]	5 [0, 12]	209 (44)	0	100	1.8%	90.9%
Opus + qcore6 hand + discovered ids, no text	98.73% [98.07, 99.30]	90 [47, 142]	260 (59)	0	0	2.2%	90.3%
Opus + qcore6 zero-config, text	99.18% [98.66, 99.60]	13 [4, 25]	198 (84)	0	52	1.7%	84.3%
Opus + qcore6 zero-config, no text	98.73% [98.07, 99.30]	109 [56, 174]	177 (40)	0	0	1.5%	87.0%
GPT alone	74.64% [70.88, 78.14]	2986 [2512, 3509]	0 (0)	0	0	0.0%	–
GPT + qcore6, text	91.78% [89.83, 93.65]	100 [58, 148]	1296 (868)	2	2020	11.0%	82.8%
GPT + qcore6, no text	74.64% [70.88, 78.14]	1611 [1362, 1871]	3086 (1375)	0	0	26.2%	82.8%
GPT + qcore6, OCR docTR	76.12% [72.29, 79.69]	1081 [894, 1280]	3780 (1731)	2	176	32.1%	82.8%
GPT + qcore6, OCR Tesseract	74.70% [70.94, 78.19]	1122 [940, 1313]	4593 (1857)	2	9	39.0%	82.8%
GPT + qcore6 discovered ids, text	91.74% [89.82, 93.55]	112 [67, 164]	1140 (861)	0	2013	9.7%	76.9%
GPT + qcore6 discovered ids, no text	74.64% [70.88, 78.14]	2482 [2091, 2899]	1070 (504)	0	0	9.1%	59.9%
GPT + qcore6 hand + discovered ids, text	91.77% [89.82, 93.62]	95 [54, 142]	1356 (874)	2	2019	11.5%	89.0%
GPT + qcore6 hand + discovered ids, no text	74.64% [70.88, 78.14]	1587 [1342, 1848]	3199 (1399)	0	0	27.2%	87.9%
GPT + qcore6 zero-config, text	75.96% [72.13, 79.52]	376 [239, 535]	3438 (2455)	0	155	29.2%	49.6%
GPT + qcore6 zero-config, no text	74.64% [70.88, 78.14]	2482 [2091, 2899]	1070 (504)	0	0	9.1%	59.9%

Table S2.5: Frozen qcore6: CORD test receipts, poor scans (100 receipts, 998 numbers)
One run per model, mode and identity setting. Fixed: wrong in the extraction, right at the end. Coverage: share of numbers in some identity used by the run. Frozen qcore5, re-run on the same inputs, gives exactly the hand-identity rows (not repeated).

Pipeline	Accuracy [95% CI]	Silent [95% CI]	Flagged (wrong)	Broken	Fixed	To human	Coverage
Haiku alone	73.75% [66.30, 80.75]	262 [187, 344]	0 (0)	0	0	0.0%	–
Haiku + qcore6, text	75.45% [68.19, 82.19]	12 [4, 22]	291 (233)	0	17	29.2%	71.5%
Haiku + qcore6, no text	73.75% [66.30, 80.75]	52 [29, 78]	250 (210)	0	0	25.1%	71.5%
Haiku + qcore6, OCR docTR	73.75% [66.30, 80.75]	48 [26, 73]	289 (214)	0	0	29.0%	71.5%
Haiku + qcore6, OCR Tesseract	73.75% [66.30, 80.75]	46 [24, 71]	310 (216)	0	0	31.1%	71.5%
Haiku + qcore6 discovered ids, text	75.15% [67.87, 81.93]	16 [6, 29]	269 (232)	0	14	27.0%	51.2%
Haiku + qcore6 discovered ids, no text	73.75% [66.30, 80.75]	77 [45, 113]	196 (185)	0	0	19.6%	45.8%
Haiku + qcore6 hand + discovered ids, text	75.55% [68.28, 82.30]	12 [4, 22]	293 (232)	0	18	29.4%	77.7%
Haiku + qcore6 hand + discovered ids, no text	73.75% [66.30, 80.75]	53 [30, 79]	253 (209)	0	0	25.4%	73.4%
Haiku + qcore6 zero-config, text	75.15% [68.00, 81.85]	16 [6, 29]	267 (232)	0	14	26.8%	50.1%
Haiku + qcore6 zero-config, no text	73.75% [66.30, 80.75]	82 [48, 122]	187 (180)	0	0	18.7%	42.1%
Opus alone	86.37% [80.59, 91.43]	136 [84, 194]	0 (0)	0	0	0.0%	–
Opus + qcore6, text	87.47% [81.89, 92.37]	7 [2, 14]	170 (118)	0	11	17.0%	71.5%
Opus + qcore6, no text	86.37% [80.59, 91.43]	67 [36, 102]	87 (69)	0	0	8.7%	71.5%
Opus + qcore6, OCR docTR	86.37% [80.59, 91.43]	67 [36, 102]	127 (69)	0	0	12.7%	71.5%
Opus + qcore6, OCR Tesseract	86.37% [80.59, 91.43]	66 [36, 100]	148 (70)	0	0	14.8%	71.5%
Opus + qcore6 discovered ids, text	87.47% [81.89, 92.37]	7 [2, 14]	160 (118)	0	11	16.0%	61.3%
Opus + qcore6 discovered ids, no text	86.37% [80.59, 91.43]	68 [37, 104]	72 (68)	0	0	7.2%	55.0%
Opus + qcore6 hand + discovered ids, text	87.47% [81.89, 92.37]	7 [2, 14]	173 (118)	0	11	17.3%	79.0%
Opus + qcore6 hand + discovered ids, no text	86.37% [80.59, 91.43]	67 [36, 102]	90 (69)	0	0	9.0%	74.0%
Opus + qcore6 zero-config, text	87.47% [81.89, 92.37]	9 [3, 16]	154 (116)	0	11	15.4%	59.2%
Opus + qcore6 zero-config, no text	86.37% [80.59, 91.43]	68 [37, 104]	72 (68)	0	0	7.2%	37.3%

Table S2.6: Frozen qcore6: SEC notes tables, held-out filers, clean pages (233 tables, 14,778 numbers)

One run per model, mode and identity setting. Fixed: wrong in the extraction, right at the end. Coverage: share of numbers in some identity used by the run. Frozen qcore5, re-run on the same inputs, gives exactly the hand-identity rows (not repeated).

Pipeline	Accuracy [95% CI]	Silent [95% CI]	Flagged (wrong)	Broken	Fixed	To human	Coverage
Haiku alone	99.68% [99.18, 100.00]	48 [0, 120]	0 (0)	0	0	0.0%	–
Haiku + qcore6, text	99.67% [99.17, 99.99]	23 [0, 66]	344 (26)	1	0	2.3%	32.5%
Haiku + qcore6, no text	99.68% [99.18, 100.00]	24 [0, 68]	30 (24)	0	0	0.2%	32.5%
Haiku + qcore6 discovered ids, text	99.68% [99.18, 100.00]	46 [0, 115]	430 (2)	0	0	2.9%	56.5%
Haiku + qcore6 discovered ids, no text	99.68% [99.18, 100.00]	48 [0, 120]	110 (0)	0	0	0.7%	57.1%
Haiku + qcore6 hand + discovered ids, text	99.68% [99.18, 100.00]	22 [0, 63]	453 (26)	0	0	3.1%	67.2%
Haiku + qcore6 hand + discovered ids, no text	99.68% [99.18, 100.00]	24 [0, 68]	136 (24)	0	0	0.9%	67.3%
Haiku + qcore6 zero-config, text	99.68% [99.18, 100.00]	46 [0, 115]	271 (2)	0	0	1.8%	55.8%
Haiku + qcore6 zero-config, no text	99.68% [99.18, 100.00]	48 [0, 120]	155 (0)	0	0	1.0%	57.3%
Opus alone	100.00% [100.00, 100.00]	0 [0, 0]	0 (0)	0	0	0.0%	–
Opus + qcore6, text	99.99% [99.98, 100.00]	1 [0, 3]	312 (0)	1	0	2.1%	32.5%
Opus + qcore6, no text	100.00% [100.00, 100.00]	0 [0, 0]	0 (0)	0	0	0.0%	32.5%
Opus + qcore6 discovered ids, text	100.00% [100.00, 100.00]	0 [0, 0]	428 (0)	0	0	2.9%	56.6%
Opus + qcore6 discovered ids, no text	100.00% [100.00, 100.00]	0 [0, 0]	110 (0)	0	0	0.7%	57.2%
Opus + qcore6 hand + discovered ids, text	100.00% [100.00, 100.00]	0 [0, 0]	421 (0)	0	0	2.8%	67.2%
Opus + qcore6 hand + discovered ids, no text	100.00% [100.00, 100.00]	0 [0, 0]	106 (0)	0	0	0.7%	67.3%
Opus + qcore6 zero-config, text	100.00% [100.00, 100.00]	0 [0, 0]	273 (0)	0	0	1.8%	55.9%
Opus + qcore6 zero-config, no text	100.00% [100.00, 100.00]	0 [0, 0]	110 (0)	0	0	0.7%	57.2%
GPT alone	86.43% [79.05, 93.84]	2005 [837, 3403]	0 (0)	0	0	0.0%	–
GPT + qcore6, text	86.48% [79.09, 93.90]	3 [0, 7]	2916 (1995)	1	8	19.7%	32.5%
GPT + qcore6, no text	86.43% [79.05, 93.84]	3 [0, 7]	2655 (2002)	0	0	18.0%	32.5%
GPT + qcore6 discovered ids, text	86.49% [79.09, 93.90]	0 [0, 0]	3029 (1997)	0	8	20.5%	48.2%
GPT + qcore6 discovered ids, no text	86.43% [79.05, 93.84]	1 [0, 3]	2689 (2004)	0	0	18.2%	48.1%
GPT + qcore6 hand + discovered ids, text	86.49% [79.09, 93.90]	0 [0, 0]	3035 (1997)	0	8	20.5%	61.5%
GPT + qcore6 hand + discovered ids, no text	86.43% [79.05, 93.84]	1 [0, 3]	2708 (2004)	0	0	18.3%	61.5%
GPT + qcore6 zero-config, text	86.43% [79.05, 93.84]	1 [0, 3]	3506 (2004)	0	0	23.7%	47.3%
GPT + qcore6 zero-config, no text	86.43% [79.05, 93.84]	1 [0, 3]	2689 (2004)	0	0	18.2%	48.1%

Table S2.7: Frozen qcore6: Real scans held-out (77 forms with truth, 1,009 numbers)

One run per model, mode and identity setting. Fixed: wrong in the extraction, right at the end. Coverage: share of numbers in some identity used by the run. Frozen qcore5, re-run on the same inputs, gives exactly the hand-identity rows (not repeated).

Pipeline	Accuracy [95% CI]	Silent [95% CI]	Flagged (wrong)	Broken	Fixed	To human	Coverage
Opus alone	96.13% [91.22, 99.42]	39 [6, 90]	0 (0)	0	0	0.0%	–
Opus + qcore6, no text	96.13% [91.22, 99.42]	2 [0, 5]	167 (37)	0	0	16.6%	98.9%
Opus + qcore6, dataset OCR	96.13% [91.22, 99.42]	2 [0, 5]	176 (37)	0	0	17.4%	98.9%
Opus + qcore6, scanner OCR layer	98.04% [95.40, 99.86]	1 [0, 3]	138 (14)	0	0	18.0%	98.9%
Opus + qcore6 discovered ids, no text	96.13% [91.22, 99.42]	39 [6, 90]	0 (0)	0	0	0.0%	41.9%
Opus + qcore6 hand + discovered ids, no text	96.13% [91.22, 99.42]	2 [0, 5]	188 (37)	0	0	18.6%	98.9%
Opus + qcore6 zero-config, no text	96.13% [91.22, 99.42]	39 [6, 90]	0 (0)	0	0	0.0%	41.9%
Haiku alone	96.13% [91.16, 99.48]	39 [5, 91]	0 (0)	0	0	0.0%	–
Haiku + qcore6, no text	96.13% [91.16, 99.48]	3 [0, 8]	139 (36)	0	0	13.8%	98.9%
Haiku + qcore6, dataset OCR	96.13% [91.16, 99.48]	3 [0, 8]	152 (36)	0	0	15.1%	98.9%
Haiku + qcore6, scanner OCR layer	97.91% [95.21, 99.86]	1 [0, 3]	138 (15)	0	0	18.0%	98.9%
Haiku + qcore6 discovered ids, no text	96.13% [91.16, 99.48]	39 [5, 91]	0 (0)	0	0	0.0%	40.9%
Haiku + qcore6 hand + discovered ids, no text	96.13% [91.16, 99.48]	3 [0, 8]	160 (36)	0	0	15.9%	98.9%
Haiku + qcore6 zero-config, no text	96.13% [91.16, 99.48]	39 [5, 91]	0 (0)	0	0	0.0%	40.9%

Table S2.8: Real scans held-out (77 forms with truth, 1,009 numbers): the models alone on the subset with an embedded scanner OCR layer (57 forms)

Pipeline	Accuracy [95% CI]	Silent [95% CI]	Flagged (wrong)	Broken	Fixed	To human	Coverage
Opus alone	98.04% [95.40, 99.86]	15 [1, 37]	0 (0)	0	0	0.0%	–
Haiku alone	97.91% [95.21, 99.86]	16 [1, 39]	0 (0)	0	0	0.0%	–

Table S2.9: Real scans: the 16 held-out forms whose own line items do not sum to the printed gross
A correct checker leaves these values as printed and routes them. Changed: any key of the identity rewritten; forced to tie: the identity ties on the final values though it fails on the truth.

Pipeline	In scope	Changed	Changed off truth	Forced to tie	Routed	Silent
Opus alone	16	0	0	0	0	16
Opus + qcore6, no text	16	0	0	0	13	3
Opus + qcore6, dataset OCR	16	0	0	0	13	3
Opus + qcore6, scanner OCR layer	11	0	0	0	9	2
Opus + qcore6 discovered ids, no text	16	0	0	0	0	16
Opus + qcore6 hand + discovered ids, no text	16	0	0	0	13	3
Opus + qcore6 zero-config, no text	16	0	0	0	0	16
Haiku alone	16	0	0	0	0	16
Haiku + qcore6, no text	16	0	0	0	14	2
Haiku + qcore6, dataset OCR	16	0	0	0	14	2
Haiku + qcore6, scanner OCR layer	11	0	0	0	9	2
Haiku + qcore6 discovered ids, no text	16	0	0	0	0	16
Haiku + qcore6 hand + discovered ids, no text	16	0	0	0	14	2
Haiku + qcore6 zero-config, no text	16	0	0	0	0	16

Table S2.10: Frozen qcore6: Synthetic domains, held-out bundles 02–05 (192 documents, 7,892 numbers)
One run per model, mode and identity setting. Fixed: wrong in the extraction, right at the end. Coverage: share of numbers in some identity used by the run. Frozen qcore5, re-run on the same inputs, gives exactly the hand-identity rows (not repeated).

Pipeline	Accuracy [95% CI]	Silent [95% CI]	Flagged (wrong)	Broken	Fixed	To human	Coverage
Haiku alone	99.97% [99.94, 100.00]	2 [0, 5]	0 (0)	0	0	0.0%	–
Haiku + qcore6, text	99.97% [99.94, 100.00]	2 [0, 5]	196 (0)	0	0	2.5%	97.8%
Haiku + qcore6, no text	99.97% [99.94, 100.00]	2 [0, 5]	122 (0)	0	0	1.5%	97.8%
Haiku + qcore6 discovered ids, text	99.77% [99.67, 99.87]	18 [10, 27]	122 (0)	16	0	1.5%	81.2%
Haiku + qcore6 discovered ids, no text	99.97% [99.94, 100.00]	2 [0, 5]	51 (0)	0	0	0.6%	78.2%
Haiku + qcore6 hand + discovered ids, text	99.97% [99.94, 100.00]	2 [0, 5]	238 (0)	0	0	3.0%	97.8%
Haiku + qcore6 hand + discovered ids, no text	99.97% [99.94, 100.00]	2 [0, 5]	149 (0)	0	0	1.9%	97.8%
Haiku + qcore6 zero-config, text	99.87% [99.80, 99.95]	10 [4, 16]	122 (0)	8	0	1.5%	81.3%
Haiku + qcore6 zero-config, no text	99.97% [99.94, 100.00]	2 [0, 5]	51 (0)	0	0	0.6%	78.2%
Opus alone	100.00% [100.00, 100.00]	0 [0, 0]	0 (0)	0	0	0.0%	–
Opus + qcore6, text	100.00% [100.00, 100.00]	0 [0, 0]	221 (0)	0	0	2.8%	97.8%
Opus + qcore6, no text	100.00% [100.00, 100.00]	0 [0, 0]	125 (0)	0	0	1.6%	97.8%
Opus + qcore6 discovered ids, text	99.80% [99.70, 99.88]	16 [9, 24]	137 (0)	16	0	1.7%	81.3%
Opus + qcore6 discovered ids, no text	100.00% [100.00, 100.00]	0 [0, 0]	53 (0)	0	0	0.7%	78.2%
Opus + qcore6 hand + discovered ids, text	100.00% [100.00, 100.00]	0 [0, 0]	263 (0)	0	0	3.3%	97.8%
Opus + qcore6 hand + discovered ids, no text	100.00% [100.00, 100.00]	0 [0, 0]	151 (0)	0	0	1.9%	97.8%
Opus + qcore6 zero-config, text	99.90% [99.83, 99.96]	8 [3, 14]	130 (0)	8	0	1.6%	81.3%
Opus + qcore6 zero-config, no text	100.00% [100.00, 100.00]	0 [0, 0]	53 (0)	0	0	0.7%	78.2%

Table S2.11: Synthetic domains: planted source-document issues (correct as printed), summed over the 12 held-out bundles

Pipeline	Broken	Planted issues	Changed	Routed	Tier 2 items
Haiku + qcore6, text	0	24	0	22	104
Haiku + qcore6, no text	0	24	0	22	40
Haiku + qcore6 discovered ids, text	16	24	0	10	47
Haiku + qcore6 discovered ids, no text	0	24	0	8	20
Haiku + qcore6 hand + discovered ids, text	0	24	0	22	119
Haiku + qcore6 hand + discovered ids, no text	0	24	0	22	50
Haiku + qcore6 zero-config, text	8	24	0	10	47
Haiku + qcore6 zero-config, no text	0	24	0	8	20
Opus + qcore6, text	0	24	0	24	114
Opus + qcore6, no text	0	24	0	24	43
Opus + qcore6 discovered ids, text	16	24	0	12	51
Opus + qcore6 discovered ids, no text	0	24	0	10	22
Opus + qcore6 hand + discovered ids, text	0	24	0	24	131
Opus + qcore6 hand + discovered ids, no text	0	24	0	24	52
Opus + qcore6 zero-config, text	8	24	0	12	50
Opus + qcore6 zero-config, no text	0	24	0	10	22

Table S2.12: Every run of frozen qcore6 with hand identities against the same model alone, and against Opus alone

Paired bootstrap of the difference in silent errors over documents (same resamples, 10,000, seed 0). A cut is a difference whose interval lies below 0. Pooled SEC rows repeat the per-set rows. The last column compares the pipeline with Opus alone on the same documents (not shown for Opus's own rows).

Set	Model	Mode	Identities	Silent	Alone	Difference [95% CI]	Broken	Fixed	To human	vs. Opus alone [95% CI]
SEC 1	Haiku	text	hand	0	109	−109 [−175, −53]	0	84	2.4%	−30 [−56, −9]
SEC 1	Haiku	no text	hand	57	109	−52 [−92, −19]	0	0	7.2%	+27 [+8, +46]
SEC 1	Haiku	OCR docTR	hand	47	109	−62 [−108, −25]	0	13	11.6%	+17 [−3, +37]
SEC 1	Haiku	OCR Tesseract	hand	38	109	−71 [−124, −28]	0	0	18.2%	+8 [−13, +28]
SEC 1	Opus	text	hand	0	30	−30 [−56, −9]	0	18	0.7%	–
SEC 1	Opus	no text	hand	13	30	−17 [−32, −5]	0	0	2.9%	–
SEC 1	Opus	OCR docTR	hand	10	30	−20 [−38, −5]	0	4	8.5%	–
SEC 1	Opus	OCR Tesseract	hand	9	30	−21 [−40, −6]	0	0	13.2%	–
SEC 1	GPT	text	hand	8	531	−523 [−697, −369]	0	413	8.7%	−22 [−51, +3]
SEC 1	GPT	no text	hand	316	531	−215 [−304, −138]	0	0	25.8%	+286 [+186, +390]
SEC 1	GPT	OCR docTR	hand	194	531	−337 [−479, −219]	2	30	32.1%	+164 [+93, +241]
SEC 1	GPT	OCR Tesseract	hand	193	531	−338 [−477, −219]	0	1	40.7%	+163 [+92, +240]
SEC 2	Haiku	text	hand	4	202	−198 [−313, −103]	0	141	2.4%	−65 [−133, −10]
SEC 2	Haiku	no text	hand	105	202	−97 [−164, −41]	0	0	5.5%	+36 [−15, +91]
SEC 2	Haiku	OCR docTR	hand	52	202	−150 [−256, −66]	0	8	13.2%	−17 [−67, +23]
SEC 2	Haiku	OCR Tesseract	hand	54	202	−148 [−252, −65]	1	0	16.7%	−15 [−65, +24]
SEC 2	Opus	text	hand	0	69	−69 [−136, −14]	0	43	1.3%	–
SEC 2	Opus	no text	hand	42	69	−27 [−59, −3]	0	0	2.4%	–
SEC 2	Opus	OCR docTR	hand	30	69	−39 [−83, −6]	1	0	8.7%	–
SEC 2	Opus	OCR Tesseract	hand	28	69	−41 [−85, −9]	0	0	12.2%	–
SEC 2	GPT	text	hand	27	1,012	−985 [−1308, −716]	0	720	8.9%	−42 [−115, +20]
SEC 2	GPT	no text	hand	541	1,012	−471 [−727, −279]	0	0	22.4%	+472 [+360, +605]
SEC 2	GPT	OCR docTR	hand	355	1,012	−657 [−951, −431]	0	87	26.8%	+286 [+222, +360]
SEC 2	GPT	OCR Tesseract	hand	377	1,012	−635 [−930, −405]	1	5	34.3%	+308 [+236, +386]
SEC 3	Haiku	text	hand	15	216	−201 [−278, −135]	0	123	3.3%	−35 [−73, −6]
SEC 3	Haiku	no text	hand	123	216	−93 [−156, −47]	0	0	7.9%	+73 [+42, +105]
SEC 3	Haiku	OCR docTR	hand	85	216	−131 [−197, −79]	2	12	16.1%	+35 [+3, +66]
SEC 3	Haiku	OCR Tesseract	hand	80	216	−136 [−206, −79]	1	0	20.3%	+30 [−8, +66]
SEC 3	Haiku	OCR docTR	hand + re-read	80	216	−136 [−203, −82]	2	12	16.6%	+30 [−3, +62]
SEC 3	Opus	text	hand	5	50	−45 [−82, −17]	0	39	1.4%	–
SEC 3	Opus	no text	hand	39	50	−11 [−26, −2]	0	0	0.7%	–
SEC 3	Opus	OCR docTR	hand	29	50	−21 [−40, −6]	0	0	9.9%	–
SEC 3	Opus	OCR Tesseract	hand	25	50	−25 [−51, −6]	0	0	12.2%	–
SEC 3	Opus	OCR docTR	hand + re-read	27	50	−23 [−45, −6]	0	0	10.1%	–
SEC 3	GPT	text	hand	65	1,443	−1378 [−1735, −1068]	2	887	13.8%	+15 [−32, +58]
SEC 3	GPT	no text	hand	754	1,443	−689 [−1000, −452]	0	0	29.6%	+704 [+541, +877]
SEC 3	GPT	OCR docTR	hand	532	1,443	−911 [−1232, −650]	0	59	36.6%	+482 [+350, +628]
SEC 3	GPT	OCR Tesseract	hand	552	1,443	−891 [−1225, −620]	1	3	42.3%	+502 [+374, +648]
SEC 1+2	Haiku	text	hand	4	311	−307 [−443, −192]	0	225	2.4%	−95 [−168, −37]
SEC 1+2	Haiku	no text	hand	162	311	−149 [−228, −82]	0	0	6.0%	+63 [+8, +122]
SEC 1+2	Haiku	OCR docTR	hand	99	311	−212 [−330, −116]	0	21	12.7%	+0 [−57, +44]
SEC 1+2	Haiku	OCR Tesseract	hand	92	311	−219 [−338, −121]	1	0	17.2%	−7 [−64, +37]
SEC 1+2	Opus	text	hand	0	99	−99 [−173, −42]	0	61	1.1%	–
SEC 1+2	Opus	no text	hand	55	99	−44 [−80, −17]	0	0	2.6%	–
SEC 1+2	Opus	OCR docTR	hand	40	99	−59 [−107, −22]	1	4	8.6%	–
SEC 1+2	Opus	OCR Tesseract	hand	37	99	−62 [−110, −25]	0	0	12.5%	–
SEC 1+2	GPT	text	hand	35	1,543	−1508 [−1878, −1193]	0	1,133	8.8%	−64 [−142, +3]

Set	Model	Mode	Identities	Silent	Alone	Difference [95% CI]	Broken	Fixed	To human	vs. Opus alone [95% CI]
SEC 1+2	GPT	no text	hand	857	1,543	−686 [−960, −475]	0	0	23.6%	+758 [+604, +929]
SEC 1+2	GPT	OCR docTR	hand	549	1,543	−994 [−1316, −728]	2	117	28.6%	+450 [+352, +555]
SEC 1+2	GPT	OCR Tesseract	hand	570	1,543	−973 [−1298, −707]	1	6	36.4%	+471 [+366, +579]
SEC 1–3	Haiku	text	hand	19	527	−508 [−664, −371]	0	348	2.8%	−130 [−211, −62]
SEC 1–3	Haiku	no text	hand	285	527	−242 [−344, −156]	0	0	6.9%	+136 [+73, +202]
SEC 1–3	Haiku	OCR docTR	hand	184	527	−343 [−477, −230]	2	33	14.2%	+35 [−29, +91]
SEC 1–3	Haiku	OCR Tesseract	hand	172	527	−355 [−492, −238]	2	0	18.5%	+23 [−45, +82]
SEC 1–3	Opus	text	hand	5	149	−144 [−224, −77]	0	100	1.2%	–
SEC 1–3	Opus	no text	hand	94	149	−55 [−93, −24]	0	0	1.8%	–
SEC 1–3	Opus	OCR docTR	hand	69	149	−80 [−131, −39]	1	4	9.2%	–
SEC 1–3	Opus	OCR Tesseract	hand	62	149	−87 [−141, −43]	0	0	12.4%	–
SEC 1–3	GPT	text	hand	100	2,986	−2886 [−3394, −2427]	2	2,020	11.0%	−49 [−141, +33]
SEC 1–3	GPT	no text	hand	1,611	2,986	−1375 [−1794, −1044]	0	0	26.2%	+1462 [+1229, +1701]
SEC 1–3	GPT	OCR docTR	hand	1,081	2,986	−1905 [−2359, −1519]	2	176	32.1%	+932 [+761, +1109]
SEC 1–3	GPT	OCR Tesseract	hand	1,122	2,986	−1864 [−2331, −1467]	2	9	39.0%	+973 [+800, +1154]
CORD	Haiku	text	hand	12	262	−250 [−330, −177]	0	17	29.2%	−124 [−182, −73]
CORD	Haiku	no text	hand	52	262	−210 [−289, −139]	0	0	25.1%	−84 [−147, −29]
CORD	Haiku	OCR docTR	hand	48	262	−214 [−294, −142]	0	0	29.0%	−88 [−151, −33]
CORD	Haiku	OCR Tesseract	hand	46	262	−216 [−297, −144]	0	0	31.1%	−90 [−153, −34]
CORD	Opus	text	hand	7	136	−129 [−187, −78]	0	11	17.0%	–
CORD	Opus	no text	hand	67	136	−69 [−121, −26]	0	0	8.7%	–
CORD	Opus	OCR docTR	hand	67	136	−69 [−121, −26]	0	0	12.7%	–
CORD	Opus	OCR Tesseract	hand	66	136	−70 [−122, −27]	0	0	14.8%	–
Notes	Haiku	text	hand	23	48	−25 [−75, +1]	1	0	2.3%	+23 [+0, +66]
Notes	Haiku	no text	hand	24	48	−24 [−72, +0]	0	0	0.2%	+24 [+0, +68]
Notes	Opus	text	hand	1	0	+1 [+0, +3]	1	0	2.1%	–
Notes	Opus	no text	hand	0	0	+0 [+0, +0]	0	0	0.0%	–
Notes	GPT	text	hand	3	2,005	−2002 [−3400, −835]	1	8	19.7%	+3 [+0, +7]
Notes	GPT	no text	hand	3	2,005	−2002 [−3400, −835]	0	0	18.0%	+3 [+0, +7]
Real scans	Haiku	no text	hand	3	39	−36 [−86, −4]	0	0	13.8%	−36 [−86, −5]
Real scans	Haiku	dataset OCR	hand	3	39	−36 [−86, −4]	0	0	15.1%	−36 [−86, −5]
Real scans	Haiku	scanner OCR layer	hand	1	16	−15 [−38, +0]	0	0	18.0%	−14 [−36, +0]
Real scans	Opus	no text	hand	2	39	−37 [−88, −4]	0	0	16.6%	–
Real scans	Opus	dataset OCR	hand	2	39	−37 [−88, −4]	0	0	17.4%	–
Real scans	Opus	scanner OCR layer	hand	1	15	−14 [−36, +0]	0	0	18.0%	–
Synthetic	Haiku	text	hand	2	2	+0 [+0, +0]	0	0	2.5%	+2 [+0, +5]
Synthetic	Haiku	no text	hand	2	2	+0 [+0, +0]	0	0	1.5%	+2 [+0, +5]
Synthetic	Opus	text	hand	0	0	+0 [+0, +0]	0	0	2.8%	–
Synthetic	Opus	no text	hand	0	0	+0 [+0, +0]	0	0	1.6%	–

Table S2.14: Every run of frozen qcore6 with discovered identities against the same model alone
As Table S2.12. Discovered: identities found in the documents, hand schema. Hand + disc.: both. Zero-config: discovered identities and an inferred schema, no file written by hand.

Set	Model	Mode	Identities	Silent	Alone	Difference [95% CI]	Broken	Fixed	To human	vs. Opus alone [95% CI]
SEC 1	Haiku	text	discovered	0	109	−109 [−175, −53]	0	84	1.8%	−30 [−56, −9]
SEC 1	Haiku	no text	discovered	80	109	−29 [−49, −12]	0	0	4.7%	+50 [+15, +94]
SEC 1	Haiku	text	hand + disc.	0	109	−109 [−175, −53]	0	84	2.3%	−30 [−56, −9]
SEC 1	Haiku	no text	hand + disc.	56	109	−53 [−93, −20]	0	0	8.2%	+26 [+8, +45]
SEC 1	Haiku	text	zero-config	0	109	−109 [−175, −53]	0	51	3.7%	−30 [−56, −9]
SEC 1	Haiku	no text	zero-config	80	109	−29 [−49, −12]	0	0	4.7%	+50 [+15, +94]
SEC 1	Opus	text	discovered	5	30	−25 [−48, −7]	0	18	0.4%	–
SEC 1	Opus	no text	discovered	18	30	−12 [−24, −3]	0	0	2.4%	–
SEC 1	Opus	text	hand + disc.	0	30	−30 [−56, −9]	0	18	1.4%	–
SEC 1	Opus	no text	hand + disc.	12	30	−18 [−34, −5]	0	0	3.8%	–
SEC 1	Opus	text	zero-config	5	30	−25 [−48, −7]	0	16	0.5%	–
SEC 1	Opus	no text	zero-config	18	30	−12 [−24, −3]	0	0	2.4%	–
SEC 1	GPT	text	discovered	10	531	−521 [−695, −367]	0	413	6.4%	−20 [−49, +6]
SEC 1	GPT	no text	discovered	484	531	−47 [−81, −24]	0	0	6.7%	+454 [+310, +613]
SEC 1	GPT	text	hand + disc.	8	531	−523 [−697, −369]	0	415	8.8%	−22 [−51, +3]
SEC 1	GPT	no text	hand + disc.	303	531	−228 [−316, −149]	0	0	27.3%	+273 [+179, +371]
SEC 1	GPT	text	zero-config	56	531	−475 [−633, −333]	0	32	24.9%	+26 [−20, +78]
SEC 1	GPT	no text	zero-config	484	531	−47 [−81, −24]	0	0	6.7%	+454 [+310, +613]
SEC 2	Haiku	text	discovered	8	202	−194 [−310, −98]	1	140	1.7%	−61 [−129, −6]
SEC 2	Haiku	no text	discovered	171	202	−31 [−46, −18]	0	0	2.9%	+102 [+49, +183]
SEC 2	Haiku	text	hand + disc.	4	202	−198 [−313, −103]	0	141	2.6%	−65 [−133, −10]
SEC 2	Haiku	no text	hand + disc.	102	202	−100 [−167, −44]	0	0	6.1%	+33 [−18, +88]
SEC 2	Haiku	text	zero-config	33	202	−169 [−262, −92]	0	40	5.8%	−36 [−120, +48]
SEC 2	Haiku	no text	zero-config	171	202	−31 [−46, −18]	0	0	2.9%	+102 [+49, +183]
SEC 2	Opus	text	discovered	0	69	−69 [−136, −14]	0	43	0.9%	–
SEC 2	Opus	no text	discovered	51	69	−18 [−37, −4]	0	0	1.9%	–
SEC 2	Opus	text	hand + disc.	0	69	−69 [−136, −14]	0	43	1.5%	–
SEC 2	Opus	no text	hand + disc.	39	69	−30 [−63, −6]	0	0	2.9%	–
SEC 2	Opus	text	zero-config	2	69	−67 [−132, −14]	0	20	1.2%	–
SEC 2	Opus	no text	zero-config	51	69	−18 [−37, −4]	0	0	1.9%	–

Set	Model	Mode	Identities	Silent	Alone	Difference [95% CI]	Broken	Fixed	To human	vs. Opus alone [95% CI]
SEC 2	GPT	text	discovered	37	1,012	−975 [−1297, −708]	0	726	7.2%	−32 [−106, +31]
SEC 2	GPT	no text	discovered	793	1,012	−219 [−447, −84]	0	0	10.4%	+724 [+551, +920]
SEC 2	GPT	text	hand + disc.	25	1,012	−987 [−1311, −717]	0	727	9.3%	−44 [−116, +17]
SEC 2	GPT	no text	hand + disc.	529	1,012	−483 [−739, −290]	0	0	24.2%	+460 [+350, +592]
SEC 2	GPT	text	zero-config	133	1,012	−879 [−1177, −637]	0	73	24.2%	+64 [−32, +184]
SEC 2	GPT	no text	zero-config	793	1,012	−219 [−447, −84]	0	0	10.4%	+724 [+551, +920]
SEC 3	Haiku	text	discovered	18	216	−198 [−272, −134]	0	121	3.7%	−32 [−71, −2]
SEC 3	Haiku	no text	discovered	170	216	−46 [−65, −28]	0	0	4.5%	+120 [+66, +187]
SEC 3	Haiku	text	hand + disc.	15	216	−201 [−278, −135]	0	121	4.0%	−35 [−73, −6]
SEC 3	Haiku	no text	hand + disc.	125	216	−91 [−149, −47]	0	0	7.9%	+75 [+43, +108]
SEC 3	Haiku	text	zero-config	43	216	−173 [−226, −124]	0	51	4.4%	−7 [−57, +47]
SEC 3	Haiku	no text	zero-config	170	216	−46 [−65, −28]	0	0	4.5%	+120 [+66, +187]
SEC 3	Opus	text	discovered	5	50	−45 [−82, −17]	0	39	2.1%	–
SEC 3	Opus	no text	discovered	40	50	−10 [−23, −2]	0	0	0.8%	–
SEC 3	Opus	text	hand + disc.	5	50	−45 [−82, −17]	0	39	2.2%	–
SEC 3	Opus	no text	hand + disc.	39	50	−11 [−26, −2]	0	0	0.9%	–
SEC 3	Opus	text	zero-config	6	50	−44 [−80, −17]	0	16	2.6%	–
SEC 3	Opus	no text	zero-config	40	50	−10 [−23, −2]	0	0	0.8%	–
SEC 3	GPT	text	discovered	65	1,443	−1378 [−1735, −1069]	0	874	13.1%	+15 [−31, +59]
SEC 3	GPT	no text	discovered	1,205	1,443	−238 [−477, −103]	0	0	9.0%	+1155 [+896, +1435]
SEC 3	GPT	text	hand + disc.	62	1,443	−1381 [−1740, −1070]	2	877	14.5%	+12 [−35, +55]
SEC 3	GPT	no text	hand + disc.	755	1,443	−688 [−1002, −452]	0	0	29.6%	+705 [+543, +877]
SEC 3	GPT	text	zero-config	187	1,443	−1256 [−1585, −982]	0	50	35.2%	+137 [+56, +232]
SEC 3	GPT	no text	zero-config	1,205	1,443	−238 [−477, −103]	0	0	9.0%	+1155 [+896, +1435]
SEC 1+2	Haiku	text	discovered	8	311	−303 [−439, −188]	1	224	1.7%	−91 [−164, −33]
SEC 1+2	Haiku	no text	discovered	251	311	−60 [−85, −38]	0	0	3.5%	+152 [+83, +243]
SEC 1+2	Haiku	text	hand + disc.	4	311	−307 [−443, −192]	0	225	2.5%	−95 [−168, −37]
SEC 1+2	Haiku	no text	hand + disc.	158	311	−153 [−232, −85]	0	0	6.8%	+59 [+4, +119]
SEC 1+2	Haiku	text	zero-config	33	311	−278 [−393, −180]	0	91	5.1%	−66 [−150, +22]
SEC 1+2	Haiku	no text	zero-config	251	311	−60 [−85, −38]	0	0	3.5%	+152 [+83, +243]
SEC 1+2	Opus	text	discovered	5	99	−94 [−167, −37]	0	61	0.7%	–
SEC 1+2	Opus	no text	discovered	69	99	−30 [−52, −12]	0	0	2.0%	–
SEC 1+2	Opus	text	hand + disc.	0	99	−99 [−173, −42]	0	61	1.5%	–
SEC 1+2	Opus	no text	hand + disc.	51	99	−48 [−86, −19]	0	0	3.2%	–
SEC 1+2	Opus	text	zero-config	7	99	−92 [−163, −37]	0	36	1.0%	–
SEC 1+2	Opus	no text	zero-config	69	99	−30 [−52, −12]	0	0	2.0%	–
SEC 1+2	GPT	text	discovered	47	1,543	−1496 [−1864, −1183]	0	1,139	6.9%	−52 [−131, +17]
SEC 1+2	GPT	no text	discovered	1,277	1,543	−266 [−499, −124]	0	0	9.2%	+1178 [+945, +1438]
SEC 1+2	GPT	text	hand + disc.	33	1,543	−1510 [−1880, −1194]	0	1,142	9.1%	−66 [−144, +0]
SEC 1+2	GPT	no text	hand + disc.	832	1,543	−711 [−985, −499]	0	0	25.2%	+733 [+586, +899]
SEC 1+2	GPT	text	zero-config	189	1,543	−1354 [−1694, −1072]	0	105	24.4%	+90 [−18, +219]
SEC 1+2	GPT	no text	zero-config	1,277	1,543	−266 [−499, −124]	0	0	9.2%	+1178 [+945, +1438]
SEC 1–3	Haiku	text	discovered	26	527	−501 [−656, −366]	1	345	2.6%	−123 [−204, −55]
SEC 1–3	Haiku	no text	discovered	421	527	−106 [−138, −77]	0	0	3.9%	+272 [+182, +385]
SEC 1–3	Haiku	text	hand + disc.	19	527	−508 [−664, −371]	0	346	3.2%	−130 [−211, −62]
SEC 1–3	Haiku	no text	hand + disc.	283	527	−244 [−343, −160]	0	0	7.3%	+134 [+70, +202]
SEC 1–3	Haiku	text	zero-config	76	527	−451 [−577, −337]	0	142	4.8%	−73 [−173, +30]
SEC 1–3	Haiku	no text	zero-config	421	527	−106 [−138, −77]	0	0	3.9%	+272 [+182, +385]
SEC 1–3	Opus	text	discovered	10	149	−139 [−219, −73]	0	100	1.3%	–
SEC 1–3	Opus	no text	discovered	109	149	−40 [−64, −19]	0	0	1.5%	–
SEC 1–3	Opus	text	hand + disc.	5	149	−144 [−224, −77]	0	100	1.8%	–
SEC 1–3	Opus	no text	hand + disc.	90	149	−59 [−99, −26]	0	0	2.2%	–
SEC 1–3	Opus	text	zero-config	13	149	−136 [−214, −72]	0	52	1.7%	–
SEC 1–3	Opus	no text	zero-config	109	149	−40 [−64, −19]	0	0	1.5%	–
SEC 1–3	GPT	text	discovered	112	2,986	−2874 [−3382, −2417]	0	2,013	9.7%	−37 [−130, +46]
SEC 1–3	GPT	no text	discovered	2,482	2,986	−504 [−849, −265]	0	0	9.1%	+2333 [+1977, +2714]
SEC 1–3	GPT	text	hand + disc.	95	2,986	−2891 [−3400, −2430]	2	2,019	11.5%	−54 [−145, +27]
SEC 1–3	GPT	no text	hand + disc.	1,587	2,986	−1399 [−1817, −1065]	0	0	27.2%	+1438 [+1210, +1675]
SEC 1–3	GPT	text	zero-config	376	2,986	−2610 [−3075, −2191]	0	155	29.2%	+227 [+82, +387]
SEC 1–3	GPT	no text	zero-config	2,482	2,986	−504 [−849, −265]	0	0	9.1%	+2333 [+1977, +2714]
CORD	Haiku	text	discovered	16	262	−246 [−326, −173]	0	14	27.0%	−120 [−179, −68]
CORD	Haiku	no text	discovered	77	262	−185 [−262, −117]	0	0	19.6%	−59 [−126, +2]
CORD	Haiku	text	hand + disc.	12	262	−250 [−330, −177]	0	18	29.4%	−124 [−182, −73]
CORD	Haiku	no text	hand + disc.	53	262	−209 [−287, −138]	0	0	25.4%	−83 [−146, −27]
CORD	Haiku	text	zero-config	16	262	−246 [−326, −173]	0	14	26.8%	−120 [−179, −68]
CORD	Haiku	no text	zero-config	82	262	−180 [−256, −113]	0	0	18.7%	−54 [−121, +6]
CORD	Opus	text	discovered	7	136	−129 [−187, −78]	0	11	16.0%	–
CORD	Opus	no text	discovered	68	136	−68 [−120, −25]	0	0	7.2%	–
CORD	Opus	text	hand + disc.	7	136	−129 [−187, −78]	0	11	17.3%	–
CORD	Opus	no text	hand + disc.	67	136	−69 [−121, −26]	0	0	9.0%	–
CORD	Opus	text	zero-config	9	136	−127 [−184, −76]	0	11	15.4%	–
CORD	Opus	no text	zero-config	68	136	−68 [−120, −25]	0	0	7.2%	–
Notes	Haiku	text	discovered	46	48	−2 [−5, +0]	0	0	2.9%	+46 [+0, +115]
Notes	Haiku	no text	discovered	48	48	+0 [+0, +0]	0	0	0.7%	+48 [+0, +120]
Notes	Haiku	text	hand + disc.	22	48	−26 [−76, +0]	0	0	3.1%	+22 [+0, +63]
Notes	Haiku	no text	hand + disc.	24	48	−24 [−72, +0]	0	0	0.9%	+24 [+0, +68]
Notes	Haiku	text	zero-config	46	48	−2 [−5, +0]	0	0	1.8%	+46 [+0, +115]
Notes	Haiku	no text	zero-config	48	48	+0 [+0, +0]	0	0	1.0%	+48 [+0, +120]
Notes	Opus	text	discovered	0	0	+0 [+0, +0]	0	0	2.9%	–
Notes	Opus	no text	discovered	0	0	+0 [+0, +0]	0	0	0.7%	–

Set	Model	Mode	Identities	Silent	Alone	Difference [95% CI]	Broken	Fixed	To human	vs. Opus alone [95% CI]
Notes	Opus	text	hand + disc.	0	0	+0 [+0, +0]	0	0	2.8%	–
Notes	Opus	no text	hand + disc.	0	0	+0 [+0, +0]	0	0	0.7%	–
Notes	Opus	text	zero-config	0	0	+0 [+0, +0]	0	0	1.8%	–
Notes	Opus	no text	zero-config	0	0	+0 [+0, +0]	0	0	0.7%	–
Notes	GPT	text	discovered	0	2,005	−2005 [−3403, −837]	0	8	20.5%	+0 [+0, +0]
Notes	GPT	no text	discovered	1	2,005	−2004 [−3402, −837]	0	0	18.2%	+1 [+0, +3]
Notes	GPT	text	hand + disc.	0	2,005	−2005 [−3403, −837]	0	8	20.5%	+0 [+0, +0]
Notes	GPT	no text	hand + disc.	1	2,005	−2004 [−3402, −837]	0	0	18.3%	+1 [+0, +3]
Notes	GPT	text	zero-config	1	2,005	−2004 [−3402, −837]	0	0	23.7%	+1 [+0, +3]
Notes	GPT	no text	zero-config	1	2,005	−2004 [−3402, −837]	0	0	18.2%	+1 [+0, +3]
Real scans	Haiku	no text	discovered	39	39	+0 [+0, +0]	0	0	0.0%	+0 [−3, +3]
Real scans	Haiku	no text	hand + disc.	3	39	−36 [−86, −4]	0	0	15.9%	−36 [−86, −5]
Real scans	Haiku	no text	zero-config	39	39	+0 [+0, +0]	0	0	0.0%	+0 [−3, +3]
Real scans	Opus	no text	discovered	39	39	+0 [+0, +0]	0	0	0.0%	–
Real scans	Opus	no text	hand + disc.	2	39	−37 [−88, −4]	0	0	18.6%	–
Real scans	Opus	no text	zero-config	39	39	+0 [+0, +0]	0	0	0.0%	–
Synthetic	Haiku	text	discovered	18	2	+16 [+9, +24]	16	0	1.5%	+18 [+10, +27]
Synthetic	Haiku	no text	discovered	2	2	+0 [+0, +0]	0	0	0.6%	+2 [+0, +5]
Synthetic	Haiku	text	hand + disc.	2	2	+0 [+0, +0]	0	0	3.0%	+2 [+0, +5]
Synthetic	Haiku	no text	hand + disc.	2	2	+0 [+0, +0]	0	0	1.9%	+2 [+0, +5]
Synthetic	Haiku	text	zero-config	10	2	+8 [+3, +14]	8	0	1.5%	+10 [+4, +16]
Synthetic	Haiku	no text	zero-config	2	2	+0 [+0, +0]	0	0	0.6%	+2 [+0, +5]
Synthetic	Opus	text	discovered	16	0	+16 [+9, +24]	16	0	1.7%	–
Synthetic	Opus	no text	discovered	0	0	+0 [+0, +0]	0	0	0.7%	–
Synthetic	Opus	text	hand + disc.	0	0	+0 [+0, +0]	0	0	3.3%	–
Synthetic	Opus	no text	hand + disc.	0	0	+0 [+0, +0]	0	0	1.9%	–
Synthetic	Opus	text	zero-config	8	0	+8 [+3, +14]	8	0	1.6%	–
Synthetic	Opus	no text	zero-config	0	0	+0 [+0, +0]	0	0	0.7%	–

Table S2.16: Discovered identities against hand identities, same model and mode (paired bootstrap of the silent difference) Positive: more silent errors than with the hand identities. Coverage: share of numbers in some identity, setting / hand.

Set	Model	Mode	Identities	Silent	Hand	Difference [95% CI]	Broken (hand)	Fixed (hand)	To human (hand)	Coverage (hand)
SEC 1	Haiku	text	discovered	0	0	+0 [+0, +0]	0 (0)	84 (84)	1.8 (2.4)%	89.2 (85.6)%
SEC 1	Haiku	no text	discovered	80	57	+23 [−1, +56]	0 (0)	0 (0)	4.7 (7.2)%	87.5 (85.6)%
SEC 1	Haiku	text	hand + disc.	0	0	+0 [+0, +0]	0 (0)	84 (84)	2.3 (2.4)%	93.8 (85.6)%
SEC 1	Haiku	no text	hand + disc.	56	57	−1 [−3, +0]	0 (0)	0 (0)	8.2 (7.2)%	92.7 (85.6)%
SEC 1	Haiku	text	zero-config	0	0	+0 [+0, +0]	0 (0)	51 (84)	3.7 (2.4)%	84.7 (85.6)%
SEC 1	Haiku	no text	zero-config	80	57	+23 [−1, +56]	0 (0)	0 (0)	4.7 (7.2)%	87.5 (85.6)%
SEC 1	Opus	text	discovered	5	0	+5 [+0, +14]	0 (0)	18 (18)	0.4 (0.7)%	90.2 (85.6)%
SEC 1	Opus	no text	discovered	18	13	+5 [+0, +14]	0 (0)	0 (0)	2.4 (2.9)%	90.6 (85.6)%
SEC 1	Opus	text	hand + disc.	0	0	+0 [+0, +0]	0 (0)	18 (18)	1.4 (0.7)%	93.8 (85.6)%
SEC 1	Opus	no text	hand + disc.	12	13	−1 [−3, +0]	0 (0)	0 (0)	3.8 (2.9)%	92.5 (85.6)%
SEC 1	Opus	text	zero-config	5	0	+5 [+0, +14]	0 (0)	16 (18)	0.5 (0.7)%	88.5 (85.6)%
SEC 1	Opus	no text	zero-config	18	13	+5 [+0, +14]	0 (0)	0 (0)	2.4 (2.9)%	90.6 (85.6)%
SEC 1	GPT	text	discovered	10	8	+2 [+0, +5]	0 (0)	413 (413)	6.4 (8.7)%	75.4 (85.6)%
SEC 1	GPT	no text	discovered	484	316	+168 [+84, +260]	0 (0)	0 (0)	6.7 (25.8)%	60.7 (85.6)%
SEC 1	GPT	text	hand + disc.	8	8	+0 [+0, +0]	0 (0)	415 (413)	8.8 (8.7)%	89.3 (85.6)%
SEC 1	GPT	no text	hand + disc.	303	316	−13 [−38, +0]	0 (0)	0 (0)	27.3 (25.8)%	90.4 (85.6)%
SEC 1	GPT	text	zero-config	56	8	+48 [+10, +96]	0 (0)	32 (413)	24.9 (8.7)%	51.1 (85.6)%
SEC 1	GPT	no text	zero-config	484	316	+168 [+84, +260]	0 (0)	0 (0)	6.7 (25.8)%	60.7 (85.6)%
SEC 2	Haiku	text	discovered	8	4	+4 [+0, +9]	1 (0)	140 (141)	1.7 (2.4)%	83.7 (82.9)%
SEC 2	Haiku	no text	discovered	171	105	+66 [+18, +124]	0 (0)	0 (0)	2.9 (5.5)%	82.7 (82.9)%
SEC 2	Haiku	text	hand + disc.	4	4	+0 [+0, +0]	0 (0)	141 (141)	2.6 (2.4)%	90.3 (82.9)%
SEC 2	Haiku	no text	hand + disc.	102	105	−3 [−7, +0]	0 (0)	0 (0)	6.1 (5.5)%	89.5 (82.9)%
SEC 2	Haiku	text	zero-config	33	4	+29 [+0, +84]	0 (0)	40 (141)	5.8 (2.4)%	77.8 (82.9)%
SEC 2	Haiku	no text	zero-config	171	105	+66 [+18, +124]	0 (0)	0 (0)	2.9 (5.5)%	82.7 (82.9)%
SEC 2	Opus	text	discovered	0	0	+0 [+0, +0]	0 (0)	43 (43)	0.9 (1.3)%	85.2 (82.9)%
SEC 2	Opus	no text	discovered	51	42	+9 [−7, +31]	0 (0)	0 (0)	1.9 (2.4)%	86.8 (82.9)%
SEC 2	Opus	text	hand + disc.	0	0	+0 [+0, +0]	0 (0)	43 (43)	1.5 (1.3)%	90.0 (82.9)%
SEC 2	Opus	no text	hand + disc.	39	42	−3 [−8, +0]	0 (0)	0 (0)	2.9 (2.4)%	90.0 (82.9)%
SEC 2	Opus	text	zero-config	2	0	+2 [+0, +6]	0 (0)	20 (43)	1.2 (1.3)%	84.3 (82.9)%
SEC 2	Opus	no text	zero-config	51	42	+9 [−7, +31]	0 (0)	0 (0)	1.9 (2.4)%	86.8 (82.9)%
SEC 2	GPT	text	discovered	37	27	+10 [+0, +22]	0 (0)	726 (720)	7.2 (8.9)%	76.0 (82.9)%
SEC 2	GPT	no text	discovered	793	541	+252 [+156, +364]	0 (0)	0 (0)	10.4 (22.4)%	63.1 (82.9)%
SEC 2	GPT	text	hand + disc.	25	27	−2 [−5, +0]	0 (0)	727 (720)	9.3 (8.9)%	89.0 (82.9)%
SEC 2	GPT	no text	hand + disc.	529	541	−12 [−26, +0]	0 (0)	0 (0)	24.2 (22.4)%	88.3 (82.9)%
SEC 2	GPT	text	zero-config	133	27	+106 [+43, +196]	0 (0)	73 (720)	24.2 (8.9)%	52.3 (82.9)%
SEC 2	GPT	no text	zero-config	793	541	+252 [+156, +364]	0 (0)	0 (0)	10.4 (22.4)%	63.1 (82.9)%
SEC 3	Haiku	text	discovered	18	15	+3 [+0, +8]	0 (0)	121 (123)	3.7 (3.3)%	84.7 (81.7)%
SEC 3	Haiku	no text	discovered	170	123	+47 [+10, +101]	0 (0)	0 (0)	4.5 (7.9)%	77.7 (81.7)%
SEC 3	Haiku	text	hand + disc.	15	15	+0 [+0, +0]	0 (0)	121 (123)	4.0 (3.3)%	89.4 (81.7)%
SEC 3	Haiku	no text	hand + disc.	125	123	+2 [+0, +6]	0 (0)	0 (0)	7.9 (7.9)%	87.9 (81.7)%
SEC 3	Haiku	text	zero-config	43	15	+28 [+1, +72]	0 (0)	51 (123)	4.4 (3.3)%	72.7 (81.7)%
SEC 3	Haiku	no text	zero-config	170	123	+47 [+10, +101]	0 (0)	0 (0)	4.5 (7.9)%	77.7 (81.7)%
SEC 3	Opus	text	discovered	5	5	+0 [+0, +0]	0 (0)	39 (39)	2.1 (1.4)%	87.3 (81.7)%
SEC 3	Opus	no text	discovered	40	39	+1 [−2, +4]	0 (0)	0 (0)	0.8 (0.7)%	85.7 (81.7)%

Set	Model	Mode	Identities	Silent	Hand	Difference [95% CI]	Broken (hand)	Fixed (hand)	To human (hand)	Coverage (hand)
SEC 3	Opus	text	hand + disc.	5	5	+0 [+0, +0]	0 (0)	39 (39)	2.2 (1.4)%	90.4 (81.7)%
SEC 3	Opus	no text	hand + disc.	39	39	+0 [+0, +0]	0 (0)	0 (0)	0.9 (0.7)%	89.6 (81.7)%
SEC 3	Opus	text	zero-config	6	5	+1 [+0, +3]	0 (0)	16 (39)	2.6 (1.4)%	82.5 (81.7)%
SEC 3	Opus	no text	zero-config	40	39	+1 [−2, +4]	0 (0)	0 (0)	0.8 (0.7)%	85.7 (81.7)%
SEC 3	GPT	text	discovered	65	65	+0 [−10, +10]	0 (2)	874 (887)	13.1 (13.8)%	78.3 (81.7)%
SEC 3	GPT	no text	discovered	1,205	754	+451 [+293, +629]	0 (0)	0 (0)	9.0 (29.6)%	56.8 (81.7)%
SEC 3	GPT	text	hand + disc.	62	65	−3 [−9, +0]	2 (2)	877 (887)	14.5 (13.8)%	88.9 (81.7)%
SEC 3	GPT	no text	hand + disc.	755	754	+1 [−16, +18]	0 (0)	0 (0)	29.6 (29.6)%	86.5 (81.7)%
SEC 3	GPT	text	zero-config	187	65	+122 [+55, +202]	0 (2)	50 (887)	35.2 (13.8)%	46.7 (81.7)%
SEC 3	GPT	no text	zero-config	1,205	754	+451 [+293, +629]	0 (0)	0 (0)	9.0 (29.6)%	56.8 (81.7)%
SEC 1+2	Haiku	text	discovered	8	4	+4 [+0, +9]	1 (0)	224 (225)	1.7 (2.4)%	85.5 (83.8)%
SEC 1+2	Haiku	no text	discovered	251	162	+89 [+33, +157]	0 (0)	0 (0)	3.5 (6.0)%	84.3 (83.8)%
SEC 1+2	Haiku	text	hand + disc.	4	4	+0 [+0, +0]	0 (0)	225 (225)	2.5 (2.4)%	91.4 (83.8)%
SEC 1+2	Haiku	no text	hand + disc.	158	162	−4 [−8, −1]	0 (0)	0 (0)	6.8 (6.0)%	90.5 (83.8)%
SEC 1+2	Haiku	text	zero-config	33	4	+29 [+0, +84]	0 (0)	91 (225)	5.1 (2.4)%	80.1 (83.8)%
SEC 1+2	Haiku	no text	zero-config	251	162	+89 [+33, +157]	0 (0)	0 (0)	3.5 (6.0)%	84.3 (83.8)%
SEC 1+2	Opus	text	discovered	5	0	+5 [+0, +14]	0 (0)	61 (61)	0.7 (1.1)%	86.9 (83.8)%
SEC 1+2	Opus	no text	discovered	69	55	+14 [−4, +37]	0 (0)	0 (0)	2.0 (2.6)%	88.1 (83.8)%
SEC 1+2	Opus	text	hand + disc.	0	0	+0 [+0, +0]	0 (0)	61 (61)	1.5 (1.1)%	91.3 (83.8)%
SEC 1+2	Opus	no text	hand + disc.	51	55	−4 [−9, +0]	0 (0)	0 (0)	3.2 (2.6)%	90.8 (83.8)%
SEC 1+2	Opus	text	zero-config	7	0	+7 [+0, +17]	0 (0)	36 (61)	1.0 (1.1)%	85.7 (83.8)%
SEC 1+2	Opus	no text	zero-config	69	55	+14 [−4, +37]	0 (0)	0 (0)	2.0 (2.6)%	88.1 (83.8)%
SEC 1+2	GPT	text	discovered	47	35	+12 [+2, +24]	0 (0)	1,139 (1,133)	6.9 (8.8)%	75.8 (83.8)%
SEC 1+2	GPT	no text	discovered	1,277	857	+420 [+286, +564]	0 (0)	0 (0)	9.2 (23.6)%	62.3 (83.8)%
SEC 1+2	GPT	text	hand + disc.	33	35	−2 [−5, +0]	0 (0)	1,142 (1,133)	9.1 (8.8)%	89.1 (83.8)%
SEC 1+2	GPT	no text	hand + disc.	832	857	−25 [−56, −4]	0 (0)	0 (0)	25.2 (23.6)%	89.0 (83.8)%
SEC 1+2	GPT	text	zero-config	189	35	+154 [+76, +255]	0 (0)	105 (1,133)	24.4 (8.8)%	51.9 (83.8)%
SEC 1+2	GPT	no text	zero-config	1,277	857	+420 [+286, +564]	0 (0)	0 (0)	9.2 (23.6)%	62.3 (83.8)%
SEC 1–3	Haiku	text	discovered	26	19	+7 [+1, +14]	1 (0)	345 (348)	2.6 (2.8)%	85.1 (82.8)%
SEC 1–3	Haiku	no text	discovered	421	285	+136 [+64, +222]	0 (0)	0 (0)	3.9 (6.9)%	81.3 (82.8)%
SEC 1–3	Haiku	text	hand + disc.	19	19	+0 [+0, +0]	0 (0)	346 (348)	3.2 (2.8)%	90.5 (82.8)%
SEC 1–3	Haiku	no text	hand + disc.	283	285	−2 [−7, +4]	0 (0)	0 (0)	7.3 (6.9)%	89.4 (82.8)%
SEC 1–3	Haiku	text	zero-config	76	19	+57 [+5, +130]	0 (0)	142 (348)	4.8 (2.8)%	76.8 (82.8)%
SEC 1–3	Haiku	no text	zero-config	421	285	+136 [+64, +222]	0 (0)	0 (0)	3.9 (6.9)%	81.3 (82.8)%
SEC 1–3	Opus	text	discovered	10	5	+5 [+0, +14]	0 (0)	100 (100)	1.3 (1.2)%	87.1 (82.8)%
SEC 1–3	Opus	no text	discovered	109	94	+15 [−3, +38]	0 (0)	0 (0)	1.5 (1.8)%	87.0 (82.8)%
SEC 1–3	Opus	text	hand + disc.	5	5	+0 [+0, +0]	0 (0)	100 (100)	1.8 (1.2)%	90.9 (82.8)%
SEC 1–3	Opus	no text	hand + disc.	90	94	−4 [−9, +0]	0 (0)	0 (0)	2.2 (1.8)%	90.3 (82.8)%
SEC 1–3	Opus	text	zero-config	13	5	+8 [+1, +19]	0 (0)	52 (100)	1.7 (1.2)%	84.3 (82.8)%
SEC 1–3	Opus	no text	zero-config	109	94	+15 [−3, +38]	0 (0)	0 (0)	1.5 (1.8)%	87.0 (82.8)%
SEC 1–3	GPT	text	discovered	112	100	+12 [−2, +27]	0 (2)	2,013 (2,020)	9.7 (11.0)%	76.9 (82.8)%
SEC 1–3	GPT	no text	discovered	2,482	1,611	+871 [+659, +1100]	0 (0)	0 (0)	9.1 (26.2)%	59.9 (82.8)%
SEC 1–3	GPT	text	hand + disc.	95	100	−5 [−13, +0]	2 (2)	2,019 (2,020)	11.5 (11.0)%	89.0 (82.8)%
SEC 1–3	GPT	no text	hand + disc.	1,587	1,611	−24 [−58, +5]	0 (0)	0 (0)	27.2 (26.2)%	87.9 (82.8)%
SEC 1–3	GPT	text	zero-config	376	100	+276 [+167, +407]	0 (2)	155 (2,020)	29.2 (11.0)%	49.6 (82.8)%
SEC 1–3	GPT	no text	zero-config	2,482	1,611	+871 [+659, +1100]	0 (0)	0 (0)	9.1 (26.2)%	59.9 (82.8)%
CORD	Haiku	text	discovered	16	12	+4 [+0, +12]	0 (0)	14 (17)	27.0 (29.2)%	51.2 (71.5)%
CORD	Haiku	no text	discovered	77	52	+25 [+8, +47]	0 (0)	0 (0)	19.6 (25.1)%	45.8 (71.5)%
CORD	Haiku	text	hand + disc.	12	12	+0 [+0, +0]	0 (0)	18 (17)	29.4 (29.2)%	77.7 (71.5)%
CORD	Haiku	no text	hand + disc.	53	52	+1 [+0, +3]	0 (0)	0 (0)	25.4 (25.1)%	73.4 (71.5)%
CORD	Haiku	text	zero-config	16	12	+4 [+0, +12]	0 (0)	14 (17)	26.8 (29.2)%	50.1 (71.5)%
CORD	Haiku	no text	zero-config	82	52	+30 [+10, +55]	0 (0)	0 (0)	18.7 (25.1)%	42.1 (71.5)%
CORD	Opus	text	discovered	7	7	+0 [+0, +0]	0 (0)	11 (11)	16.0 (17.0)%	61.3 (71.5)%
CORD	Opus	no text	discovered	68	67	+1 [+0, +3]	0 (0)	0 (0)	7.2 (8.7)%	55.0 (71.5)%
CORD	Opus	text	hand + disc.	7	7	+0 [+0, +0]	0 (0)	11 (11)	17.3 (17.0)%	79.0 (71.5)%
CORD	Opus	no text	hand + disc.	67	67	+0 [+0, +0]	0 (0)	0 (0)	9.0 (8.7)%	74.0 (71.5)%
CORD	Opus	text	zero-config	9	7	+2 [+0, +5]	0 (0)	11 (11)	15.4 (17.0)%	59.2 (71.5)%
CORD	Opus	no text	zero-config	68	67	+1 [+0, +3]	0 (0)	0 (0)	7.2 (8.7)%	37.3 (71.5)%
Notes	Haiku	text	discovered	46	23	+23 [−3, +72]	0 (1)	0 (0)	2.9 (2.3)%	56.5 (32.5)%
Notes	Haiku	no text	discovered	48	24	+24 [+0, +72]	0 (0)	0 (0)	0.7 (0.2)%	57.1 (32.5)%
Notes	Haiku	text	hand + disc.	22	23	−1 [−3, +0]	0 (1)	0 (0)	3.1 (2.3)%	67.2 (32.5)%
Notes	Haiku	no text	hand + disc.	24	24	+0 [+0, +0]	0 (0)	0 (0)	0.9 (0.2)%	67.3 (32.5)%
Notes	Haiku	text	zero-config	46	23	+23 [−3, +72]	0 (1)	0 (0)	1.8 (2.3)%	55.8 (32.5)%
Notes	Haiku	no text	zero-config	48	24	+24 [+0, +72]	0 (0)	0 (0)	1.0 (0.2)%	57.3 (32.5)%
Notes	Opus	text	discovered	0	1	−1 [−3, +0]	0 (1)	0 (0)	2.9 (2.1)%	56.6 (32.5)%
Notes	Opus	no text	discovered	0	0	+0 [+0, +0]	0 (0)	0 (0)	0.7 (0.0)%	57.2 (32.5)%
Notes	Opus	text	hand + disc.	0	1	−1 [−3, +0]	0 (1)	0 (0)	2.8 (2.1)%	67.2 (32.5)%
Notes	Opus	no text	hand + disc.	0	0	+0 [+0, +0]	0 (0)	0 (0)	0.7 (0.0)%	67.3 (32.5)%
Notes	Opus	text	zero-config	0	1	−1 [−3, +0]	0 (1)	0 (0)	1.8 (2.1)%	55.9 (32.5)%
Notes	Opus	no text	zero-config	0	0	+0 [+0, +0]	0 (0)	0 (0)	0.7 (0.0)%	57.2 (32.5)%
Notes	GPT	text	discovered	0	3	−3 [−7, +0]	0 (1)	8 (8)	20.5 (19.7)%	48.2 (32.5)%
Notes	GPT	no text	discovered	1	3	−2 [−5, +0]	0 (0)	0 (0)	18.2 (18.0)%	48.1 (32.5)%
Notes	GPT	text	hand + disc.	0	3	−3 [−7, +0]	0 (1)	8 (8)	20.5 (19.7)%	61.5 (32.5)%
Notes	GPT	no text	hand + disc.	1	3	−2 [−5, +0]	0 (0)	0 (0)	18.3 (18.0)%	61.5 (32.5)%
Notes	GPT	text	zero-config	1	3	−2 [−6, +2]	0 (1)	0 (8)	23.7 (19.7)%	47.3 (32.5)%
Notes	GPT	no text	zero-config	1	3	−2 [−5, +0]	0 (0)	0 (0)	18.2 (18.0)%	48.1 (32.5)%
Real scans	Haiku	no text	discovered	39	3	+36 [+4, +86]	0 (0)	0 (0)	0.0 (13.8)%	40.9 (98.9)%
Real scans	Haiku	no text	hand + disc.	3	3	+0 [+0, +0]	0 (0)	0 (0)	15.9 (13.8)%	98.9 (98.9)%
Real scans	Haiku	no text	zero-config	39	3	+36 [+4, +86]	0 (0)	0 (0)	0.0 (13.8)%	40.9 (98.9)%
Real scans	Opus	no text	discovered	39	2	+37 [+4, +88]	0 (0)	0 (0)	0.0 (16.6)%	41.9 (98.9)%

Set	Model	Mode	Identities	Silent	Hand	Difference [95% CI]	Broken (hand)	Fixed (hand)	To human (hand)	Coverage (hand)
Real scans	Opus	no text	hand + disc.	2	2	+0 [+0, +0]	0 (0)	0 (0)	18.6 (16.6)%	98.9 (98.9)%
Real scans	Opus	no text	zero-config	39	2	+37 [+4, +88]	0 (0)	0 (0)	0.0 (16.6)%	41.9 (98.9)%
Synthetic	Haiku	text	discovered	18	2	+16 [+9, +24]	16 (0)	0 (0)	1.5 (2.5)%	81.2 (97.8)%
Synthetic	Haiku	no text	discovered	2	2	+0 [+0, +0]	0 (0)	0 (0)	0.6 (1.5)%	78.2 (97.8)%
Synthetic	Haiku	text	hand + disc.	2	2	+0 [+0, +0]	0 (0)	0 (0)	3.0 (2.5)%	97.8 (97.8)%
Synthetic	Haiku	no text	hand + disc.	2	2	+0 [+0, +0]	0 (0)	0 (0)	1.9 (1.5)%	97.8 (97.8)%
Synthetic	Haiku	text	zero-config	10	2	+8 [+3, +14]	8 (0)	0 (0)	1.5 (2.5)%	81.3 (97.8)%
Synthetic	Haiku	no text	zero-config	2	2	+0 [+0, +0]	0 (0)	0 (0)	0.6 (1.5)%	78.2 (97.8)%
Synthetic	Opus	text	discovered	16	0	+16 [+9, +24]	16 (0)	0 (0)	1.7 (2.8)%	81.3 (97.8)%
Synthetic	Opus	no text	discovered	0	0	+0 [+0, +0]	0 (0)	0 (0)	0.7 (1.6)%	78.2 (97.8)%
Synthetic	Opus	text	hand + disc.	0	0	+0 [+0, +0]	0 (0)	0 (0)	3.3 (2.8)%	97.8 (97.8)%
Synthetic	Opus	no text	hand + disc.	0	0	+0 [+0, +0]	0 (0)	0 (0)	1.9 (1.6)%	97.8 (97.8)%
Synthetic	Opus	text	zero-config	8	0	+8 [+3, +14]	8 (0)	0 (0)	1.6 (2.8)%	81.3 (97.8)%
Synthetic	Opus	no text	zero-config	0	0	+0 [+0, +0]	0 (0)	0 (0)	0.7 (1.6)%	78.2 (97.8)%

Table S3.1: Development data, not held out: frozen qcore6 → frozen qcore7, zero-config, on every set already spent (the qcore6 held-out sets of Section 5 and qcore6's development sets), pooled over models and over text and no-text modes; legacy scoring, as in Section 5. qcore7 was tuned on these sets. *inj.*: extractions with injected errors. qcore6 dev sets (pooled): SEC, notes, CORD train and validation, real-scan and synthetic 00–01 development data. With hand identities qcore7 returned qcore6's fixes and tier 2 items in all 190 development runs. Every set and identity setting: Section S4.

Set (zero-config)	Numbers	Silent	Broken	To a person %
SEC set 1	12,864	643 → 605	0 → 0	7.1 → 5.8
SEC set 2	26,544	1,183 → 1,080	0 → 0	7.7 → 5.9
SEC set 3	31,236	1,651 → 1,536	0 → 0	9.4 → 7.7
CORD test	3,992	175 → 153	0 → 0	17.0 → 19.3
Notes tables	88,668	96 → 101	0 → 6	7.9 → 7.8
Real scans	2,018	78 → 5	0 → 0	0.0 → 20.6
Synthetic 02–05	31,568	20 → 4	16 → 0	1.1 → 1.4
qcore6 dev sets	98,796	278 → 244	0 → 0	5.0 → 5.8
qcore6 dev sets inj.	55,010	673 → 543	0 → 0	2.8 → 4.0

S3 Frozen qcore7 on the fresh synthetic held-out set: every run

Material moved from Appendix D of the paper; the run protocol is described there.

Table S3.2: Frozen qcore6 and qcore7 on the fresh synthetic held-out set, every run (strict scoring)
Silent [95% CI], broken [95% CI], fixed, flagged, % of numbers to a person, identity coverage. Model alone: clean 0, low 89, high 296 wrong of 5,920. Source: `out/qcore7_heldout/results.md`.

Extraction	Mode	Identities	Version	Silent [CI]	Broken	Fixed	Flagged	To a person	Coverage
clean	text	hand	qcore6	0 [0, 0]	0	0	174	2.9%	98%
clean	text	hand	qcore7	0 [0, 0]	0	0	174	2.9%	98%
clean	text	discovered	qcore6	16 [9, 24]	16	0	123	2.1%	82%
clean	text	discovered	qcore7	0 [0, 0]	0	0	123	2.1%	82%
clean	text	hand + disc.	qcore6	0 [0, 0]	0	0	218	3.7%	98%
clean	text	hand + disc.	qcore7	0 [0, 0]	0	0	186	3.1%	98%
clean	text	zero-config	qcore6	8 [3, 14]	8	0	116	2.0%	82%
clean	text	zero-config	qcore7	0 [0, 0]	0	0	142	2.4%	82%
clean	no text	hand	qcore6	0 [0, 0]	0	0	80	1.4%	98%
clean	no text	hand	qcore7	0 [0, 0]	0	0	80	1.4%	98%
clean	no text	discovered	qcore6	0 [0, 0]	0	0	34	0.6%	79%
clean	no text	discovered	qcore7	0 [0, 0]	0	0	34	0.6%	79%
clean	no text	hand + disc.	qcore6	0 [0, 0]	0	0	116	2.0%	98%
clean	no text	hand + disc.	qcore7	0 [0, 0]	0	0	116	2.0%	98%
clean	no text	zero-config	qcore6	0 [0, 0]	0	0	34	0.6%	79%
clean	no text	zero-config	qcore7	0 [0, 0]	0	0	34	0.6%	79%
low, 1.5%	text	hand	qcore6	5 [1, 10]	0	55	328	5.5%	98%
low, 1.5%	text	hand	qcore7	5 [1, 10]	0	55	328	5.5%	98%
low, 1.5%	text	discovered	qcore6	32 [22, 42]	16	46	243	4.1%	81%
low, 1.5%	text	discovered	qcore7	14 [7, 21]	0	37	273	4.6%	81%
low, 1.5%	text	hand + disc.	qcore6	5 [1, 10]	0	54	381	6.4%	98%
low, 1.5%	text	hand + disc.	qcore7	5 [1, 10]	0	50	382	6.5%	98%
low, 1.5%	text	zero-config	qcore6	26 [17, 35]	8	39	251	4.2%	81%
low, 1.5%	text	zero-config	qcore7	15 [8, 23]	0	31	288	4.9%	81%
low, 1.5%	no text	hand	qcore6	6 [2, 11]	0	4	429	7.2%	98%
low, 1.5%	no text	hand	qcore7	6 [2, 11]	0	4	429	7.2%	98%
low, 1.5%	no text	discovered	qcore6	32 [21, 44]	0	2	237	4.0%	76%
low, 1.5%	no text	discovered	qcore7	27 [17, 38]	0	1	289	4.9%	76%
low, 1.5%	no text	hand + disc.	qcore6	6 [2, 11]	0	3	480	8.1%	98%
low, 1.5%	no text	hand + disc.	qcore7	6 [2, 11]	0	3	503	8.5%	98%
low, 1.5%	no text	zero-config	qcore6	32 [21, 44]	0	0	239	4.0%	76%
low, 1.5%	no text	zero-config	qcore7	27 [17, 38]	0	0	290	4.9%	77%
high, 5%	text	hand	qcore6	10 [4, 18]	0	171	791	13.4%	98%
high, 5%	text	hand	qcore7	10 [4, 18]	0	171	791	13.4%	98%
high, 5%	text	discovered	qcore6	77 [55, 101]	15	149	413	7.0%	75%
high, 5%	text	discovered	qcore7	54 [36, 74]	0	123	451	7.6%	75%
high, 5%	text	hand + disc.	qcore6	10 [4, 18]	0	169	847	14.3%	98%
high, 5%	text	hand + disc.	qcore7	8 [2, 16]	0	161	815	13.8%	98%
high, 5%	text	zero-config	qcore6	77 [56, 100]	8	121	455	7.7%	75%
high, 5%	text	zero-config	qcore7	60 [41, 81]	0	91	503	8.5%	74%
high, 5%	no text	hand	qcore6	17 [8, 27]	0	23	1214	20.5%	98%
high, 5%	no text	hand	qcore7	17 [8, 27]	0	23	1214	20.5%	98%
high, 5%	no text	discovered	qcore6	168 [140, 198]	0	11	446	7.5%	64%
high, 5%	no text	discovered	qcore7	147 [120, 176]	0	4	515	8.7%	65%
high, 5%	no text	hand + disc.	qcore6	17 [8, 27]	0	23	1260	21.3%	98%
high, 5%	no text	hand + disc.	qcore7	17 [8, 27]	0	23	1260	21.3%	98%
high, 5%	no text	zero-config	qcore6	168 [140, 198]	0	0	457	7.7%	63%
high, 5%	no text	zero-config	qcore7	147 [120, 176]	0	0	522	8.8%	65%

Table S3.4: Paired differences on the fresh held-out set (strict), sums over documents, 95% bootstrap intervals

Extraction	Mode	Identities	qcore7−qcore6 silent	qcore7−qcore6 broken	qcore7−qcore6 flagged	qcore6−hand silent	qcore7−hand silent
clean	text	hand	0 [0, 0]	0 [0, 0]	0 [0, 0]	–	–
clean	text	discovered	−16 [−24, −9]	−16 [−24, −9]	0 [0, 0]	+16 [+9, +24]	0 [0, 0]
clean	text	hand + disc.	0 [0, 0]	0 [0, 0]	−32 [−48, −18]	0 [0, 0]	0 [0, 0]
clean	text	zero-config	−8 [−14, −3]	−8 [−14, −3]	+26 [+10, +45]	+8 [+3, +14]	0 [0, 0]
clean	no text	hand	0 [0, 0]	0 [0, 0]	0 [0, 0]	–	–
clean	no text	discovered	0 [0, 0]	0 [0, 0]	0 [0, 0]	0 [0, 0]	0 [0, 0]
clean	no text	hand + disc.	0 [0, 0]	0 [0, 0]	0 [0, 0]	0 [0, 0]	0 [0, 0]
clean	no text	zero-config	0 [0, 0]	0 [0, 0]	0 [0, 0]	0 [0, 0]	0 [0, 0]
low, 1.5%	text	hand	0 [0, 0]	0 [0, 0]	0 [0, 0]	–	–
low, 1.5%	text	discovered	−18 [−26, −11]	−16 [−24, −9]	+30 [+5, +63]	+27 [+18, +37]	+9 [+4, +15]
low, 1.5%	text	hand + disc.	0 [0, 0]	0 [0, 0]	+1 [−26, +33]	0 [0, 0]	0 [0, 0]
low, 1.5%	text	zero-config	−11 [−18, −5]	−8 [−14, −3]	+37 [−4, +79]	+21 [+13, +30]	+10 [+4, +16]
low, 1.5%	no text	hand	0 [0, 0]	0 [0, 0]	0 [0, 0]	–	–
low, 1.5%	no text	discovered	−5 [−11, 0]	0 [0, 0]	+52 [+15, +97]	+26 [+17, +36]	+21 [+13, +30]
low, 1.5%	no text	hand + disc.	0 [0, 0]	0 [0, 0]	+23 [−4, +58]	0 [0, 0]	0 [0, 0]
low, 1.5%	no text	zero-config	−5 [−11, 0]	0 [0, 0]	+51 [+14, +96]	+26 [+17, +36]	+21 [+13, +30]
high, 5%	text	hand	0 [0, 0]	0 [0, 0]	0 [0, 0]	–	–
high, 5%	text	discovered	−23 [−32, −14]	−15 [−22, −8]	+38 [+22, +57]	+67 [+46, +90]	+44 [+27, +62]
high, 5%	text	hand + disc.	−2 [−5, 0]	0 [0, 0]	−32 [−56, −10]	0 [−3, +3]	−2 [−5, 0]
high, 5%	text	zero-config	−17 [−25, −10]	−8 [−14, −3]	+48 [+22, +74]	+67 [+46, +89]	+50 [+32, +69]
high, 5%	no text	hand	0 [0, 0]	0 [0, 0]	0 [0, 0]	–	–
high, 5%	no text	discovered	−21 [−32, −11]	0 [0, 0]	+69 [+30, +115]	+151 [+124, +180]	+130 [+104, +157]
high, 5%	no text	hand + disc.	0 [0, 0]	0 [0, 0]	0 [0, 0]	0 [0, 0]	0 [0, 0]
high, 5%	no text	zero-config	−21 [−32, −11]	0 [0, 0]	+65 [+24, +115]	+151 [+124, +180]	+130 [+104, +157]

Table S3.6: Legacy scoring (the rule of every earlier report: $|v-t| \le \max(0.5, 10^{-6}|t|)$) on the fresh held-out set: silent errors, qcore6 / qcore7 Model alone under this rule: clean 0, low 81, high 278 wrong. Broken counts are the same as under strict scoring.

Extraction	Mode	Hand	Discovered	Hand + disc.	Zero-config
clean	text	0 / 0	16 / 0	0 / 0	8 / 0
clean	no text	0 / 0	0 / 0	0 / 0	0 / 0
low, 1.5%	text	4 / 4	26 / 8	4 / 4	20 / 9
low, 1.5%	no text	5 / 5	26 / 21	5 / 5	26 / 21
high, 5%	text	7 / 7	63 / 40	8 / 6	63 / 46
high, 5%	no text	15 / 15	151 / 132	15 / 15	151 / 132

Table S3.7: Injected errors by kind, high extraction (strict): fixed / flagged / silent

Mode	Identities	Version	blank	column shift	digit sub	drop digit	extra digit	label as value	row shift	sign flip	transpose
text	hand	qcore6	12/6/0	21/22/1	34/28/3	16/9/0	17/4/2	9/14/0	11/19/0	30/2/2	22/11/2
text	hand	qcore7	12/6/0	21/22/1	34/28/3	16/9/0	17/4/2	9/14/0	11/19/0	30/2/2	22/11/2
text	zero-config	qcore6	9/9/0	15/15/14	21/26/18	12/7/6	11/9/3	9/11/3	10/11/9	18/6/10	17/12/6
text	zero-config	qcore7	4/14/0	13/17/14	14/35/16	9/11/5	7/13/3	8/14/1	11/13/6	13/12/9	13/16/6
no text	hand	qcore6	1/17/0	0/42/2	0/59/6	0/24/1	0/19/4	0/23/0	0/30/0	23/9/2	0/33/2
no text	hand	qcore7	1/17/0	0/42/2	0/59/6	0/24/1	0/19/4	0/23/0	0/30/0	23/9/2	0/33/2
no text	zero-config	qcore6	1/17/0	0/19/25	0/17/48	0/11/14	0/11/12	0/13/10	0/15/15	0/15/19	0/10/25
no text	zero-config	qcore7	1/17/0	0/20/24	0/20/45	0/11/14	0/14/9	0/22/1	0/16/14	0/18/16	0/11/24

Table S3.8: By domain, high extraction (strict): silent / broken / flagged

Mode	Identities	Version	Reports (2,290)	Invoices (1,444)	Bulletins (2,186)
text	hand	qcore6	5 / 0 / 190	4 / 0 / 78	1 / 0 / 523
text	hand	qcore7	5 / 0 / 190	4 / 0 / 78	1 / 0 / 523
text	discovered	qcore6	17 / 0 / 137	16 / 0 / 25	44 / 15 / 251
text	discovered	qcore7	17 / 0 / 148	12 / 0 / 25	25 / 0 / 278
text	zero-config	qcore6	17 / 0 / 151	24 / 0 / 54	36 / 8 / 250
text	zero-config	qcore7	16 / 0 / 173	19 / 0 / 56	25 / 0 / 274
no text	hand	qcore6	13 / 0 / 419	3 / 0 / 272	1 / 0 / 523
no text	hand	qcore7	13 / 0 / 419	3 / 0 / 272	1 / 0 / 523
no text	discovered	qcore6	55 / 0 / 179	65 / 0 / 19	48 / 0 / 248
no text	discovered	qcore7	52 / 0 / 188	53 / 0 / 63	42 / 0 / 264
no text	zero-config	qcore6	55 / 0 / 185	65 / 0 / 19	48 / 0 / 253
no text	zero-config	qcore7	52 / 0 / 188	53 / 0 / 63	42 / 0 / 271

S4 Development grid of frozen qcore7

Development data only. Every set below was spent before qcore7 was written, and qcore7 was tuned on them. Frozen qcore7 and frozen qcore6 were run on every (set, extraction, mode, identity setting) of the grid, 682 pairs of runs, none missing; for the qcore6 held-out sets the qcore6 runs are its held-out runs. Pooled over extractions and modes, legacy scoring (the rule of the qcore6 results). Hand-identity check: in all 190 hand runs qcore7 returned exactly qcore6's fixes (190) and tier 2 items (190). Source: `out/qcore7_work/dev_summary7.md`, `dev_summary7.json`.

Set	Identities	Extractions	Numbers	Silent qcore6 / qcore7	Broken	Fixed	To a person
SEC set 1	hand	real	25,728	885 / 885	2 / 2	563 / 563	14.3 / 14.3%
SEC set 1	discovered	real	12,864	597 / 597	0 / 0	515 / 476	3.7 / 4.0%
SEC set 1	hand + disc.	real	12,864	379 / 379	0 / 0	517 / 518	8.6 / 8.6%
SEC set 1	zero-config	real	12,864	643 / 605	0 / 0	99 / 260	7.1 / 5.8%
SEC set 2	hand	real	53,088	1,615 / 1,615	3 / 3	1,001 / 1,001	12.9 / 12.9%
SEC set 2	discovered	real	26,544	1,060 / 1,058	1 / 0	906 / 804	4.2 / 4.5%
SEC set 2	hand + disc.	real	26,544	699 / 699	0 / 0	908 / 871	7.8 / 7.8%
SEC set 2	zero-config	real	26,544	1,183 / 1,080	0 / 0	133 / 397	7.7 / 5.9%
SEC set 3	hand	real	62,472	2,304 / 2,304	6 / 6	1,108 / 1,108	16.2 / 16.2%
SEC set 3	discovered	real	31,236	1,503 / 1,502	0 / 0	1,019 / 955	5.5 / 5.7%
SEC set 3	hand + disc.	real	31,236	1,001 / 999	2 / 0	1,022 / 1,011	9.9 / 9.9%
SEC set 3	zero-config	real	31,236	1,651 / 1,536	0 / 0	117 / 481	9.4 / 7.7%
CORD test	hand	real	7,984	365 / 365	0 / 0	28 / 28	20.9 / 20.9%
CORD test	discovered	real	3,992	168 / 151	0 / 0	25 / 25	17.5 / 19.5%
CORD test	hand + disc.	real	3,992	139 / 134	0 / 0	29 / 29	20.3 / 21.2%
CORD test	zero-config	real	3,992	175 / 153	0 / 0	25 / 25	17.0 / 19.3%
Notes tables	hand	real	88,668	54 / 54	3 / 3	8 / 8	7.1 / 7.1%
Notes tables	discovered	real	88,668	95 / 95	0 / 0	8 / 1	7.7 / 7.7%
Notes tables	hand + disc.	real	88,668	47 / 47	0 / 0	8 / 1	7.7 / 7.8%
Notes tables	zero-config	real	88,668	96 / 101	0 / 6	0 / 0	7.9 / 7.8%
Real scans	hand	real	6,054	13 / 13	0 / 0	0 / 0	16.2 / 16.2%
Real scans	discovered	real	2,018	78 / 5	0 / 0	0 / 0	0.0 / 20.6%
Real scans	hand + disc.	real	2,018	5 / 5	0 / 0	0 / 0	17.2 / 22.6%
Real scans	zero-config	real	2,018	78 / 5	0 / 0	0 / 0	0.0 / 20.6%
Synthetic 02–05	hand	real	31,568	4 / 4	0 / 0	0 / 0	2.1 / 2.1%
Synthetic 02–05	discovered	real	31,568	36 / 4	32 / 0	0 / 0	1.1 / 1.1%
Synthetic 02–05	hand + disc.	real	31,568	4 / 4	0 / 0	0 / 0	2.5 / 2.3%
Synthetic 02–05	zero-config	real	31,568	20 / 4	16 / 0	0 / 0	1.1 / 1.4%
SEC dev	hand	real	19,560	113 / 113	0 / 0	100 / 100	8.9 / 8.9%
SEC dev	discovered	real	19,560	147 / 147	0 / 0	91 / 71	7.7 / 7.8%
SEC dev	hand + disc.	real	19,560	113 / 113	0 / 0	100 / 100	6.9 / 6.9%
SEC dev	zero-config	real	19,560	165 / 157	0 / 0	54 / 50	7.0 / 7.6%
SEC dev	hand	injected	4,890	17 / 17	0 / 0	54 / 54	3.2 / 3.2%
SEC dev	discovered	injected	4,890	30 / 30	0 / 0	54 / 39	3.1 / 3.4%
SEC dev	hand + disc.	injected	4,890	13 / 13	0 / 0	54 / 51	4.0 / 4.0%
SEC dev	zero-config	injected	4,890	41 / 33	0 / 0	38 / 27	3.3 / 3.6%
Notes dev	hand	real	39,534	36 / 36	0 / 0	16 / 16	6.6 / 6.6%
Notes dev	discovered	real	39,534	36 / 36	0 / 0	6 / 4	6.9 / 6.9%
Notes dev	hand + disc.	real	39,534	36 / 36	0 / 0	18 / 4	7.2 / 7.2%
Notes dev	zero-config	real	39,534	46 / 40	0 / 0	0 / 0	7.5 / 7.1%
Notes dev	hand	injected	13,178	67 / 67	0 / 0	18 / 18	1.2 / 1.2%
Notes dev	discovered	injected	13,178	61 / 61	0 / 0	17 / 12	2.0 / 2.0%
Notes dev	hand + disc.	injected	13,178	40 / 40	0 / 0	18 / 15	2.4 / 2.4%
Notes dev	zero-config	injected	13,178	67 / 65	0 / 0	14 / 11	1.8 / 1.8%
CORD train	hand	real	16,810	0 / 0	0 / 0	0 / 0	2.3 / 2.3%
CORD train	discovered	real	16,810	0 / 0	0 / 0	0 / 0	1.1 / 3.1%
CORD train	hand + disc.	real	16,810	0 / 0	0 / 0	0 / 0	2.5 / 4.2%
CORD train	zero-config	real	16,810	0 / 0	0 / 0	0 / 0	1.0 / 2.9%
CORD train	hand	injected	16,810	33 / 33	0 / 0	63 / 63	3.5 / 3.5%
CORD train	discovered	injected	16,810	142 / 81	0 / 0	36 / 33	1.6 / 4.1%
CORD train	hand + disc.	injected	16,810	48 / 42	0 / 0	55 / 54	4.0 / 5.7%

Set	Identities	Extractions	Numbers	Silent qcore6 / qcore7	Broken	Fixed	To a person
CORD train	zero-config	injected	16,810	147 / 86	0 / 0	34 / 31	1.5 / 3.9%
CORD val.	hand	real	3,768	21 / 21	0 / 0	2 / 2	8.5 / 8.5%
CORD val.	discovered	real	3,768	35 / 22	0 / 0	3 / 3	5.9 / 9.4%
CORD val.	hand + disc.	real	3,768	21 / 20	0 / 0	3 / 3	8.8 / 11.4%
CORD val.	zero-config	real	3,768	37 / 25	0 / 0	1 / 1	5.5 / 8.7%
CORD val.	hand	injected	3,768	43 / 43	0 / 0	34 / 34	12.3 / 12.3%
CORD val.	discovered	injected	3,768	141 / 108	0 / 0	17 / 15	7.1 / 9.4%
CORD val.	hand + disc.	injected	3,768	43 / 42	0 / 0	36 / 36	12.5 / 13.0%
CORD val.	zero-config	injected	3,768	145 / 112	0 / 0	10 / 8	6.8 / 8.9%
Real-scan dev	hand	real	2,760	8 / 8	0 / 0	0 / 0	24.9 / 24.9%
Real-scan dev	discovered	real	2,760	24 / 16	0 / 0	0 / 0	7.5 / 19.2%
Real-scan dev	hand + disc.	real	2,760	8 / 0	0 / 0	0 / 0	25.1 / 36.8%
Real-scan dev	zero-config	real	2,760	24 / 16	0 / 0	0 / 0	6.4 / 18.3%
Synthetic 00–01	hand	real	16,364	6 / 6	0 / 0	0 / 0	1.4 / 1.4%
Synthetic 00–01	discovered	real	16,364	6 / 6	0 / 0	0 / 0	0.6 / 0.7%
Synthetic 00–01	hand + disc.	real	16,364	6 / 6	0 / 0	0 / 0	1.5 / 1.6%
Synthetic 00–01	zero-config	real	16,364	6 / 6	0 / 0	0 / 0	0.6 / 0.7%
Synthetic 00–01	hand	injected	16,364	16 / 16	0 / 0	354 / 354	7.3 / 7.3%
Synthetic 00–01	discovered	injected	16,364	258 / 232	0 / 0	254 / 202	3.5 / 4.3%
Synthetic 00–01	hand + disc.	injected	16,364	16 / 16	0 / 0	352 / 352	7.9 / 7.9%
Synthetic 00–01	zero-config	injected	16,364	273 / 247	0 / 0	195 / 150	3.9 / 4.6%

Table S4.2: Development data: the correct numbers frozen qcore7 breaks with zero-config on the notes tables (set E), where qcore6 broke none Every break is silent. Analysis: `out/qcore7_work/E_zc_breaks.md` and Section 5.4.

Run	Cell (document | concept | column)	Truth	Changed to
Haiku, text	CRM_2026-07-31_t036 | DebtInstrumentCarryingAmount | Outstanding Principal as of July 31, 2026	39,500 M	39,288 M
Haiku, text	MRK_2026-06-30_t036 | DefinedBenefitPlanAmortizationOfPriorServiceCostCredit [International pension] | Three Months Ended June 30, 2026 International	-4 M	-9 M
Opus, text	CRM_2026-07-31_t036 | DebtInstrumentCarryingAmount | Outstanding Principal as of July 31, 2026	39,500 M	39,288 M
Opus, text	MRK_2026-06-30_t036 | DefinedBenefitPlanAmortizationOfPriorServiceCostCredit [International pension] | Three Months Ended June 30, 2026 International	-4 M	-9 M
GPT, text	CRM_2026-07-31_t036 | DebtInstrumentCarryingAmount | Outstanding Principal as of July 31, 2026	39,500 M	39,288 M
GPT, text	MRK_2026-06-30_t036 | DefinedBenefitPlanAmortizationOfPriorServiceCostCredit [International pension] | Three Months Ended June 30, 2026 International	-4 M	-9 M

S5 Frozen qcore9, qcore8 and qcore6 on SEC set 4: every run

Material moved from Appendix E of the paper; the run protocol is described there.

Table S5.1: Frozen qcore9, qcore8 and qcore6 on SEC set 4, every run (legacy scoring)
5,351 numbers, 72 statements. Model alone: Haiku 168, Opus 25, GPT 1,090 wrong (a GPT blank always counts as wrong). Silent [95% CI], broken, fixed, flagged (numbers named by a tier 2 item), % of numbers to a person, identity coverage; *strict*: silent errors of the qcore9 run under strict scoring (half a printed unit). Source: `out/qcore9_heldout/results.md`.

Model	Mode	Identities	Version	Silent [CI]	Broken	Fixed	Flagged	To a person	Coverage	Strict
Haiku	text	hand	qcore9	9 [0, 26]	0	124	101	1.9%	85.5%	9
Haiku	text	hand	qcore8	13 [1, 32]	0	124	97	1.8%	85.5%	
Haiku	text	hand	qcore6	13 [1, 32]	0	124	97	1.8%	85.5%	
Haiku	text	discovered	qcore9	13 [1, 33]	3	124	87	1.6%	86.2%	13
Haiku	text	discovered	qcore8	16 [3, 36]	3	114	87	1.6%	85.8%	
Haiku	text	discovered	qcore6	16 [3, 36]	4	124	74	1.4%	83.8%	
Haiku	text	hand + disc.	qcore9	9 [0, 26]	0	124	122	2.3%	92.1%	9
Haiku	text	hand + disc.	qcore8	13 [1, 32]	0	122	113	2.1%	91.9%	
Haiku	text	hand + disc.	qcore6	13 [1, 32]	0	124	109	2.0%	91.2%	
Haiku	text	zero-config	qcore9	16 [3, 36]	3	88	135	2.5%	84.0%	16
Haiku	text	zero-config	qcore8	19 [5, 39]	3	77	138	2.6%	83.7%	
Haiku	text	zero-config	qcore6	31 [6, 67]	0	33	127	2.4%	77.2%	
Haiku	no text	hand	qcore9	67 [45, 93]	0	0	418	7.8%	85.5%	70
Haiku	no text	hand	qcore8	67 [45, 93]	0	0	418	7.8%	85.5%	
Haiku	no text	hand	qcore6	67 [45, 93]	0	0	418	7.8%	85.5%	
Haiku	no text	discovered	qcore9	120 [66, 191]	0	0	231	4.3%	83.1%	123
Haiku	no text	discovered	qcore8	124 [68, 198]	0	0	211	3.9%	82.4%	
Haiku	no text	discovered	qcore6	125 [69, 199]	0	0	205	3.8%	81.6%	
Haiku	no text	hand + disc.	qcore9	56 [35, 81]	0	0	496	9.3%	91.8%	59
Haiku	no text	hand + disc.	qcore8	56 [35, 81]	0	0	487	9.1%	91.7%	
Haiku	no text	hand + disc.	qcore6	56 [35, 81]	0	0	483	9.0%	91.1%	
Haiku	no text	zero-config	qcore9	120 [66, 191]	0	0	231	4.3%	83.0%	123
Haiku	no text	zero-config	qcore8	125 [68, 200]	0	0	207	3.9%	82.2%	
Haiku	no text	zero-config	qcore6	124 [68, 199]	0	0	210	3.9%	81.8%	
Haiku	OCR docTR	hand	qcore9	52 [32, 76]	1	17	644	12.0%	85.5%	56
Haiku	OCR docTR	hand	qcore8	53 [33, 77]	1	17	639	11.9%	85.5%	
Haiku	OCR docTR	hand	qcore6	53 [33, 77]	1	17	639	11.9%	85.5%	
Haiku	OCR Tesseract	hand	qcore9	39 [23, 58]	1	3	930	17.4%	85.5%	43
Haiku	OCR Tesseract	hand	qcore8	39 [23, 58]	1	3	928	17.3%	85.5%	
Haiku	OCR Tesseract	hand	qcore6	39 [23, 58]	1	3	928	17.3%	85.5%	
Opus	text	hand	qcore9	0 [0, 0]	0	20	54	1.0%	85.5%	0
Opus	text	hand	qcore8	0 [0, 0]	0	20	54	1.0%	85.5%	
Opus	text	hand	qcore6	0 [0, 0]	0	20	54	1.0%	85.5%	
Opus	text	discovered	qcore9	3 [0, 9]	3	20	49	0.9%	88.2%	3
Opus	text	discovered	qcore8	3 [0, 9]	3	18	44	0.8%	87.9%	
Opus	text	discovered	qcore6	2 [0, 5]	4	20	43	0.8%	86.4%	
Opus	text	hand + disc.	qcore9	0 [0, 0]	0	20	74	1.4%	92.5%	0
Opus	text	hand + disc.	qcore8	0 [0, 0]	0	20	67	1.3%	92.4%	
Opus	text	hand + disc.	qcore6	0 [0, 0]	0	20	65	1.2%	91.6%	
Opus	text	zero-config	qcore9	3 [0, 9]	3	20	63	1.2%	88.3%	3
Opus	text	zero-config	qcore8	3 [0, 9]	3	18	60	1.1%	88.0%	
Opus	text	zero-config	qcore6	1 [0, 3]	0	11	25	0.5%	85.7%	
Opus	no text	hand	qcore9	17 [5, 32]	0	0	38	0.7%	85.5%	17
Opus	no text	hand	qcore8	17 [5, 32]	0	0	38	0.7%	85.5%	
Opus	no text	hand	qcore6	17 [5, 32]	0	0	38	0.7%	85.5%	
Opus	no text	discovered	qcore9	20 [6, 37]	0	0	59	1.1%	89.4%	20
Opus	no text	discovered	qcore8	20 [6, 37]	0	0	55	1.0%	89.4%	
Opus	no text	discovered	qcore6	20 [6, 37]	0	0	39	0.7%	88.6%	
Opus	no text	hand + disc.	qcore9	17 [5, 32]	0	0	68	1.3%	92.6%	17
Opus	no text	hand + disc.	qcore8	17 [5, 32]	0	0	63	1.2%	92.6%	
Opus	no text	hand + disc.	qcore6	17 [5, 32]	0	0	47	0.9%	91.9%	
Opus	no text	zero-config	qcore9	20 [6, 37]	0	0	59	1.1%	89.4%	20
Opus	no text	zero-config	qcore8	20 [6, 37]	0	0	55	1.0%	89.4%	
Opus	no text	zero-config	qcore6	20 [6, 37]	0	0	39	0.7%	88.6%	
Opus	OCR docTR	hand	qcore9	11 [2, 24]	0	0	376	7.0%	85.5%	11
Opus	OCR docTR	hand	qcore8	11 [2, 24]	0	0	372	7.0%	85.5%	
Opus	OCR docTR	hand	qcore6	11 [2, 24]	0	0	372	7.0%	85.5%	
Opus	OCR Tesseract	hand	qcore9	14 [3, 28]	0	0	539	10.1%	85.5%	14
Opus	OCR Tesseract	hand	qcore8	14 [3, 28]	0	0	538	10.1%	85.5%	
Opus	OCR Tesseract	hand	qcore6	14 [3, 28]	0	0	538	10.1%	85.5%	
GPT	text	hand	qcore9	15 [2, 34]	2	739	514	9.6%	85.5%	19
GPT	text	hand	qcore8	26 [12, 42]	1	737	516	9.6%	85.5%	
GPT	text	hand	qcore6	26 [12, 42]	1	737	516	9.6%	85.5%	
GPT	text	discovered	qcore9	27 [11, 48]	3	716	440	8.2%	80.3%	31
GPT	text	discovered	qcore8	38 [19, 61]	3	701	436	8.1%	79.8%	
GPT	text	discovered	qcore6	42 [23, 65]	5	739	380	7.1%	77.2%	
GPT	text	hand + disc.	qcore9	15 [2, 35]	1	730	559	10.4%	91.4%	19
GPT	text	hand + disc.	qcore8	24 [10, 40]	0	726	558	10.4%	91.3%	
GPT	text	hand + disc.	qcore6	25 [11, 41]	1	735	554	10.4%	90.9%	
GPT	text	zero-config	qcore9	37 [17, 63]	3	370	797	14.9%	69.8%	41
GPT	text	zero-config	qcore8	51 [28, 78]	3	356	786	14.7%	69.2%	

Model	Mode	Identities	Version	Silent [CI]	Broken	Fixed	Flagged	To a person	Coverage	Strict
GPT	text	zero-config	qcore6	113 [55, 199]	0	118	1,254	23.4%	52.0%	
GPT	no text	hand	qcore9	585 [467, 718]	0	0	1,288	24.1%	85.5%	597
GPT	no text	hand	qcore8	585 [467, 718]	0	0	1,288	24.1%	85.5%	
GPT	no text	hand	qcore6	585 [467, 718]	0	0	1,288	24.1%	85.5%	
GPT	no text	discovered	qcore9	927 [721, 1,151]	0	0	519	9.7%	66.6%	940
GPT	no text	discovered	qcore8	946 [736, 1,175]	0	0	426	8.0%	65.3%	
GPT	no text	discovered	qcore6	959 [750, 1,185]	0	0	392	7.3%	62.7%	
GPT	no text	hand + disc.	qcore9	554 [440, 683]	0	0	1,367	25.5%	90.1%	566
GPT	no text	hand + disc.	qcore8	561 [443, 695]	0	0	1,351	25.2%	90.0%	
GPT	no text	hand + disc.	qcore6	562 [444, 695]	0	0	1,349	25.2%	89.3%	
GPT	no text	zero-config	qcore9	927 [721, 1,151]	0	0	519	9.7%	66.6%	940
GPT	no text	zero-config	qcore8	942 [736, 1,168]	0	0	431	8.1%	65.7%	
GPT	no text	zero-config	qcore6	955 [748, 1,179]	0	0	397	7.4%	63.0%	
GPT	OCR docTR	hand	qcore9	343 [264, 433]	0	76	1,532	28.6%	85.5%	356
GPT	OCR docTR	hand	qcore8	344 [265, 434]	0	76	1,529	28.6%	85.5%	
GPT	OCR docTR	hand	qcore6	344 [265, 434]	0	76	1,529	28.6%	85.5%	
GPT	OCR Tesseract	hand	qcore9	408 [314, 516]	0	16	1,887	35.3%	85.5%	420
GPT	OCR Tesseract	hand	qcore8	408 [314, 516]	0	16	1,886	35.2%	85.5%	
GPT	OCR Tesseract	hand	qcore6	408 [314, 516]	0	16	1,886	35.2%	85.5%	

Table S5.3: Hand identities: does qcore9 return qcore8's fixes and tier 2 items? (SEC set 4, run by run)
qcore8 and qcore6 returned the same fixes and tier 2 items in all 12 hand runs. qcore9 differs only where it reads page text, through change 3 (`page.find_value`, Section 3.5): counts of fixes and tier 2 items only in qcore9 or only in qcore8, and the change in silent and broken numbers.

Model	Mode	Same fixes	Same tier 2	Fixes only qcore9 / qcore8	Tier 2 only qcore9 / qcore8	Silent qcore8 → qcore9	Broken qcore8 → qcore9
Haiku	text	yes	no	0 / 0	4 / 0	13 → 9	0 → 0
Haiku	no text	yes	yes	0 / 0	0 / 0	67 → 67	0 → 0
Haiku	OCR docTR	yes	no	0 / 0	5 / 0	53 → 52	1 → 1
Haiku	OCR Tesseract	yes	no	0 / 0	2 / 0	39 → 39	1 → 1
Opus	text	yes	yes	0 / 0	0 / 0	0 → 0	0 → 0
Opus	no text	yes	yes	0 / 0	0 / 0	17 → 17	0 → 0
Opus	OCR docTR	yes	no	0 / 0	4 / 0	11 → 11	0 → 0
Opus	OCR Tesseract	yes	no	0 / 0	1 / 0	14 → 14	0 → 0
GPT	text	no	no	3 / 0	14 / 3	26 → 15	1 → 2
GPT	no text	yes	yes	0 / 0	0 / 0	585 → 585	0 → 0
GPT	OCR docTR	yes	no	0 / 0	3 / 0	344 → 343	0 → 0
GPT	OCR Tesseract	yes	no	0 / 0	1 / 0	408 → 408	0 → 0

S6 Development grid of frozen qcore9: every set

Material moved from Appendix F of the paper.

Table S6.1: Frozen qcore8 / qcore9 on every development set (pooled over models, extractions and modes)

Set	Identities	Extractions, scoring	Numbers	Silent qcore8 / qcore9	Broken	Fixed	To a person
SEC set 1	hand	real, legacy	25,728	885 / 885	2 / 2	563 / 563	14.3 / 14.4%
SEC set 1	discovered	real, legacy	12,864	586 / 568	0 / 0	477 / 503	4.3 / 4.6%
SEC set 1	hand + disc.	real, legacy	12,864	374 / 373	0 / 0	518 / 520	8.8 / 8.8%
SEC set 1	zero-config	real, legacy	12,864	593 / 575	0 / 0	259 / 284	6.0 / 6.3%
SEC set 2	hand	real, legacy	53,088	1,615 / 1,614	3 / 3	1,001 / 1,001	12.9 / 12.9%
SEC set 2	discovered	real, legacy	26,544	1,048 / 1,030	0 / 0	801 / 833	4.6 / 4.8%
SEC set 2	hand + disc.	real, legacy	26,544	694 / 690	0 / 0	871 / 876	7.9 / 8.0%
SEC set 2	zero-config	real, legacy	26,544	1,072 / 1,054	0 / 0	381 / 406	6.1 / 6.3%
SEC set 3	hand	real, legacy	62,472	2,304 / 2,241	6 / 6	1,108 / 1,108	16.2 / 16.3%
SEC set 3	discovered	real, legacy	31,236	1,497 / 1,439	0 / 0	955 / 981	5.9 / 6.2%
SEC set 3	hand + disc.	real, legacy	31,236	1,002 / 948	0 / 0	1,011 / 1,015	9.9 / 10.2%
SEC set 3	zero-config	real, legacy	31,236	1,525 / 1,460	0 / 0	449 / 473	8.0 / 8.3%
CORD test	hand	real, legacy	7,984	365 / 365	0 / 0	28 / 28	20.9 / 20.9%
CORD test	discovered	real, legacy	3,992	142 / 142	0 / 0	25 / 25	21.6 / 22.4%
CORD test	hand + disc.	real, legacy	3,992	126 / 126	0 / 0	29 / 29	23.4 / 24.1%
CORD test	zero-config	real, legacy	3,992	144 / 144	0 / 0	25 / 25	21.5 / 22.2%
Notes tables	hand	real, legacy	88,668	54 / 54	3 / 3	8 / 8	7.1 / 7.1%
Notes tables	discovered	real, legacy	88,668	95 / 96	0 / 0	1 / 2	8.4 / 8.5%
Notes tables	hand + disc.	real, legacy	88,668	47 / 48	0 / 0	1 / 2	8.5 / 8.5%
Notes tables	zero-config	real, legacy	88,668	95 / 96	0 / 0	0 / 1	8.6 / 8.7%
Real scans	hand	real, legacy	6,054	13 / 13	0 / 0	0 / 0	16.2 / 16.2%
Real scans	discovered	real, legacy	2,018	5 / 5	0 / 0	0 / 0	20.6 / 20.7%
Real scans	hand + disc.	real, legacy	2,018	5 / 5	0 / 0	0 / 0	22.8 / 22.9%
Real scans	zero-config	real, legacy	2,018	5 / 5	0 / 0	0 / 0	20.6 / 20.6%
Synthetic 02–05	hand	real, legacy	31,568	4 / 4	0 / 0	0 / 0	2.1 / 2.1%
Synthetic 02–05	discovered	real, legacy	31,568	4 / 4	0 / 0	0 / 0	2.9 / 2.9%
Synthetic 02–05	hand + disc.	real, legacy	31,568	4 / 4	0 / 0	0 / 0	4.0 / 4.0%
Synthetic 02–05	zero-config	real, legacy	31,568	4 / 4	0 / 0	0 / 0	2.9 / 2.9%
Set H (qcore7 fresh)	hand	H clean, legacy	11,840	0 / 0	0 / 0	0 / 0	2.1 / 2.1%
Set H (qcore7 fresh)	discovered	H clean, legacy	11,840	0 / 0	0 / 0	0 / 0	3.3 / 3.4%
Set H (qcore7 fresh)	hand + disc.	H clean, legacy	11,840	0 / 0	0 / 0	0 / 0	4.5 / 4.5%
Set H (qcore7 fresh)	zero-config	H clean, legacy	11,840	0 / 0	0 / 0	0 / 0	3.4 / 3.4%
Set H (qcore7 fresh)	hand	H clean, strict	11,840	0 / 0	0 / 0	0 / 0	2.1 / 2.1%
Set H (qcore7 fresh)	discovered	H clean, strict	11,840	0 / 0	0 / 0	0 / 0	3.3 / 3.4%
Set H (qcore7 fresh)	hand + disc.	H clean, strict	11,840	0 / 0	0 / 0	0 / 0	4.5 / 4.5%
Set H (qcore7 fresh)	zero-config	H clean, strict	11,840	0 / 0	0 / 0	0 / 0	3.4 / 3.4%
Set H (qcore7 fresh)	hand	H high, legacy	11,840	22 / 22	0 / 0	194 / 194	16.9 / 16.9%
Set H (qcore7 fresh)	discovered	H high, legacy	11,840	82 / 80	0 / 0	150 / 162	14.4 / 14.5%
Set H (qcore7 fresh)	hand + disc.	H high, legacy	11,840	9 / 9	0 / 0	185 / 195	19.2 / 19.1%
Set H (qcore7 fresh)	zero-config	H high, legacy	11,840	88 / 82	0 / 0	108 / 129	15.0 / 15.0%
Set H (qcore7 fresh)	hand	H high, strict	11,840	27 / 27	0 / 0	194 / 194	16.9 / 16.9%
Set H (qcore7 fresh)	discovered	H high, strict	11,840	90 / 88	0 / 0	150 / 162	14.4 / 14.5%
Set H (qcore7 fresh)	hand + disc.	H high, strict	11,840	11 / 11	0 / 0	185 / 195	19.2 / 19.1%
Set H (qcore7 fresh)	zero-config	H high, strict	11,840	96 / 90	0 / 0	108 / 129	15.0 / 15.0%
Set H (qcore7 fresh)	hand	H low, legacy	11,840	9 / 8	0 / 0	58 / 58	6.4 / 6.4%
Set H (qcore7 fresh)	discovered	H low, legacy	11,840	7 / 6	0 / 0	40 / 46	6.9 / 6.9%
Set H (qcore7 fresh)	hand + disc.	H low, legacy	11,840	0 / 0	0 / 0	50 / 53	9.7 / 9.6%
Set H (qcore7 fresh)	zero-config	H low, legacy	11,840	8 / 7	0 / 0	30 / 38	7.0 / 7.0%
Set H (qcore7 fresh)	hand	H low, strict	11,840	11 / 10	0 / 0	59 / 59	6.4 / 6.4%
Set H (qcore7 fresh)	discovered	H low, strict	11,840	13 / 12	0 / 0	40 / 46	6.9 / 6.9%
Set H (qcore7 fresh)	hand + disc.	H low, strict	11,840	0 / 0	0 / 0	50 / 53	9.7 / 9.6%
Set H (qcore7 fresh)	zero-config	H low, strict	11,840	14 / 13	0 / 0	30 / 38	7.0 / 7.0%
Set I (qcore8 fresh)	hand	I clean, legacy	27,736	0 / 0	0 / 0	0 / 0	1.8 / 1.8%
Set I (qcore8 fresh)	discovered	I clean, legacy	27,736	0 / 0	0 / 0	0 / 0	6.3 / 6.1%
Set I (qcore8 fresh)	hand + disc.	I clean, legacy	27,736	0 / 0	0 / 0	0 / 0	6.8 / 6.6%
Set I (qcore8 fresh)	zero-config	I clean, legacy	27,736	2 / 0	2 / 0	0 / 0	6.3 / 5.5%
Set I (qcore8 fresh)	hand	I clean, strict	27,736	0 / 0	0 / 0	0 / 0	1.8 / 1.8%
Set I (qcore8 fresh)	discovered	I clean, strict	27,736	0 / 0	0 / 0	0 / 0	6.3 / 6.1%
Set I (qcore8 fresh)	hand + disc.	I clean, strict	27,736	0 / 0	0 / 0	0 / 0	6.8 / 6.6%
Set I (qcore8 fresh)	zero-config	I clean, strict	27,736	2 / 0	2 / 0	0 / 0	6.3 / 5.5%
Set I (qcore8 fresh)	hand	I high, legacy	27,736	44 / 43	1 / 0	478 / 478	15.3 / 15.3%
Set I (qcore8 fresh)	discovered	I high, legacy	27,736	233 / 224	1 / 0	322 / 366	14.2 / 14.2%
Set I (qcore8 fresh)	hand + disc.	I high, legacy	27,736	29 / 28	1 / 0	454 / 467	19.8 / 19.6%
Set I (qcore8 fresh)	zero-config	I high, legacy	27,736	259 / 236	12 / 0	269 / 323	14.5 / 14.4%
Set I (qcore8 fresh)	hand	I high, strict	27,736	57 / 56	1 / 0	478 / 478	15.3 / 15.3%
Set I (qcore8 fresh)	discovered	I high, strict	27,736	290 / 279	1 / 0	321 / 365	14.2 / 14.2%
Set I (qcore8 fresh)	hand + disc.	I high, strict	27,736	37 / 36	1 / 0	453 / 466	19.8 / 19.6%
Set I (qcore8 fresh)	zero-config	I high, strict	27,736	317 / 292	12 / 0	268 / 322	14.5 / 14.4%
Set I (qcore8 fresh)	hand	I low, legacy	27,736	23 / 23	0 / 0	144 / 144	6.4 / 6.4%
Set I (qcore8 fresh)	discovered	I low, legacy	27,736	61 / 59	2 / 0	120 / 134	8.3 / 8.1%
Set I (qcore8 fresh)	hand + disc.	I low, legacy	27,736	11 / 11	0 / 0	137 / 140	10.8 / 10.4%
Set I (qcore8 fresh)	zero-config	I low, legacy	27,736	70 / 68	2 / 0	99 / 113	8.1 / 7.9%
Set I (qcore8 fresh)	hand	I low, strict	27,736	34 / 34	0 / 0	144 / 144	6.4 / 6.4%
Set I (qcore8 fresh)	discovered	I low, strict	27,736	76 / 74	2 / 0	120 / 134	8.3 / 8.1%
Set I (qcore8 fresh)	hand + disc.	I low, strict	27,736	16 / 16	0 / 0	137 / 140	10.8 / 10.4%

Set	Identities	Extractions, scoring	Numbers	Silent qcore8 / qcore9	Broken	Fixed	To a person
Set I (qcore8 fresh)	zero-config	I low, strict	27,736	85 / 83	2 / 0	99 / 113	8.1 / 7.9%
SEC dev	hand	injected, legacy	4,890	17 / 17	0 / 0	54 / 54	3.2 / 3.2%
SEC dev	discovered	injected, legacy	4,890	30 / 28	0 / 1	39 / 54	3.4 / 3.3%
SEC dev	hand + disc.	injected, legacy	4,890	13 / 13	0 / 1	51 / 54	4.0 / 4.0%
SEC dev	zero-config	injected, legacy	4,890	30 / 27	0 / 0	35 / 51	3.5 / 3.3%
SEC dev	hand	real, legacy	19,560	113 / 113	0 / 0	100 / 100	8.9 / 8.9%
SEC dev	discovered	real, legacy	19,560	142 / 135	0 / 2	71 / 79	7.3 / 7.4%
SEC dev	hand + disc.	real, legacy	19,560	109 / 111	0 / 2	100 / 100	6.9 / 7.0%
SEC dev	zero-config	real, legacy	19,560	151 / 142	0 / 0	54 / 67	7.2 / 7.3%
Notes dev	hand	injected, legacy	13,178	67 / 67	0 / 0	18 / 18	1.2 / 1.2%
Notes dev	discovered	injected, legacy	13,178	52 / 48	0 / 0	12 / 16	2.8 / 2.8%
Notes dev	hand + disc.	injected, legacy	13,178	34 / 33	0 / 0	15 / 18	3.2 / 3.1%
Notes dev	zero-config	injected, legacy	13,178	56 / 52	0 / 0	11 / 14	2.6 / 2.7%
Notes dev	hand	real, legacy	39,534	36 / 36	0 / 0	16 / 16	6.6 / 6.6%
Notes dev	discovered	real, legacy	39,534	36 / 36	0 / 0	4 / 4	7.7 / 7.7%
Notes dev	hand + disc.	real, legacy	39,534	36 / 36	0 / 0	4 / 4	8.0 / 8.0%
Notes dev	zero-config	real, legacy	39,534	40 / 40	0 / 0	0 / 0	7.9 / 7.9%
CORD train	hand	injected, legacy	16,810	33 / 33	0 / 0	63 / 63	3.5 / 3.5%
CORD train	discovered	injected, legacy	16,810	68 / 68	0 / 0	33 / 33	5.2 / 5.2%
CORD train	hand + disc.	injected, legacy	16,810	42 / 42	0 / 0	56 / 56	6.6 / 6.6%
CORD train	zero-config	injected, legacy	16,810	73 / 73	0 / 0	31 / 31	5.1 / 5.1%
CORD train	hand	real, legacy	16,810	0 / 0	0 / 0	0 / 0	2.3 / 2.3%
CORD train	discovered	real, legacy	16,810	0 / 0	0 / 0	0 / 0	4.0 / 4.0%
CORD train	hand + disc.	real, legacy	16,810	0 / 0	0 / 0	0 / 0	5.0 / 5.0%
CORD train	zero-config	real, legacy	16,810	0 / 0	0 / 0	0 / 0	3.8 / 3.8%
CORD val.	hand	injected, legacy	3,768	43 / 43	0 / 0	34 / 34	12.3 / 12.3%
CORD val.	discovered	injected, legacy	3,768	80 / 80	0 / 0	17 / 17	11.2 / 11.2%
CORD val.	hand + disc.	injected, legacy	3,768	43 / 43	0 / 0	35 / 35	13.3 / 13.3%
CORD val.	zero-config	injected, legacy	3,768	84 / 84	0 / 0	9 / 9	10.9 / 10.9%
CORD val.	hand	real, legacy	3,768	21 / 21	0 / 0	2 / 2	8.5 / 8.5%
CORD val.	discovered	real, legacy	3,768	28 / 28	0 / 0	2 / 2	10.4 / 10.4%
CORD val.	hand + disc.	real, legacy	3,768	20 / 20	0 / 0	2 / 2	12.2 / 12.2%
CORD val.	zero-config	real, legacy	3,768	31 / 31	0 / 0	0 / 0	9.6 / 9.6%
Real-scan dev	hand	real, legacy	2,760	8 / 8	0 / 0	0 / 0	24.9 / 24.9%
Real-scan dev	discovered	real, legacy	2,760	16 / 16	0 / 0	0 / 0	21.6 / 21.8%
Real-scan dev	hand + disc.	real, legacy	2,760	0 / 0	0 / 0	0 / 0	37.0 / 37.1%
Real-scan dev	zero-config	real, legacy	2,760	16 / 16	0 / 0	0 / 0	18.6 / 18.6%
Synthetic 00–01	hand	injected, legacy	16,364	16 / 16	0 / 0	354 / 354	7.3 / 7.3%
Synthetic 00–01	discovered	injected, legacy	16,364	144 / 136	0 / 0	234 / 242	8.1 / 8.1%
Synthetic 00–01	hand + disc.	injected, legacy	16,364	8 / 8	0 / 0	350 / 350	10.1 / 10.1%
Synthetic 00–01	zero-config	injected, legacy	16,364	160 / 148	0 / 0	163 / 200	8.4 / 8.3%
Synthetic 00–01	hand	real, legacy	16,364	6 / 6	0 / 0	0 / 0	1.4 / 1.4%
Synthetic 00–01	discovered	real, legacy	16,364	6 / 6	0 / 0	0 / 0	2.8 / 2.8%
Synthetic 00–01	hand + disc.	real, legacy	16,364	6 / 6	0 / 0	0 / 0	3.5 / 3.5%
Synthetic 00–01	zero-config	real, legacy	16,364	6 / 6	0 / 0	0 / 0	2.7 / 2.7%

Table S7.1: Development data, not held out: frozen qcore7 → frozen qcore8, zero-config, on every set already spent, pooled over models and over text and no-text modes; legacy scoring except set H (qcore7's fresh set, strict, as in its held-out run). qcore8 was tuned on these sets. qcore6 dev sets: SEC, notes, CORD train and validation, real-scan and synthetic 00–01 development data; *inj.*: with injected errors. With hand identities qcore8 returned qcore7's fixes and tier 2 items in all 244 development runs. Every row: Section S8.

Set (zero-config)	Numbers	Silent	Broken	To a person %
SEC set 1	12,864	605 → 593	0 → 0	5.8 → 6.0
SEC set 2	26,544	1,080 → 1,072	0 → 0	5.9 → 6.1
SEC set 3	31,236	1,536 → 1,525	0 → 0	7.7 → 8.0
CORD test	3,992	153 → 144	0 → 0	19.3 → 21.5
Notes tables	88,668	101 → 95	6 → 0	7.8 → 8.6
Real scans	2,018	5 → 5	0 → 0	20.6 → 20.6
Synthetic 02–05	31,568	4 → 4	0 → 0	1.4 → 2.9
Set H, clean	11,840	0 → 0	0 → 0	1.5 → 3.4
Set H, low	11,840	42 → 14	0 → 0	4.9 → 7.0
Set H, high	11,840	207 → 96	0 → 0	8.7 → 15.0
qcore6 dev sets	98,796	244 → 244	0 → 0	5.8 → 6.6
of which CORD val.	3,768	25 → 31	0 → 0	8.7 → 9.6
qcore6 dev sets inj.	55,010	543 → 403	0 → 0	4.0 → 5.7

S7 Frozen qcore8 on the second fresh held-out set: every run

Material moved from Appendix G of the paper; the run protocol is described there.

Table S7.2: Frozen qcore6, qcore7 and qcore8 on the second fresh held-out set, all 18 bundles, every run (strict scoring)
Silent [95% CI], broken [95% CI], fixed, flagged, % of numbers to a person, identity coverage. 13,868 numbers, 199 documents. Model alone: clean 0, low 210, high 691 wrong. Hand rows are identical in the three versions and are shown once. Source: `out/qcore8_heldout/results.md`.

Extraction	Mode	Identities	Version	Silent [CI]	Broken [CI]	Fixed	Flagged	To a person	Coverage
clean	text	hand	all three	0 [0, 0]	0 [0, 0]	0	337	2.4%	94%
clean	text	discovered	qcore6	0 [0, 0]	0 [0, 0]	0	345	2.5%	70%
clean	text	discovered	qcore7	0 [0, 0]	0 [0, 0]	0	345	2.5%	70%
clean	text	discovered	qcore8	0 [0, 0]	0 [0, 0]	0	980	7.1%	83%
clean	text	hand + disc.	qcore6	0 [0, 0]	0 [0, 0]	0	482	3.5%	97%
clean	text	hand + disc.	qcore7	0 [0, 0]	0 [0, 0]	0	481	3.5%	97%
clean	text	hand + disc.	qcore8	0 [0, 0]	0 [0, 0]	0	982	7.1%	98%
clean	text	zero-config	qcore6	9 [1, 21]	9 [1, 21]	0	349	2.5%	70%
clean	text	zero-config	qcore7	7 [1, 16]	7 [1, 16]	0	457	3.3%	70%
clean	text	zero-config	qcore8	2 [0, 5]	2 [0, 5]	0	975	7.0%	83%
clean	no text	hand	all three	0 [0, 0]	0 [0, 0]	0	173	1.2%	94%
clean	no text	discovered	qcore6	0 [0, 0]	0 [0, 0]	0	87	0.6%	69%
clean	no text	discovered	qcore7	0 [0, 0]	0 [0, 0]	0	87	0.6%	69%
clean	no text	discovered	qcore8	0 [0, 0]	0 [0, 0]	0	772	5.6%	82%
clean	no text	hand + disc.	qcore6	0 [0, 0]	0 [0, 0]	0	239	1.7%	97%
clean	no text	hand + disc.	qcore7	0 [0, 0]	0 [0, 0]	0	239	1.7%	97%
clean	no text	hand + disc.	qcore8	0 [0, 0]	0 [0, 0]	0	900	6.5%	98%
clean	no text	zero-config	qcore6	0 [0, 0]	0 [0, 0]	0	87	0.6%	69%
clean	no text	zero-config	qcore7	0 [0, 0]	0 [0, 0]	0	87	0.6%	69%
clean	no text	zero-config	qcore8	0 [0, 0]	0 [0, 0]	0	772	5.6%	82%
low, 1.5%	text	hand	all three	12 [5, 21]	0 [0, 0]	137	686	4.9%	94%
low, 1.5%	text	discovered	qcore6	35 [23, 48]	1 [0, 3]	130	404	2.9%	67%
low, 1.5%	text	discovered	qcore7	32 [21, 44]	1 [0, 3]	117	414	3.0%	67%
low, 1.5%	text	discovered	qcore8	18 [10, 28]	2 [0, 5]	114	1128	8.1%	81%
low, 1.5%	text	hand + disc.	qcore6	7 [2, 12]	0 [0, 0]	136	753	5.4%	97%
low, 1.5%	text	hand + disc.	qcore7	6 [2, 11]	0 [0, 0]	132	754	5.4%	97%
low, 1.5%	text	hand + disc.	qcore8	4 [1, 8]	0 [0, 0]	129	1269	9.2%	98%
low, 1.5%	text	zero-config	qcore6	53 [37, 70]	9 [1, 21]	114	370	2.7%	66%
low, 1.5%	text	zero-config	qcore7	51 [36, 68]	11 [2, 23]	98	488	3.5%	67%
low, 1.5%	text	zero-config	qcore8	27 [16, 39]	2 [0, 5]	99	1060	7.6%	81%
low, 1.5%	no text	hand	all three	22 [12, 34]	0 [0, 0]	7	1079	7.8%	94%
low, 1.5%	no text	discovered	qcore6	109 [85, 134]	0 [0, 0]	5	431	3.1%	65%
low, 1.5%	no text	discovered	qcore7	87 [67, 108]	0 [0, 0]	4	583	4.2%	66%
low, 1.5%	no text	discovered	qcore8	58 [42, 76]	0 [0, 0]	6	1168	8.4%	79%
low, 1.5%	no text	hand + disc.	qcore6	16 [9, 24]	0 [0, 0]	8	1174	8.5%	97%
low, 1.5%	no text	hand + disc.	qcore7	14 [7, 21]	0 [0, 0]	8	1202	8.7%	97%
low, 1.5%	no text	hand + disc.	qcore8	12 [6, 19]	0 [0, 0]	8	1717	12.4%	98%
low, 1.5%	no text	zero-config	qcore6	109 [85, 134]	0 [0, 0]	0	442	3.2%	65%
low, 1.5%	no text	zero-config	qcore7	87 [67, 108]	0 [0, 0]	0	589	4.2%	66%
low, 1.5%	no text	zero-config	qcore8	58 [42, 76]	0 [0, 0]	0	1174	8.5%	80%
high, 5%	text	hand	all three	15 [8, 23]	1 [0, 3]	438	1442	10.4%	94%
high, 5%	text	discovered	qcore6	180 [139, 225]	4 [0, 9]	366	581	4.2%	57%
high, 5%	text	discovered	qcore7	158 [120, 200]	3 [0, 8]	322	682	4.9%	58%
high, 5%	text	discovered	qcore8	79 [56, 103]	1 [0, 3]	316	1711	12.3%	80%
high, 5%	text	hand + disc.	qcore6	11 [5, 19]	1 [0, 3]	439	1529	11.0%	97%

Extraction	Mode	Identities	Version	Silent [CI]	Broken [CI]	Fixed	Flagged	To a person	Coverage
high, 5%	text	hand + disc.	qcore7	11 [5, 19]	1 [0, 3]	426	1536	11.1%	97%
high, 5%	text	hand + disc.	qcore8	9 [3, 16]	1 [0, 3]	414	1987	14.3%	98%
high, 5%	text	zero-config	qcore6	208 [161, 259]	12 [2, 25]	323	556	4.0%	57%
high, 5%	text	zero-config	qcore7	182 [137, 233]	13 [2, 29]	263	752	5.4%	58%
high, 5%	text	zero-config	qcore8	106 [73, 145]	12 [1, 30]	268	1786	12.9%	80%
high, 5%	no text	hand	all three	42 [28, 58]	0 [0, 0]	40	2797	20.2%	94%
high, 5%	no text	discovered	qcore6	501 [425, 584]	0 [0, 0]	11	594	4.3%	48%
high, 5%	no text	discovered	qcore7	413 [348, 486]	0 [0, 0]	3	872	6.3%	50%
high, 5%	no text	discovered	qcore8	211 [168, 257]	0 [0, 0]	5	2220	16.0%	74%
high, 5%	no text	hand + disc.	qcore6	41 [27, 57]	0 [0, 0]	40	2841	20.5%	95%
high, 5%	no text	hand + disc.	qcore7	38 [25, 53]	0 [0, 0]	40	2885	20.8%	95%
high, 5%	no text	hand + disc.	qcore8	28 [16, 42]	0 [0, 0]	39	3497	25.2%	98%
high, 5%	no text	zero-config	qcore6	501 [425, 584]	0 [0, 0]	0	611	4.4%	48%
high, 5%	no text	zero-config	qcore7	413 [348, 486]	0 [0, 0]	0	883	6.4%	50%
high, 5%	no text	zero-config	qcore8	211 [168, 257]	0 [0, 0]	0	2236	16.1%	75%

Table S7.4: NEW layouts and OLD domains separately: hand and zero-config (strict)
NEW: budget, census, inventory, utilbank (12 bundles, 103 documents, 9,860 numbers; model alone: low 148, high 492 wrong). OLD: corp_fin, invoices, public_stats 09–10 (6 bundles, 96 documents, 4,008 numbers; low 62, high 199 wrong). Silent [CI] / broken / % to a person / coverage.

Group	Extraction	Mode	Hand	ZC qcore7	ZC qcore8
NEW layouts	clean	text	0 [0, 0] / 0 / 2.6% / 93%	7 [1, 16] / 7 / 4.0% / 65%	2 [0, 5] / 2 / 8.4% / 82%
NEW layouts	clean	no text	0 [0, 0] / 0 / 0.9% / 93%	0 [0, 0] / 0 / 0.6% / 65%	0 [0, 0] / 0 / 6.4% / 80%
NEW layouts	low, 1.5%	text	10 [3, 19] / 0 / 4.8% / 93%	36 [22, 51] / 11 / 3.5% / 61%	18 [9, 28] / 2 / 8.2% / 79%
NEW layouts	low, 1.5%	no text	20 [10, 32] / 0 / 7.7% / 93%	65 [48, 83] / 0 / 4.6% / 62%	46 [32, 61] / 0 / 9.0% / 76%
NEW layouts	high, 5%	text	11 [5, 18] / 1 / 10.2% / 93%	141 [98, 189] / 13 / 4.9% / 50%	83 [52, 121] / 12 / 13.4% / 77%
NEW layouts	high, 5%	no text	34 [21, 49] / 0 / 20.6% / 93%	326 [271, 387] / 0 / 5.3% / 42%	168 [130, 211] / 0 / 16.4% / 69%
OLD domains	clean	text	0 [0, 0] / 0 / 1.9% / 98%	0 [0, 0] / 0 / 1.5% / 82%	0 [0, 0] / 0 / 3.7% / 87%
OLD domains	clean	no text	0 [0, 0] / 0 / 2.1% / 98%	0 [0, 0] / 0 / 0.6% / 79%	0 [0, 0] / 0 / 3.5% / 89%
OLD domains	low, 1.5%	text	2 [0, 5] / 0 / 5.3% / 98%	15 [8, 23] / 0 / 3.5% / 82%	9 [3, 16] / 0 / 6.3% / 87%
OLD domains	low, 1.5%	no text	2 [0, 5] / 0 / 8.0% / 98%	22 [13, 31] / 0 / 3.5% / 78%	12 [6, 19] / 0 / 7.3% / 89%
OLD domains	high, 5%	text	4 [1, 8] / 0 / 11.0% / 98%	41 [27, 56] / 0 / 6.8% / 78%	23 [14, 34] / 0 / 11.6% / 86%
OLD domains	high, 5%	no text	8 [3, 14] / 0 / 19.0% / 98%	87 [68, 106] / 0 / 9.1% / 70%	43 [30, 57] / 0 / 15.5% / 87%

Table S7.6: Paired differences on the second fresh held-out set (strict), sums over documents, 95% bootstrap intervals

Group	Extraction	Mode	Identities	qcore8−qcore7 silent	qcore8−qcore7 broken	qcore8−qcore7 flagged	qcore7−hand silent	qcore8−hand silent
all 18	clean	text	discovered	0 [0, 0]	0 [0, 0]	+635 [+391, +911]	0 [0, 0]	0 [0, 0]
all 18	clean	text	hand + disc.	0 [0, 0]	0 [0, 0]	+501 [+310, +719]	0 [0, 0]	0 [0, 0]
all 18	clean	text	zero-config	−5 [−13, 0]	−5 [−13, 0]	+518 [+287, +788]	+7 [+1, +16]	+2 [0, +5]
all 18	clean	no text	discovered	0 [0, 0]	0 [0, 0]	+685 [+424, +982]	0 [0, 0]	0 [0, 0]
all 18	clean	no text	hand + disc.	0 [0, 0]	0 [0, 0]	+661 [+412, +944]	0 [0, 0]	0 [0, 0]
all 18	clean	no text	zero-config	0 [0, 0]	0 [0, 0]	+685 [+424, +982]	0 [0, 0]	0 [0, 0]
all 18	low, 1.5%	text	discovered	−14 [−22, −6]	+1 [0, +3]	+714 [+472, +990]	+20 [+8, +32]	+6 [−5, +17]
all 18	low, 1.5%	text	hand + disc.	−2 [−5, 0]	0 [0, 0]	+515 [+342, +714]	−6 [−14, −1]	−8 [−16, −2]
all 18	low, 1.5%	text	zero-config	−24 [−38, −13]	−9 [−21, 0]	+572 [+349, +832]	+39 [+23, +57]	+15 [+2, +28]
all 18	low, 1.5%	no text	discovered	−29 [−42, −17]	0 [0, 0]	+585 [+374, +829]	+65 [+47, +84]	+36 [+20, +52]
all 18	low, 1.5%	no text	hand + disc.	−2 [−5, 0]	0 [0, 0]	+515 [+320, +740]	−8 [−18, −1]	−10 [−20, −3]
all 18	low, 1.5%	no text	zero-config	−29 [−42, −17]	0 [0, 0]	+585 [+374, +829]	+65 [+47, +84]	+36 [+20, +52]
all 18	high, 5%	text	discovered	−79 [−110, −52]	−2 [−7, +2]	+1,029 [+683, +1,424]	+143 [+106, +185]	+64 [+43, +88]
all 18	high, 5%	text	hand + disc.	−2 [−5, 0]	0 [0, 0]	+451 [+292, +635]	−4 [−9, 0]	−6 [−12, −1]
all 18	high, 5%	text	zero-config	−76 [−107, −50]	−1 [−6, +3]	+1,034 [+687, +1,433]	+167 [+121, +220]	+91 [+59, +130]
all 18	high, 5%	no text	discovered	−202 [−256, −155]	0 [0, 0]	+1,348 [+988, +1,748]	+371 [+309, +441]	+169 [+128, +215]
all 18	high, 5%	no text	hand + disc.	−10 [−18, −3]	0 [0, 0]	+612 [+414, +846]	−4 [−8, −1]	−14 [−23, −7]
all 18	high, 5%	no text	zero-config	−202 [−256, −155]	0 [0, 0]	+1,353 [+995, +1,755]	+371 [+309, +441]	+169 [+128, +215]
NEW layouts	clean	text	discovered	0 [0, 0]	0 [0, 0]	+522 [+285, +788]	0 [0, 0]	0 [0, 0]
NEW layouts	clean	text	hand + disc.	0 [0, 0]	0 [0, 0]	+394 [+213, +599]	0 [0, 0]	0 [0, 0]
NEW layouts	clean	text	zero-config	−5 [−13, 0]	−5 [−13, 0]	+430 [+199, +684]	+7 [+1, +16]	+2 [0, +5]
NEW layouts	clean	no text	discovered	0 [0, 0]	0 [0, 0]	+571 [+319, +855]	0 [0, 0]	0 [0, 0]
NEW layouts	clean	no text	hand + disc.	0 [0, 0]	0 [0, 0]	+550 [+308, +817]	0 [0, 0]	0 [0, 0]
NEW layouts	clean	no text	zero-config	0 [0, 0]	0 [0, 0]	+571 [+319, +855]	0 [0, 0]	0 [0, 0]
NEW layouts	low, 1.5%	text	discovered	−8 [−15, −2]	+1 [0, +3]	+573 [+335, +839]	+9 [−1, +19]	+1 [−9, +10]
NEW layouts	low, 1.5%	text	hand + disc.	0 [0, 0]	0 [0, 0]	+408 [+241, +594]	−6 [−14, −1]	−6 [−14, −1]
NEW layouts	low, 1.5%	text	zero-config	−18 [−31, −7]	−9 [−21, 0]	+458 [+240, +704]	+26 [+11, +41]	+8 [−3, +19]
NEW layouts	low, 1.5%	no text	discovered	−19 [−30, −10]	0 [0, 0]	+435 [+232, +669]	+45 [+29, +61]	+26 [+12, +39]
NEW layouts	low, 1.5%	no text	hand + disc.	0 [0, 0]	0 [0, 0]	+408 [+219, +622]	−8 [−18, −1]	−8 [−18, −1]
NEW layouts	low, 1.5%	no text	zero-config	−19 [−30, −10]	0 [0, 0]	+434 [+230, +668]	+45 [+29, +61]	+26 [+12, +39]
NEW layouts	high, 5%	text	discovered	−60 [−89, −35]	−2 [−7, +2]	+837 [+504, +1,206]	+111 [+76, +149]	+51 [+31, +73]
NEW layouts	high, 5%	text	hand + disc.	−1 [−3, 0]	0 [0, 0]	+356 [+204, +528]	−4 [−9, 0]	−5 [−10, 0]
NEW layouts	high, 5%	text	zero-config	−58 [−87, −33]	−1 [−6, +3]	+840 [+507, +1,216]	+130 [+87, +177]	+72 [+41, +109]
NEW layouts	high, 5%	no text	discovered	−158 [−207, −114]	0 [0, 0]	+1,086 [+744, +1,456]	+292 [+237, +350]	+134 [+98, +174]
NEW layouts	high, 5%	no text	hand + disc.	−8 [−15, −1]	0 [0, 0]	+501 [+305, +721]	−3 [−7, 0]	−11 [−19, −4]

Group	Extraction	Mode	Identities	qcore8−qcore7 silent	qcore8−qcore7 broken	qcore8−qcore7 flagged	qcore7−hand silent	qcore8−hand silent
NEW layouts	high, 5%	no text	zero-config	−158 [−207, −114]	0 [0, 0]	+1,095 [+755, +1,466]	+292 [+237, +350]	+134 [+98, +174]
OLD domains	clean	text	discovered	0 [0, 0]	0 [0, 0]	+113 [+65, +167]	0 [0, 0]	0 [0, 0]
OLD domains	clean	text	hand + disc.	0 [0, 0]	0 [0, 0]	+107 [+59, +161]	0 [0, 0]	0 [0, 0]
OLD domains	clean	text	zero-config	0 [0, 0]	0 [0, 0]	+88 [+40, +140]	0 [0, 0]	0 [0, 0]
OLD domains	clean	no text	discovered	0 [0, 0]	0 [0, 0]	+114 [+67, +166]	0 [0, 0]	0 [0, 0]
OLD domains	clean	no text	hand + disc.	0 [0, 0]	0 [0, 0]	+111 [+64, +164]	0 [0, 0]	0 [0, 0]
OLD domains	clean	no text	zero-config	0 [0, 0]	0 [0, 0]	+114 [+67, +166]	0 [0, 0]	0 [0, 0]
OLD domains	low, 1.5%	text	discovered	−6 [−11, −2]	0 [0, 0]	+141 [+88, +198]	+11 [+5, +18]	+5 [−1, +12]
OLD domains	low, 1.5%	text	hand + disc.	−2 [−5, 0]	0 [0, 0]	+107 [+60, +159]	0 [0, 0]	−2 [−5, 0]
OLD domains	low, 1.5%	text	zero-config	−6 [−11, −2]	0 [0, 0]	+114 [+59, +174]	+13 [+6, +21]	+7 [+1, +14]
OLD domains	low, 1.5%	no text	discovered	−10 [−17, −4]	0 [0, 0]	+150 [+94, +212]	+20 [+12, +29]	+10 [+3, +18]
OLD domains	low, 1.5%	no text	hand + disc.	−2 [−5, 0]	0 [0, 0]	+107 [+61, +159]	0 [0, 0]	−2 [−5, 0]
OLD domains	low, 1.5%	no text	zero-config	−10 [−17, −4]	0 [0, 0]	+151 [+95, +213]	+20 [+12, +29]	+10 [+3, +18]
OLD domains	high, 5%	text	discovered	−19 [−30, −9]	0 [0, 0]	+192 [+123, +267]	+32 [+20, +46]	+13 [+6, +21]
OLD domains	high, 5%	text	hand + disc.	−1 [−3, 0]	0 [0, 0]	+95 [+52, +144]	0 [0, 0]	−1 [−3, 0]
OLD domains	high, 5%	text	zero-config	−18 [−28, −9]	0 [0, 0]	+194 [+127, +268]	+37 [+24, +51]	+19 [+11, +28]
OLD domains	high, 5%	no text	discovered	−44 [−61, −29]	0 [0, 0]	+262 [+187, +343]	+79 [+62, +98]	+35 [+23, +48]
OLD domains	high, 5%	no text	hand + disc.	−2 [−5, 0]	0 [0, 0]	+111 [+68, +160]	−1 [−3, 0]	−3 [−7, 0]
OLD domains	high, 5%	no text	zero-config	−44 [−61, −29]	0 [0, 0]	+258 [+182, +339]	+79 [+62, +98]	+35 [+23, +48]

Table S7.8: Legacy scoring (the rule of every earlier report) on the second fresh held-out set, all 18 bundles: silent errors, qcore7 / qcore8
Model alone under this rule: clean 0, low 191, high 640 wrong.

Extraction	Mode	Hand	Discovered	Hand + disc.	Zero-config
clean	text	0 / 0	0 / 0	0 / 0	7 / 2
clean	no text	0 / 0	0 / 0	0 / 0	0 / 0
low, 1.5%	text	6 / 6	17 / 10	2 / 1	35 / 19
low, 1.5%	no text	17 / 17	72 / 51	11 / 10	72 / 51
high, 5%	text	8 / 8	114 / 49	7 / 5	136 / 75
high, 5%	no text	36 / 36	365 / 184	32 / 24	365 / 184

Table S7.9: Injected errors by kind, high extraction, all 18 bundles (strict): fixed / flagged / silent

Mode	Identities	Version	blank	column shift	decimal drop	digit sub	drop digit	extra digit	label as value	pct scale	row shift	sign flip	transpose
text	hand	all	18/15/0	41/26/0	2/8/0	122/84/8	40/13/0	26/7/1	47/26/0	0/1/0	28/26/0	70/7/2	44/26/3
text	zero-config	qcore7	9/24/0	34/16/17	2/6/2	83/70/61	18/24/11	14/12/8	36/36/1	0/0/1	19/16/19	22/29/28	26/26/21
text	zero-config	qcore8	7/26/0	31/27/9	2/8/0	82/96/36	21/26/6	15/14/5	32/40/1	0/0/1	20/22/12	30/33/16	28/37/8
no text	hand	all	0/33/0	0/63/4	0/9/1	0/199/15	0/48/5	0/32/2	0/69/4	0/1/0	0/51/3	40/37/2	0/67/6
no text	zero-config	qcore7	0/33/0	0/17/50	0/2/8	0/58/156	0/16/37	0/13/21	0/69/4	0/0/1	0/22/32	0/28/51	0/20/53
no text	zero-config	qcore8	0/33/0	0/41/26	0/7/3	0/124/90	0/36/17	0/26/8	0/71/2	0/0/1	0/37/17	0/56/23	0/49/24

Table S7.10: By domain, high extraction (strict): silent / broken / flagged
The first four domains are the NEW layouts. Numbers per domain in the header.

Mode	Identities	Version	budget (3,738)	census (2,208)	inventory (2,568)	utilbank (1,346)	corp fin (1,536)	invoices (986)	public stats (1,486)
text	hand	all	2 / 1 / 652	0 / 0 / 163	3 / 0 / 142	6 / 0 / 45	1 / 0 / 99	3 / 0 / 58	0 / 0 / 283
text	zero-config	qcore7	48 / 0 / 238	17 / 0 / 109	43 / 6 / 114	33 / 7 / 19	11 / 0 / 80	12 / 0 / 33	18 / 0 / 159
text	zero-config	qcore8	16 / 0 / 920	10 / 0 / 175	29 / 7 / 181	28 / 5 / 44	2 / 0 / 156	10 / 0 / 42	11 / 0 / 268
no text	hand	all	0 / 0 / 874	5 / 0 / 549	13 / 0 / 389	16 / 0 / 224	4 / 0 / 298	4 / 0 / 160	0 / 0 / 303
no text	zero-config	qcore7	94 / 0 / 317	68 / 0 / 157	109 / 0 / 23	55 / 0 / 22	27 / 0 / 163	30 / 0 / 47	30 / 0 / 154
no text	zero-config	qcore8	23 / 0 / 1081	51 / 0 / 246	48 / 0 / 237	46 / 0 / 50	8 / 0 / 301	20 / 0 / 77	15 / 0 / 244

Table S7.11: Discovered identities by source, high extraction, zero-config, summed over the 18 bundles

vsum, hsum: row and column sums; *product*; *equal*: the same number printed twice; *aggtie*: total = sum of a field over items (qcore7); new in qcore8: *accum* (year to date = previous year to date + period), *chain* (a cell carried to a later document), *rowprod* (rate row × base row), *lagtie* (roll-forward over the previous column), *vsum_sparse* (non-contiguous sums).

Version	Mode	vsum	hsum	product	equal	aggtie	accum	chain	rowprod	lagtie	vsum_sparse
qcore6	text	1,237	674	457	371	0	0	0	0	0	0
qcore6	no text	1,188	516	100	307	0	0	0	0	0	0
qcore7	text	1,254	663	457	374	40	0	0	0	0	0
qcore7	no text	1,241	522	100	307	6	0	0	0	0	0
qcore8	text	1,493	1,268	1,060	374	40	1,006	439	65	48	7
qcore8	no text	1,385	1,211	914	307	6	984	409	63	48	32

S8 Development grid of frozen qcore8

Development data only. Every set below was spent before qcore8 was written, and qcore8 was tuned on them, set H (qcore7's fresh synthetic held-out set, corp_fin, invoices and public_stats 06–08) included. Frozen qcore8 was run on every (set, model, extraction, mode, identity setting) of the grid (898 runs, all completed); the qcore7 runs are qcore7's own frozen grid and, for set H, its held-out run. Pooled over models, extractions and modes; legacy scoring, plus strict scoring for set H. Hand-identity check: in all 244 hand runs qcore8 returned exactly qcore7's fixes (244) and tier 2 items (244). Every number qcore8 breaks on these sets is a hand-mode break that qcore7 makes too, on the same cell with the same new value (OCR look-alike repairs on SEC sets 1–3, the Snap-on swap and the Salesforce cell, Section 5.2); with discovered identities or zero-config qcore8 breaks none. Source: `out/qcore8_work/dev_summary8.md`, `dev_report8.md`, `dev_report8.json`.

Set	Identities	Extractions, scoring	Numbers	Silent qcore7 / qcore8	Broken	Fixed	To a person
SEC set 1	hand	real, legacy	25,728	885 / 885	2 / 2	563 / 563	14.3 / 14.3%
SEC set 1	discovered	real, legacy	12,864	597 / 586	0 / 0	476 / 477	4.0 / 4.3%
SEC set 1	hand + disc.	real, legacy	12,864	379 / 374	0 / 0	518 / 518	8.6 / 8.8%
SEC set 1	zero-config	real, legacy	12,864	605 / 593	0 / 0	260 / 259	5.8 / 6.0%
SEC set 2	hand	real, legacy	53,088	1,615 / 1,615	3 / 3	1,001 / 1,001	12.9 / 12.9%
SEC set 2	discovered	real, legacy	26,544	1,058 / 1,048	0 / 0	804 / 801	4.5 / 4.6%
SEC set 2	hand + disc.	real, legacy	26,544	699 / 694	0 / 0	871 / 871	7.8 / 7.9%
SEC set 2	zero-config	real, legacy	26,544	1,080 / 1,072	0 / 0	397 / 381	5.9 / 6.1%
SEC set 3	hand	real, legacy	62,472	2,304 / 2,304	6 / 6	1,108 / 1,108	16.2 / 16.2%
SEC set 3	discovered	real, legacy	31,236	1,502 / 1,497	0 / 0	955 / 955	5.7 / 5.9%
SEC set 3	hand + disc.	real, legacy	31,236	999 / 1,002	0 / 0	1,011 / 1,011	9.9 / 9.9%
SEC set 3	zero-config	real, legacy	31,236	1,536 / 1,525	0 / 0	481 / 449	7.7 / 8.0%
CORD test	hand	real, legacy	7,984	365 / 365	0 / 0	28 / 28	20.9 / 20.9%
CORD test	discovered	real, legacy	3,992	151 / 142	0 / 0	25 / 25	19.5 / 21.6%
CORD test	hand + disc.	real, legacy	3,992	134 / 126	0 / 0	29 / 29	21.2 / 23.4%
CORD test	zero-config	real, legacy	3,992	153 / 144	0 / 0	25 / 25	19.3 / 21.5%
Notes tables	hand	real, legacy	88,668	54 / 54	3 / 3	8 / 8	7.1 / 7.1%
Notes tables	discovered	real, legacy	88,668	95 / 95	0 / 0	1 / 1	7.7 / 8.4%
Notes tables	hand + disc.	real, legacy	88,668	47 / 47	0 / 0	1 / 1	7.8 / 8.5%
Notes tables	zero-config	real, legacy	88,668	101 / 95	6 / 0	0 / 0	7.8 / 8.6%
Real scans	hand	real, legacy	6,054	13 / 13	0 / 0	0 / 0	16.2 / 16.2%
Real scans	discovered	real, legacy	2,018	5 / 5	0 / 0	0 / 0	20.6 / 20.6%
Real scans	hand + disc.	real, legacy	2,018	5 / 5	0 / 0	0 / 0	22.6 / 22.8%
Real scans	zero-config	real, legacy	2,018	5 / 5	0 / 0	0 / 0	20.6 / 20.6%
Synthetic 02–05	hand	real, legacy	31,568	4 / 4	0 / 0	0 / 0	2.1 / 2.1%
Synthetic 02–05	discovered	real, legacy	31,568	4 / 4	0 / 0	0 / 0	1.1 / 2.9%
Synthetic 02–05	hand + disc.	real, legacy	31,568	4 / 4	0 / 0	0 / 0	2.3 / 4.0%
Synthetic 02–05	zero-config	real, legacy	31,568	4 / 4	0 / 0	0 / 0	1.4 / 2.9%
Set H	hand	clean, legacy	11,840	0 / 0	0 / 0	0 / 0	2.1 / 2.1%
Set H	discovered	clean, legacy	11,840	0 / 0	0 / 0	0 / 0	1.3 / 3.3%
Set H	hand + disc.	clean, legacy	11,840	0 / 0	0 / 0	0 / 0	2.6 / 4.5%
Set H	zero-config	clean, legacy	11,840	0 / 0	0 / 0	0 / 0	1.5 / 3.4%
Set H	hand	clean, strict	11,840	0 / 0	0 / 0	0 / 0	2.1 / 2.1%
Set H	discovered	clean, strict	11,840	0 / 0	0 / 0	0 / 0	1.3 / 3.3%
Set H	hand + disc.	clean, strict	11,840	0 / 0	0 / 0	0 / 0	2.6 / 4.5%
Set H	zero-config	clean, strict	11,840	0 / 0	0 / 0	0 / 0	1.5 / 3.4%
Set H	hand	high, legacy	11,840	22 / 22	0 / 0	194 / 194	16.9 / 16.9%
Set H	discovered	high, legacy	11,840	172 / 82	0 / 0	127 / 150	8.2 / 14.4%
Set H	hand + disc.	high, legacy	11,840	21 / 9	0 / 0	184 / 185	17.5 / 19.2%
Set H	zero-config	high, legacy	11,840	178 / 88	0 / 0	91 / 108	8.7 / 15.0%
Set H	hand	high, strict	11,840	27 / 27	0 / 0	194 / 194	16.9 / 16.9%
Set H	discovered	high, strict	11,840	201 / 90	0 / 0	127 / 150	8.2 / 14.4%
Set H	hand + disc.	high, strict	11,840	25 / 11	0 / 0	184 / 185	17.5 / 19.2%
Set H	zero-config	high, strict	11,840	207 / 96	0 / 0	91 / 108	8.7 / 15.0%
Set H	hand	low, legacy	11,840	9 / 9	0 / 0	58 / 58	6.4 / 6.4%
Set H	discovered	low, legacy	11,840	29 / 7	0 / 0	38 / 40	4.7 / 6.9%
Set H	hand + disc.	low, legacy	11,840	9 / 0	0 / 0	53 / 50	7.5 / 9.7%
Set H	zero-config	low, legacy	11,840	30 / 8	0 / 0	31 / 30	4.9 / 7.0%
Set H	hand	low, strict	11,840	11 / 11	0 / 0	59 / 59	6.4 / 6.4%
Set H	discovered	low, strict	11,840	41 / 13	0 / 0	38 / 40	4.7 / 6.9%
Set H	hand + disc.	low, strict	11,840	11 / 0	0 / 0	53 / 50	7.5 / 9.7%
Set H	zero-config	low, strict	11,840	42 / 14	0 / 0	31 / 30	4.9 / 7.0%

Set	Identities	Extractions, scoring	Numbers	Silent qcore7 / qcore8	Broken	Fixed	To a person
SEC dev	hand	injected, legacy	4,890	17 / 17	0 / 0	54 / 54	3.2 / 3.2%
SEC dev	discovered	injected, legacy	4,890	30 / 30	0 / 0	39 / 39	3.4 / 3.4%
SEC dev	hand + disc.	injected, legacy	4,890	13 / 13	0 / 0	51 / 51	4.0 / 4.0%
SEC dev	zero-config	injected, legacy	4,890	33 / 30	0 / 0	27 / 35	3.6 / 3.5%
SEC dev	hand	real, legacy	19,560	113 / 113	0 / 0	100 / 100	8.9 / 8.9%
SEC dev	discovered	real, legacy	19,560	147 / 142	0 / 0	71 / 71	7.8 / 7.3%
SEC dev	hand + disc.	real, legacy	19,560	113 / 109	0 / 0	100 / 100	6.9 / 6.9%
SEC dev	zero-config	real, legacy	19,560	157 / 151	0 / 0	50 / 54	7.6 / 7.2%
Notes dev	hand	injected, legacy	13,178	67 / 67	0 / 0	18 / 18	1.2 / 1.2%
Notes dev	discovered	injected, legacy	13,178	61 / 52	0 / 0	12 / 12	2.0 / 2.8%
Notes dev	hand + disc.	injected, legacy	13,178	40 / 34	0 / 0	15 / 15	2.4 / 3.2%
Notes dev	zero-config	injected, legacy	13,178	65 / 56	0 / 0	11 / 11	1.8 / 2.6%
Notes dev	hand	real, legacy	39,534	36 / 36	0 / 0	16 / 16	6.6 / 6.6%
Notes dev	discovered	real, legacy	39,534	36 / 36	0 / 0	4 / 4	6.9 / 7.7%
Notes dev	hand + disc.	real, legacy	39,534	36 / 36	0 / 0	4 / 4	7.2 / 8.0%
Notes dev	zero-config	real, legacy	39,534	40 / 40	0 / 0	0 / 0	7.1 / 7.9%
CORD train	hand	injected, legacy	16,810	33 / 33	0 / 0	63 / 63	3.5 / 3.5%
CORD train	discovered	injected, legacy	16,810	81 / 68	0 / 0	33 / 33	4.1 / 5.2%
CORD train	hand + disc.	injected, legacy	16,810	42 / 42	0 / 0	54 / 56	5.7 / 6.6%
CORD train	zero-config	injected, legacy	16,810	86 / 73	0 / 0	31 / 31	3.9 / 5.1%
CORD train	hand	real, legacy	16,810	0 / 0	0 / 0	0 / 0	2.3 / 2.3%
CORD train	discovered	real, legacy	16,810	0 / 0	0 / 0	0 / 0	3.1 / 4.0%
CORD train	hand + disc.	real, legacy	16,810	0 / 0	0 / 0	0 / 0	4.2 / 5.0%
CORD train	zero-config	real, legacy	16,810	0 / 0	0 / 0	0 / 0	2.9 / 3.8%
CORD val.	hand	injected, legacy	3,768	43 / 43	0 / 0	34 / 34	12.3 / 12.3%
CORD val.	discovered	injected, legacy	3,768	108 / 80	0 / 0	15 / 17	9.4 / 11.2%
CORD val.	hand + disc.	injected, legacy	3,768	42 / 43	0 / 0	36 / 35	13.0 / 13.3%
CORD val.	zero-config	injected, legacy	3,768	112 / 84	0 / 0	8 / 9	8.9 / 10.9%
CORD val.	hand	real, legacy	3,768	21 / 21	0 / 0	2 / 2	8.5 / 8.5%
CORD val.	discovered	real, legacy	3,768	22 / 28	0 / 0	3 / 2	9.4 / 10.4%
CORD val.	hand + disc.	real, legacy	3,768	20 / 20	0 / 0	3 / 2	11.4 / 12.2%
CORD val.	zero-config	real, legacy	3,768	25 / 31	0 / 0	1 / 0	8.7 / 9.6%
Real-scan dev	hand	real, legacy	2,760	8 / 8	0 / 0	0 / 0	24.9 / 24.9%
Real-scan dev	discovered	real, legacy	2,760	16 / 16	0 / 0	0 / 0	19.2 / 21.6%
Real-scan dev	hand + disc.	real, legacy	2,760	0 / 0	0 / 0	0 / 0	36.8 / 37.0%
Real-scan dev	zero-config	real, legacy	2,760	16 / 16	0 / 0	0 / 0	18.3 / 18.6%
Synthetic 00–01	hand	injected, legacy	16,364	16 / 16	0 / 0	354 / 354	7.3 / 7.3%
Synthetic 00–01	discovered	injected, legacy	16,364	232 / 144	0 / 0	202 / 234	4.3 / 8.1%
Synthetic 00–01	hand + disc.	injected, legacy	16,364	16 / 8	0 / 0	352 / 350	7.9 / 10.1%
Synthetic 00–01	zero-config	injected, legacy	16,364	247 / 160	0 / 0	150 / 163	4.6 / 8.4%
Synthetic 00–01	hand	real, legacy	16,364	6 / 6	0 / 0	0 / 0	1.4 / 1.4%
Synthetic 00–01	discovered	real, legacy	16,364	6 / 6	0 / 0	0 / 0	0.7 / 2.8%
Synthetic 00–01	hand + disc.	real, legacy	16,364	6 / 6	0 / 0	0 / 0	1.6 / 3.5%
Synthetic 00–01	zero-config	real, legacy	16,364	6 / 6	0 / 0	0 / 0	0.7 / 2.7%

S9 Further SEC results

Material moved from Appendix I of the paper.

Table S9.1: Filers' calculation-linkbase identities against set 1's identities, frozen qcore3, held-out set 1 poor scans
Layout: set 1's 484 held-out identities (layout-inferred and semantic). Linkbase: 486 identities from the filers' calculation linkbases, semantic relations kept, layout rules kept for BRK-B, MSFT and PLD (no linkbase file). Over all 20 set-1 filings the linkbases give 752 identities, all holding on the truth, and match 621 of the 1,054 earlier ones.

Identities	Run	Errors	Fixed	To tier 2	Missed	Silent	Flagged (wrong)
Layout	Opus, text	30	18	12	0	0	22 (12)
Linkbase	Opus, text	30	18	12	0	0	19 (12)
Layout	Opus, no text	30	0	17	13	13	62 (17)
Linkbase	Opus, no text	30	0	16	14	14	58 (16)
Layout	Haiku, text	109	82	27	0	0	73 (27)
Linkbase	Haiku, text	109	82	27	0	0	61 (27)
Layout	Haiku, no text	109	0	52	57	57	155 (52)
Linkbase	Haiku, no text	109	0	51	58	58	155 (51)

Table S9.2: qcore3 against qcore4 on the SEC development filings only (30 statements, 2,445 numbers), poor scans
Accuracy after tier 1 / silent / flagged (wrong) / broken. qcore3 has no OCR mode; its OCR rows feed it the OCR words as if they were the text layer. Accuracy before the checker: Haiku 96.07%, Opus 99.47%.

Run, mode	qcore3	qcore4
Haiku, text	99.02% / 0 / 106 (24) / 0	99.02% / 0 / 61 (24) / 0
Haiku, OCR docTR	91.04% / 28 / 1,753 (191) / 126	96.73% / 23 / 424 (57) / 0
Haiku, OCR Tesseract	92.56% / 22 / 1,900 (160) / 86	96.20% / 29 / 477 (64) / 0
Haiku, no text	96.07% / 49 / 200 (47) / 0	96.07% / 49 / 200 (47) / 0
Opus, text	99.88% / 0 / 9 (3) / 0	99.88% / 0 / 7 (3) / 0
Opus, OCR docTR	93.95% / 29 / 1,722 (119) / 135	99.47% / 2 / 261 (11) / 0
Opus, OCR Tesseract	95.75% / 23 / 1,880 (81) / 91	99.47% / 3 / 268 (10) / 0
Opus, no text	99.47% / 7 / 25 (6) / 0	99.47% / 7 / 25 (6) / 0

S10 Changes made on development data, version by version

The full record behind the versions table of Appendix J of the paper.

qcore2 (synthetic dev bundles 00–01). A lone dash followed by words is not a zero. A prose reader takes only the number printed right after the label phrase. Digit-bearing tokens in a label or entity (invoice numbers, dates) must be printed on the matched line. A heading must print no values, labels that name the entity win, and a column read must sit under its header cell. New schema options: a `dot_thousands` number format and a per-field scale. Misread repair works at the extracted value's decimal precision, and tier 2 suggestions are never applied internally. On dev, text-mode breaks fell from 82 to 0 and tier 2 items from 207 to 37.

qcore3 (CORD train and validation; 10 SEC dev filings). A `grouped` number format; unit statements in parentheses; a page with no unit statement inherits the nearest earlier page's; date and stacked headers; a more specific label claims its line; a label that finds its line but cannot read it vetoes the field; label-less entity rows are not read; swaps and flips are confirmed on the fields' own lines; and the optional `label_read: backed`. A CORD adapter assigns each shared receipt-level label to the field that carries it most often on TRAIN. On dev, qcore3 broke 0 numbers in every group and mode, where qcore2 broke 1,817 on CORD train gold extractions and 11 on each clean SEC dev extraction.

qcore4 (SEC development filings: their clean and poor-scan extractions, OCR of their poor scans, injected errors; CORD train and validation; regression on synthetic dev bundles and the PE validation). An OCR mode with the two-reader rule of Section 3.4, a per-document scale vote, and OCR cleanup (rows fitted along the page slope, specks dropped, words snapped to the label vocabulary, header years snapped, missing parentheses restored, a punctuation mark before three final digits read as a thousands comma). Product terms in identities, and the CORD adapter's line identity rewritten from price $-\,c\cdot$ unit price with the true count c to price $-$ count $\times$ unit price. A row read for label-less entity rows. No sign flip away from a declared sign. Identity-localised flags. A threshold to flag high-confidence OCR disagreements was tried at 0.9 and 0.97, added flags without catching an error, and is off. On the development filings qcore4 broke no number in any mode, where qcore3 fed OCR as a text layer broke 86 to 135 per run (Table S9.2); in text mode its fixes, silent and broken counts equal qcore3's and only the flags shrink. On CORD development data it made no wrong or broken change; a broken count found during tuning (a count read from the previous item's continuation row) was fixed by preferring the continuation row below.

qcore5 (development tables of the notes benchmark, 10 filers, 76 tables, 6,589 numbers, with the Opus, Haiku and GPT extractions and injected errors; SEC development poor scans, their OCR and injected errors; CORD train and validation; synthetic dev bundles 00–01; PE validation). It changes only the page read. A read changes a value only under rule (a), identity-backed, or rule (b), unambiguous and corroborated (Section 3.2); every other read is a tier 2 item with the reason it was not taken. The guard runs jointly over all documents, and when an identity that tied breaks, the one other input whose page cell makes it tie again completes the correction. A read never changes a value by a power of 1000. Unit statements in billions and unit statements broken over two lines are read, and a line running across the row-label

edge ends a stacked header. Tried and rejected on development data: rule (b) alone with qcore4's per-document guard (notes breaks 0, but PE validation fixes fell from 4,718 to 1,411), and using a voted page scale to read values in text mode (12 wrong changes on the PE validation).
qcore6 (the qcore5 development data, plus the real-scan development split and real re-reads of development crops by Haiku and Opus readers, two rounds). The weaknesses addressed were low identity coverage, OCR and a scan's text layer posing as a trustworthy page, errors that agree or cancel, and the hand-built files every domain needed. Changes: zero-config identity discovery and schema inference (Section 3.5); the re-read tier (Section 3.6); a scanned PDF's text layer treated as an OCR (on the real-scan development split qcore5 broke 4 correct numbers per model in text mode, qcore6 none); identity coverage reported per run. Tried and not kept: a uniqueness filter on discovery's page reads (cut SEC coverage from 82% to 78%); the backed label read for every zero-config run (SEC zero-config fixes fell from 57 to 20); equality ties without the year check (paired columns of different years); zero-config on CORD with the replace-style label read and free signs (23 of 8,405 correct numbers broken on CORD train); exempting every page-confirmed label from the per-document check (12 broken on the notes development tables); and the first re-read gate (Section 5.5). With hand files, qcore6 gives qcore5's results on every development set except scanned text layers. Source: `qcore6/CHANGES.md`, `out/qcore6_work/dev_summary6.md`.
qcore7 (every qcore6 held-out set, now spent: SEC sets 1–3, CORD test, the notes tables, the real scans, synthetic 02–05; and qcore6's development grid). The weaknesses addressed, measured on those sets: zero-config broke correct numbers (age bands read as values; a discovered identity that held by accident backing a change) and left more numbers silent than hand identities (fewer identities discovered; without page text, on scanned forms and receipts, almost no flags). The changes are the four of Section 3.5, all gated on something being discovered; nine tests were added. Costs on development data: more flags on scanned forms, unchanged no-text results on SEC filings, and fewer fixes, because changes resting only on unbacked discovered identities become suggestions. Source: `qcore7/CHANGES.md`, `out/qcore7_work/dev_summary7.md` (Section S4).
qcore8 (everything above, plus set H, qcore7's fresh synthetic set, spent by qcore7's run). The weaknesses addressed, measured on those sets: the two notes-table breaks of zero-config qcore7, and zero-config leaving 3 to 9 times as many numbers silent as hand identities on set H, mostly because discovery missed long totals in error-heavy documents, cross-document ties and products with rates. The changes are the five of Section 3.5, all in zero-config code; six tests were added and one qcore7 test changed on purpose (a digit misread under a weak label now goes to tier 2). Three guards in the inferred schema's exact-label rule were added after development breaks: a long label whose words a shorter line also prints (two SEC development numbers, before the first freeze), a footnote marker after the label (one notes-table number per model, first freeze), and fewer than three significant digits (two SEC set 3 numbers, second freeze). Tried and not kept: naming every other unconfirmed input of a discovered identity when a misread suggestion rests on discovered identities alone (set H high, no text, 73 to 68 silent, but more flags on clean extractions). Costs on development data: CORD validation zero-config silent 25 to 31, fewer fixes on SEC sets 2 and 3, more flags on clean data, and slower discovery (322 to 350 s for the largest development run against 133 s). Source: `qcore8/CHANGES.md`, `out/qcore8_work/dev_summary8.md` (Section S8).
qcore9 (everything above, plus set I, qcore8's fresh set, spent by qcore8's run). The weaknesses addressed, measured there: qcore8's zero-config breaks (9 bank-statement checks printed twice with opposite signs; 7 cells of an inventory row read from a neighbour whose numbers include the record's identifier; one week-total completion on the wrong day), one hand-mode and one discovered-identity break from loose scaled page matches (a budget report storing values at the page's own unit), and the gap to hand identities without page text on the SEC development sets. The changes are the five of Section 3.5; eight tests were added. Measured one at a time on SEC development sets in zero-config (silent / false flags over 12 runs per mode): slip admission, without page text -13 / $+11$ (with it -7 / $+109$, so off with page text); tight tolerance -28 / $+116$ (with page text -3 / $+56$, off); section-total sums -7 / $+10$; equal ties -17 / $+15$; keeping a rule-(b) read's page basis raised text-mode fixes from 1,161 to 1,276. Tried and not kept: the two discovery checks with page text; the tight tolerance at 2.5 units (-3 silent); a first version of change 3 that required every scaled match to be at least one page unit, which stopped confirming SEC sub-unit lines and added 54 hand-mode flags on development sets. Not done: a generic rule for the two hand-mode breaks qcore8 already had on development data (the Snap-on swap on SEC set 3 and the Salesforce cell on the notes tables). Source: `qcore9/CHANGES.md`, `out/qcore9_work/FROZEN.md`.

Table S10.1: qcore4 against qcore5 on development data only

Accuracy after tier 1 / silent / flagged (wrong) / broken / fixed (source: `out/qcore5_work/dev_summary5.md`). Accuracy before: notes Opus 100.00%, Haiku 99.76%, GPT 83.81%; SEC poor scans Haiku 96.07%, Opus 99.47%.

Run, mode	qcore4	qcore5
Notes, Opus, text	99.15% / 52 / 731 (4) / 56 / 0	100.00% / 0 / 33 (0) / 0 / 0
Notes, Haiku, text	99.03% / 52 / 739 (12) / 56 / 8	99.76% / 0 / 49 (16) / 0 / 0
Notes, GPT, text	83.52% / 60 / 1,861 (1,026) / 56 / 37	84.05% / 8 / 1,280 (1,043) / 0 / 16
Notes, Opus + injected errors, text	98.20% / 234 / 2,316 (122) / 164 / 48	99.04% / 82 / 286 (107) / 0 / 51
SEC, Haiku poor scans, text	99.02% / 0 / 61 (24) / 0 / 72	99.06% / 0 / 60 (23) / 0 / 73
SEC, Haiku poor scans, OCR docTR	96.73% / 23 / 424 (57) / 0 / 16	96.65% / 23 / 438 (59) / 0 / 14
SEC, Opus poor scans, text	99.88% / 0 / 7 (3) / 0 / 10	99.88% / 0 / 7 (3) / 0 / 10
SEC, Opus poor scans, OCR docTR	99.47% / 2 / 261 (11) / 0 / 0	99.47% / 2 / 259 (11) / 0 / 0

No-text runs are identical under both versions; CORD and synthetic dev results are within four fixes of qcore4, with no wrong or broken change.

S11 CORD and SEC, all extractors

Table S11.1: Frozen qcore3, text mode, held-out data

Data	Extractor	Errors	Fixed	To tier 2	Missed	Broken	Accuracy before → after
CORD test	Opus	4	0	2	2	0	99.60% → 99.60%
	Haiku	15	1	12	2	0	98.50% → 98.60%
	Donut-CORD	150	1	128	21	0	84.97% → 85.07%
	Opus, poor scan	136	5	120	11	0	86.37% → 86.87%
	Haiku, poor scan	262	10	236	16	0	73.75% → 74.75%
SEC set 1 held-out	Opus / Haiku, clean	0 / 0	–	–	–	0 / 0	100% → 100%
	Opus, poor scan	30	18	12	0	0	98.60% → 99.44%
	Haiku, poor scan	109	82	27	0	0	94.92% → 98.74%

In no-text mode qcore3 gives the same output as qcore2, since none of its changes apply without a text layer. On the poor scans, tier 1 then fixes nothing and flags 69 of Opus's and 213 of Haiku's CORD errors, and 17 of Opus's and 52 of Haiku's SEC errors.